



US012396586B2

(12) **United States Patent**  
**Wexler et al.**

(10) **Patent No.:** **US 12,396,586 B2**  
(45) **Date of Patent:** **\*Aug. 26, 2025**

(54) **HANGABLE APPARATUS AND SYSTEMS  
AND METHODS THEREFOR**

*A47H 1/104* (2006.01)  
*A47H 1/142* (2006.01)  
*A47H 1/144* (2006.01)  
*A47H 13/01* (2006.01)  
*A47H 13/04* (2006.01)

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(US)

(\*) Notice: Subject to any disclaimer, the term of this  
patent is extended or adjusted under 35  
U.S.C. 154(b) by 0 days.

This patent is subject to a terminal dis-  
claimer.

(52) **U.S. Cl.**

CPC ..... *A47H 1/14* (2013.01); *A47H 1/04*  
(2013.01); *A47H 1/104* (2013.01); *A47H 1/142*  
(2013.01); *A47H 1/144* (2013.01); *A47H 13/01*  
(2013.01); *A47H 13/04* (2013.01)

(58) **Field of Classification Search**

CPC ..... *A47H 1/102*; *A47H 1/104*; *A47H 1/14*;  
*A47H 1/142*; *A47H 1/144*; *A47H 1/04*  
USPC ..... 16/87.2, 94 R, 94 D  
See application file for complete search history.

(56)

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(21) Appl. No.: **18/597,687**

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(22) Filed: **Mar. 6, 2024**

(65) **Prior Publication Data**

US 2024/0206664 A1 Jun. 27, 2024

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O'Brien, Jeffrey, "Notice of Allowance", U.S. Appl. No. 18/141,551,  
filed May 1, 2023; Mailed Dec. 6, 2023.

(Continued)

**Related U.S. Application Data**

(63) Continuation of application No. 18/141,551, filed on  
May 1, 2023, now Pat. No. 11,944,225, which is a  
continuation of application No. 17/684,282, filed on  
Mar. 1, 2022, now Pat. No. 11,723,485, which is a  
continuation of application No. 16/913,815, filed on  
Jun. 26, 2020, now Pat. No. 11,291,325, which is a  
continuation-in-part of application No. 16/588,771,  
filed on Sep. 30, 2019, now Pat. No. 10,945,548,  
which is a continuation-in-part of application No.  
(Continued)

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(74) *Attorney, Agent, or Firm* — Philip H. Burrus, IV

(57)

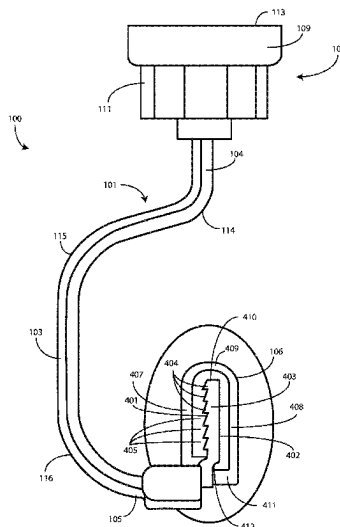
**ABSTRACT**

A track hanger (100) includes a hook (103) and an extension  
(104), a base member (102), and a key (403). The key is  
detachable from the hook, and the base member is detach-  
able from the extension. The key can include a first major  
surface (401) and a second major surface (402). The first  
major surface can define one or more barbs (404) that  
facilitate one-way insertion of the key into a track (106).

**20 Claims, 33 Drawing Sheets**

(51) **Int. Cl.**

*A47H 1/14* (2006.01)  
*A47H 1/04* (2006.01)



**Related U.S. Application Data**

16/045,391, filed on Jul. 25, 2018, now Pat. No.  
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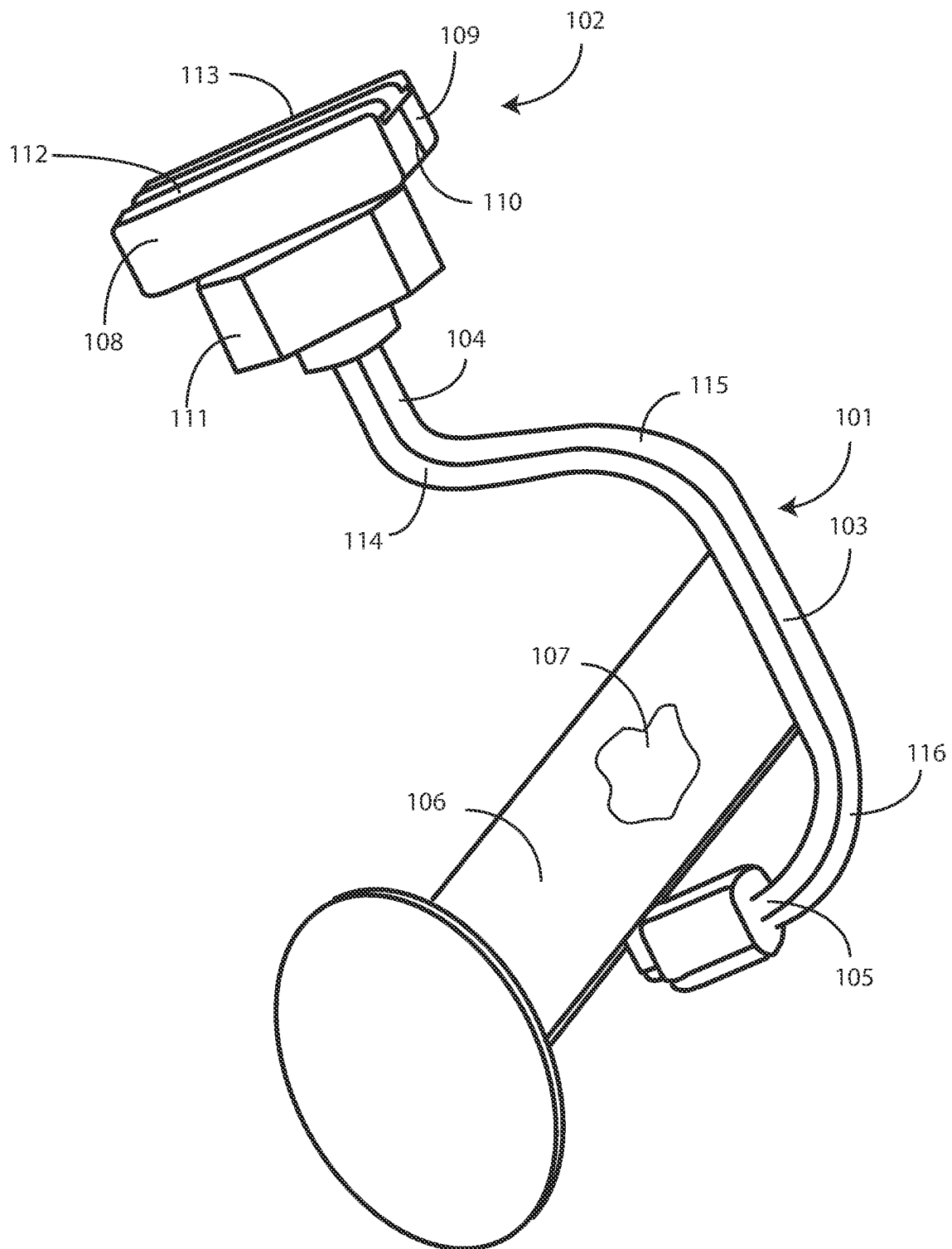
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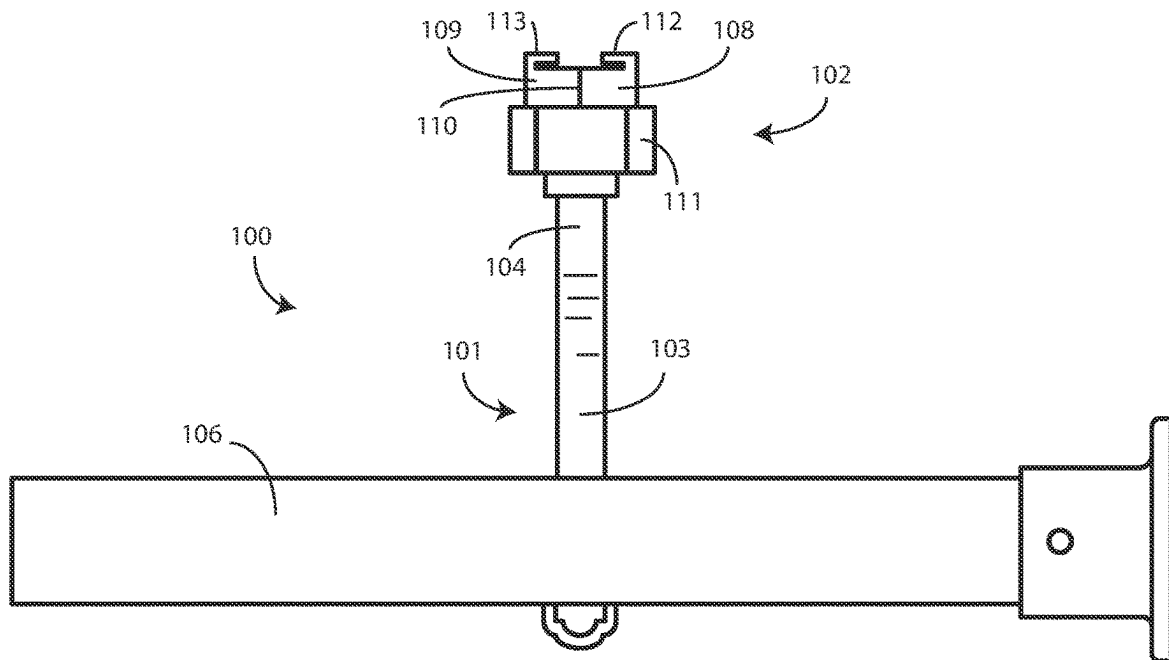
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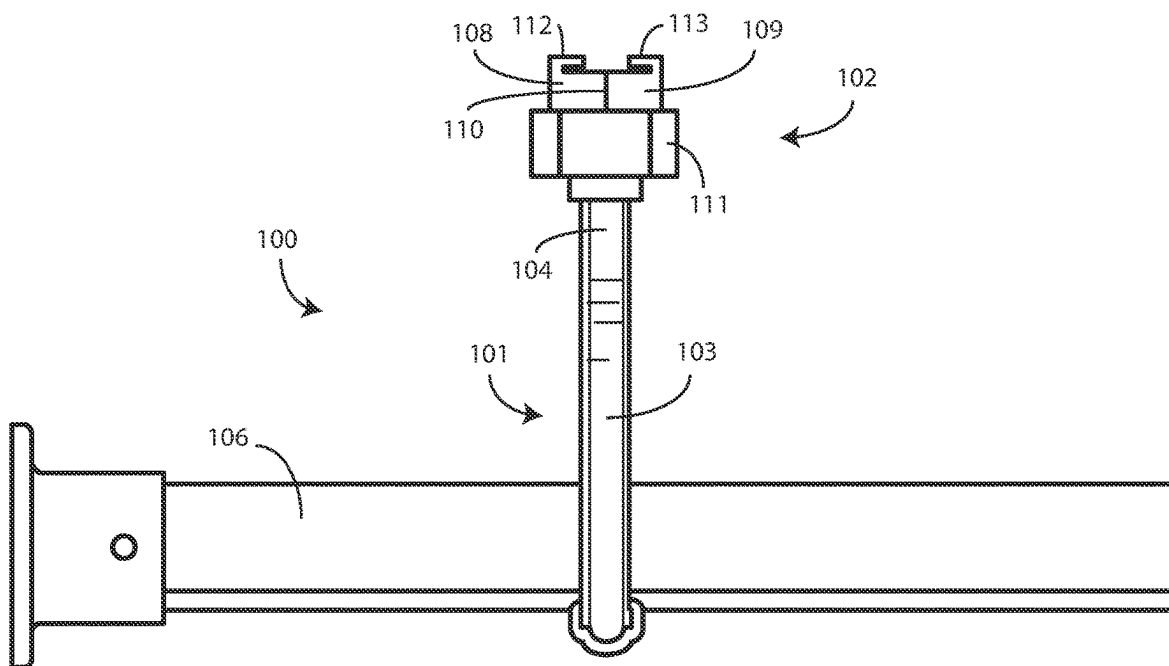
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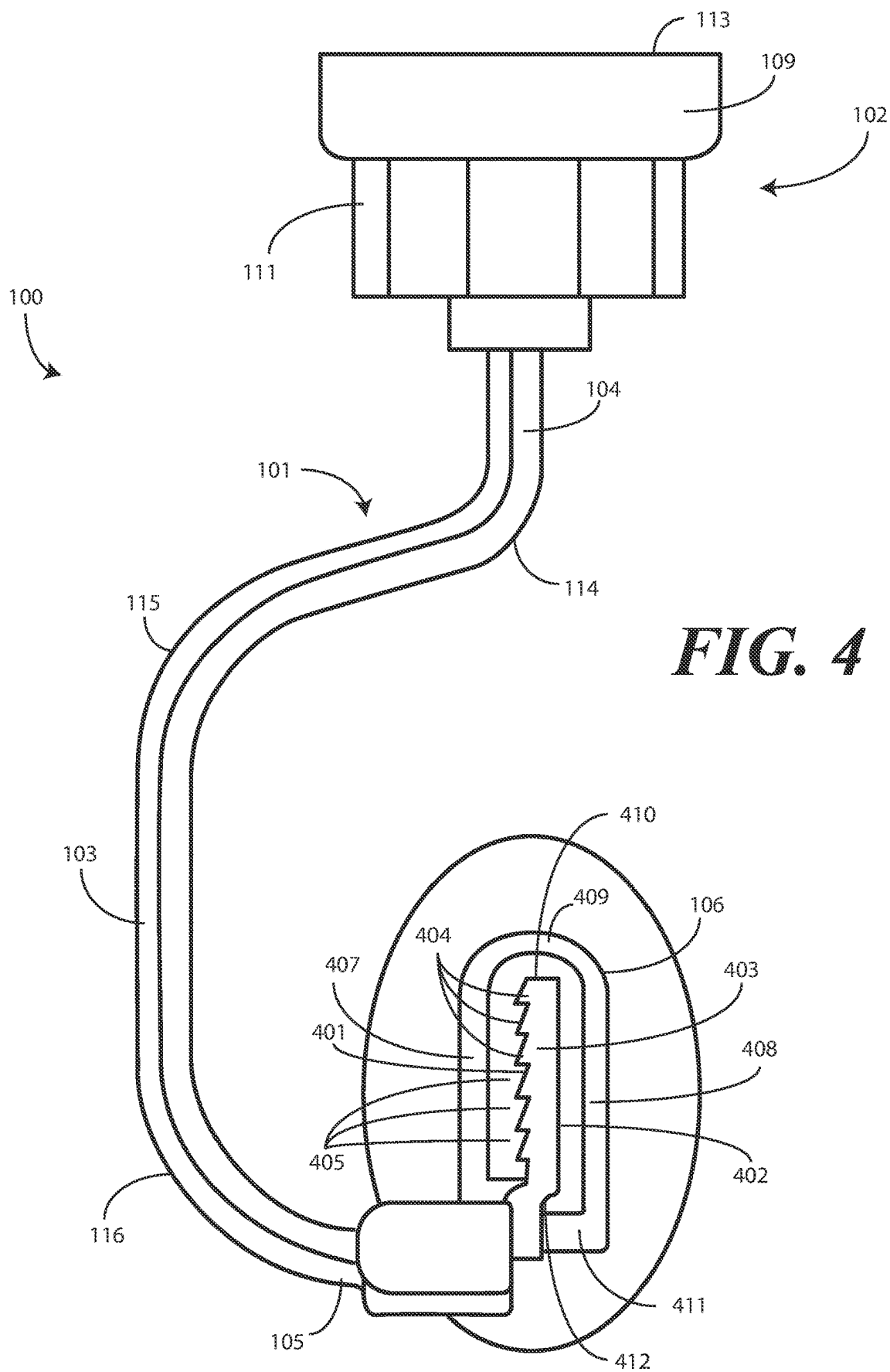
**FIG. 1**

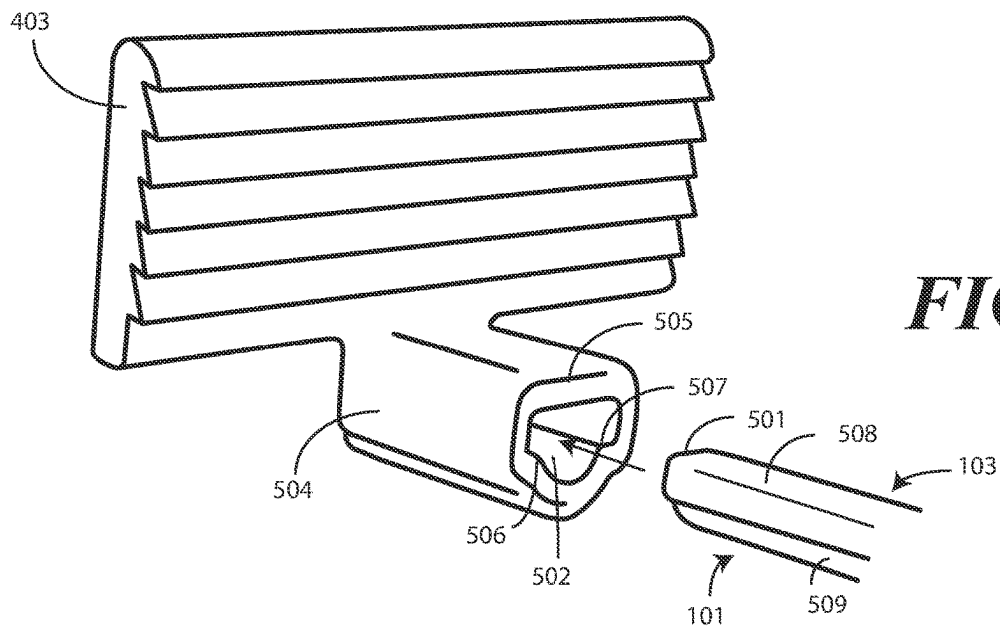


**FIG. 2**

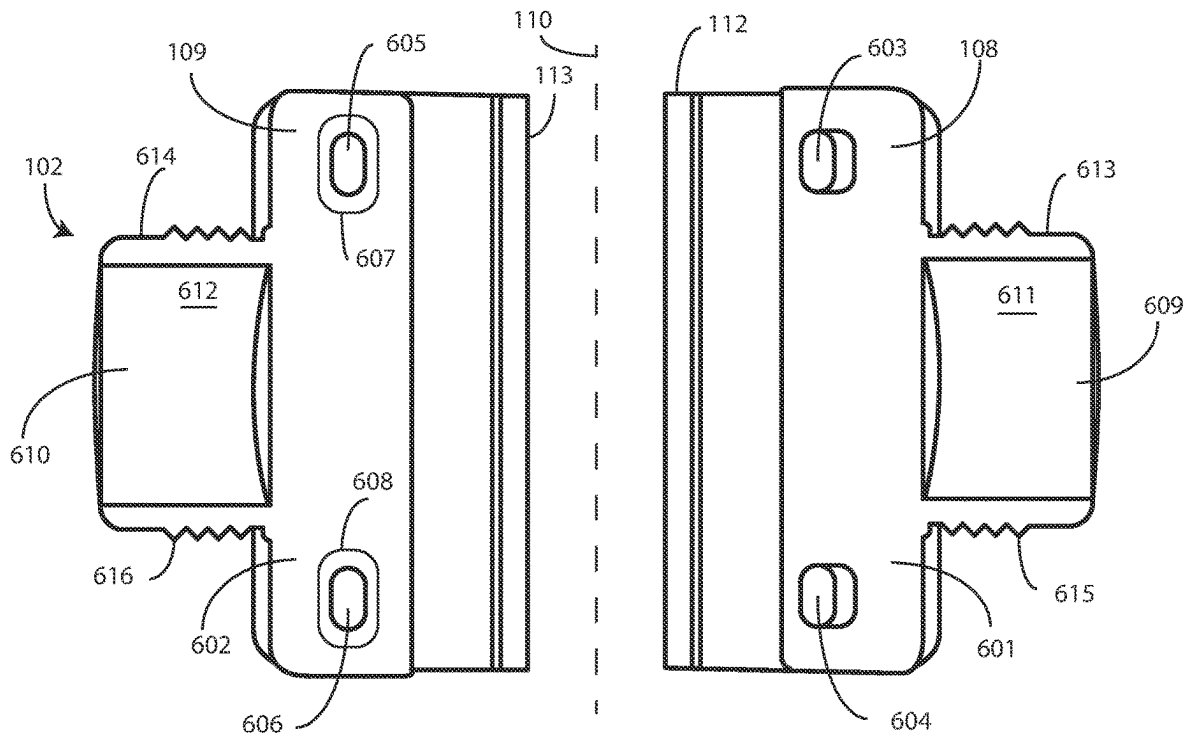


**FIG. 3**



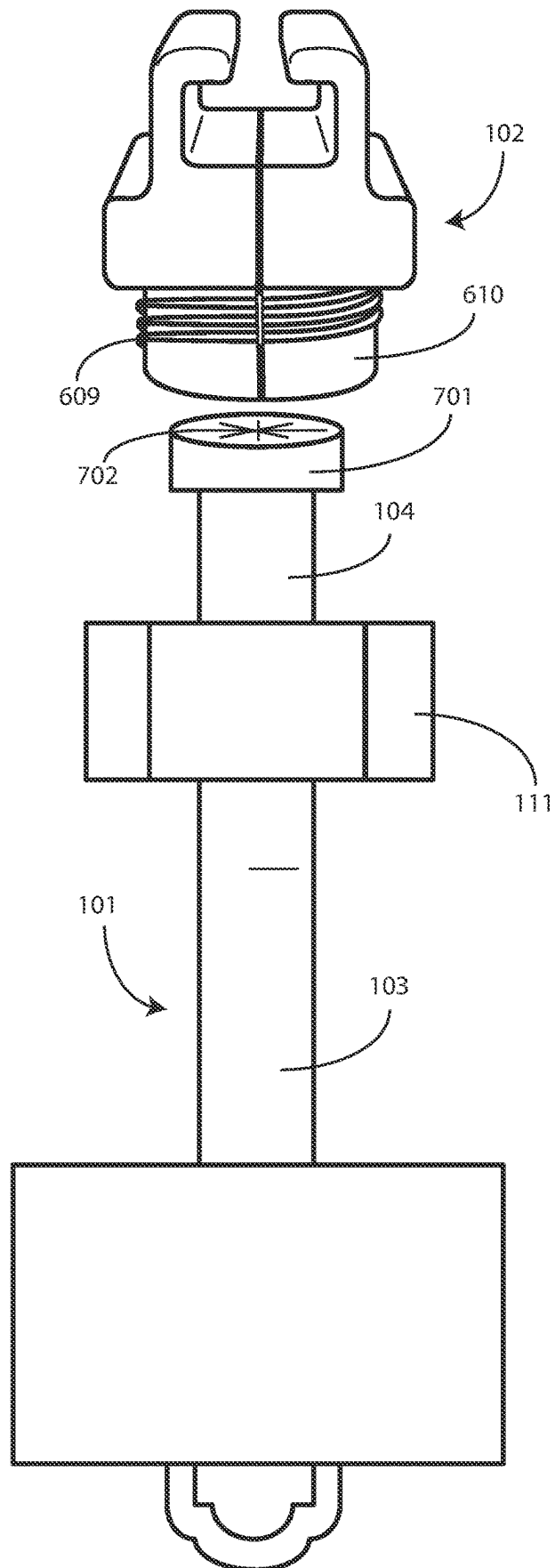


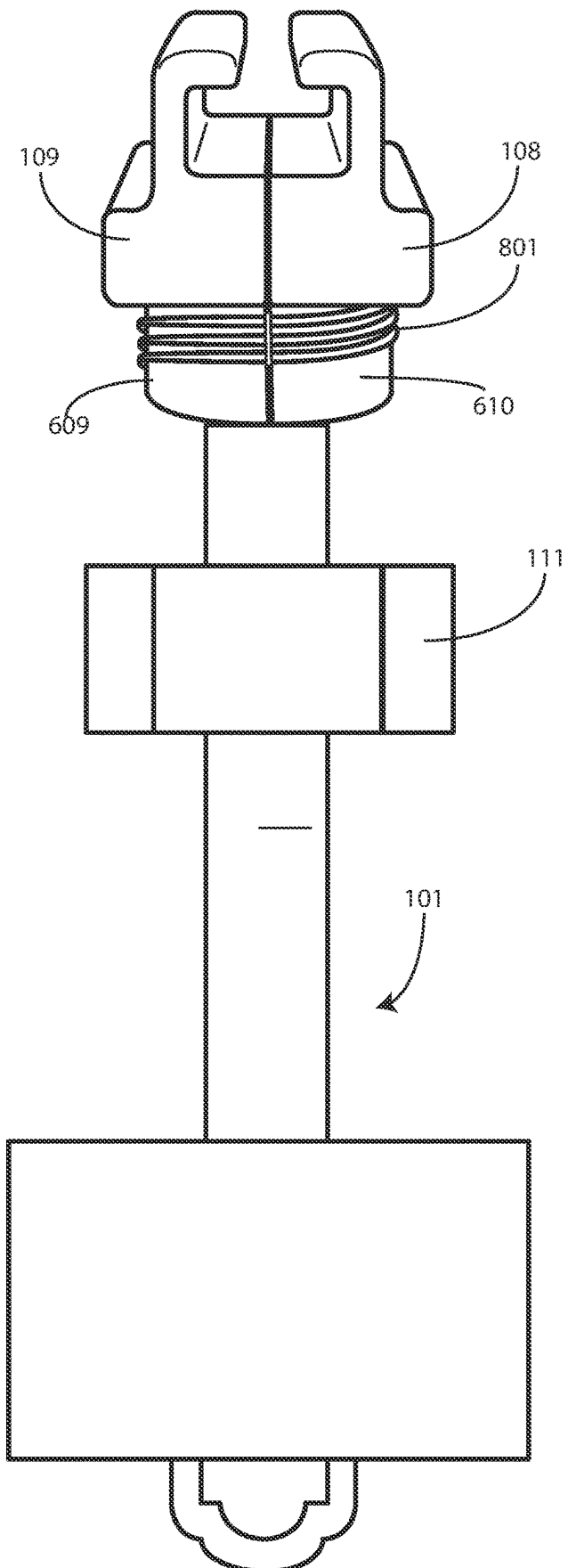
**FIG. 5**



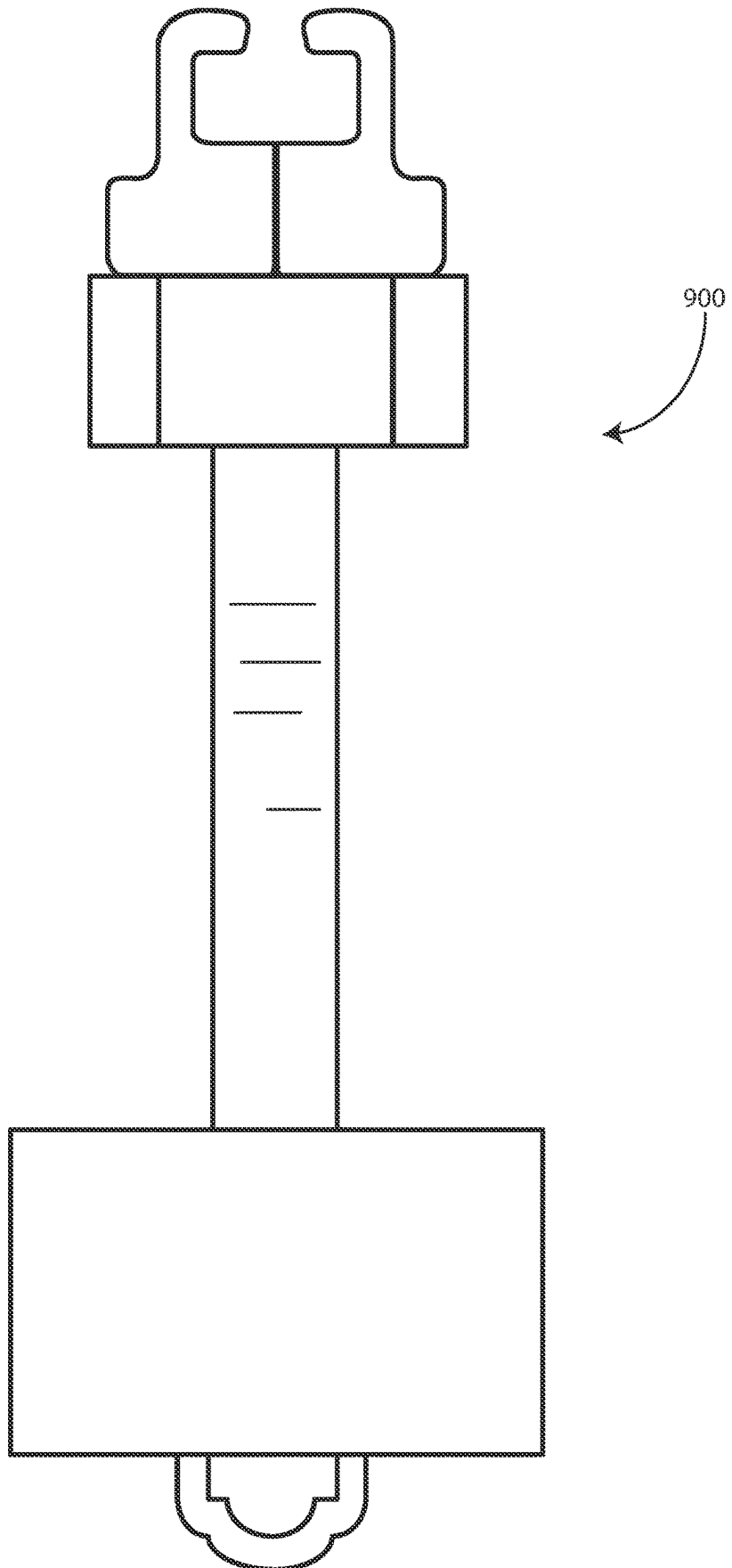
**FIG. 6**

**FIG. 7**

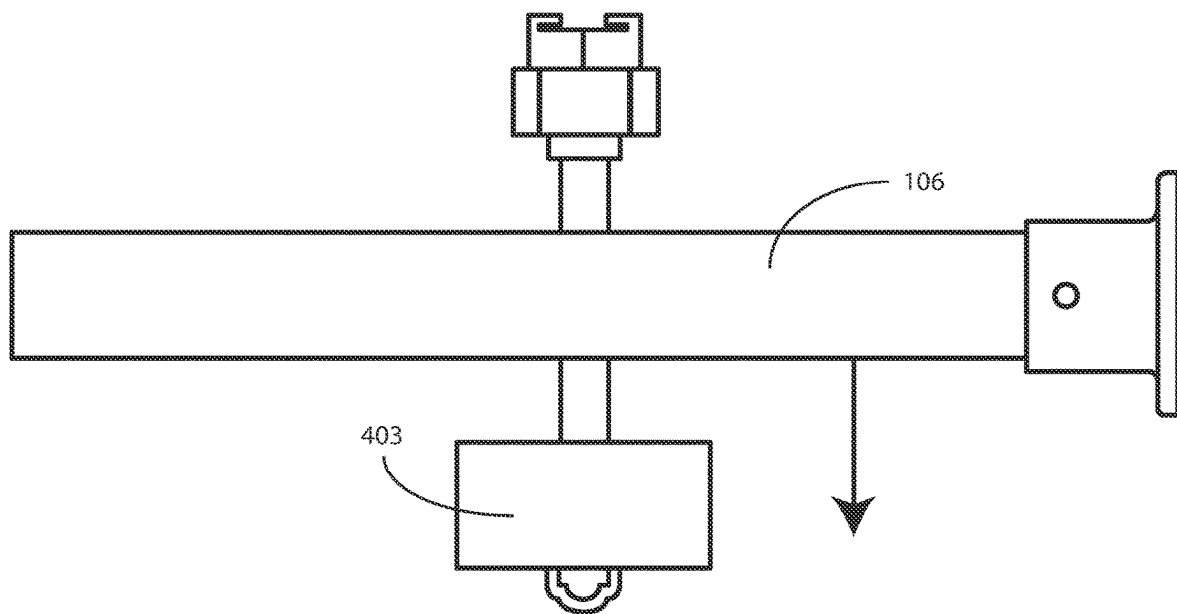




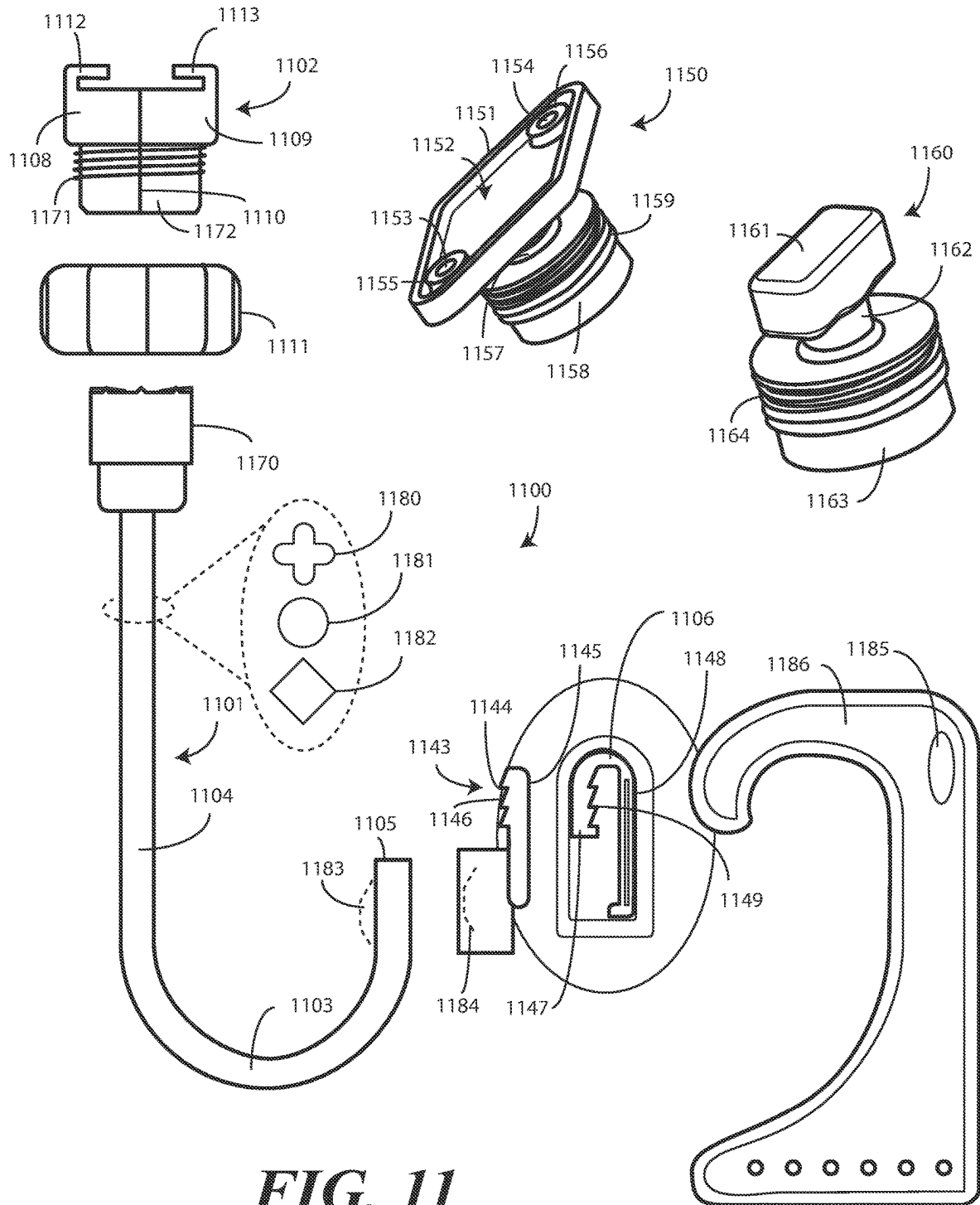


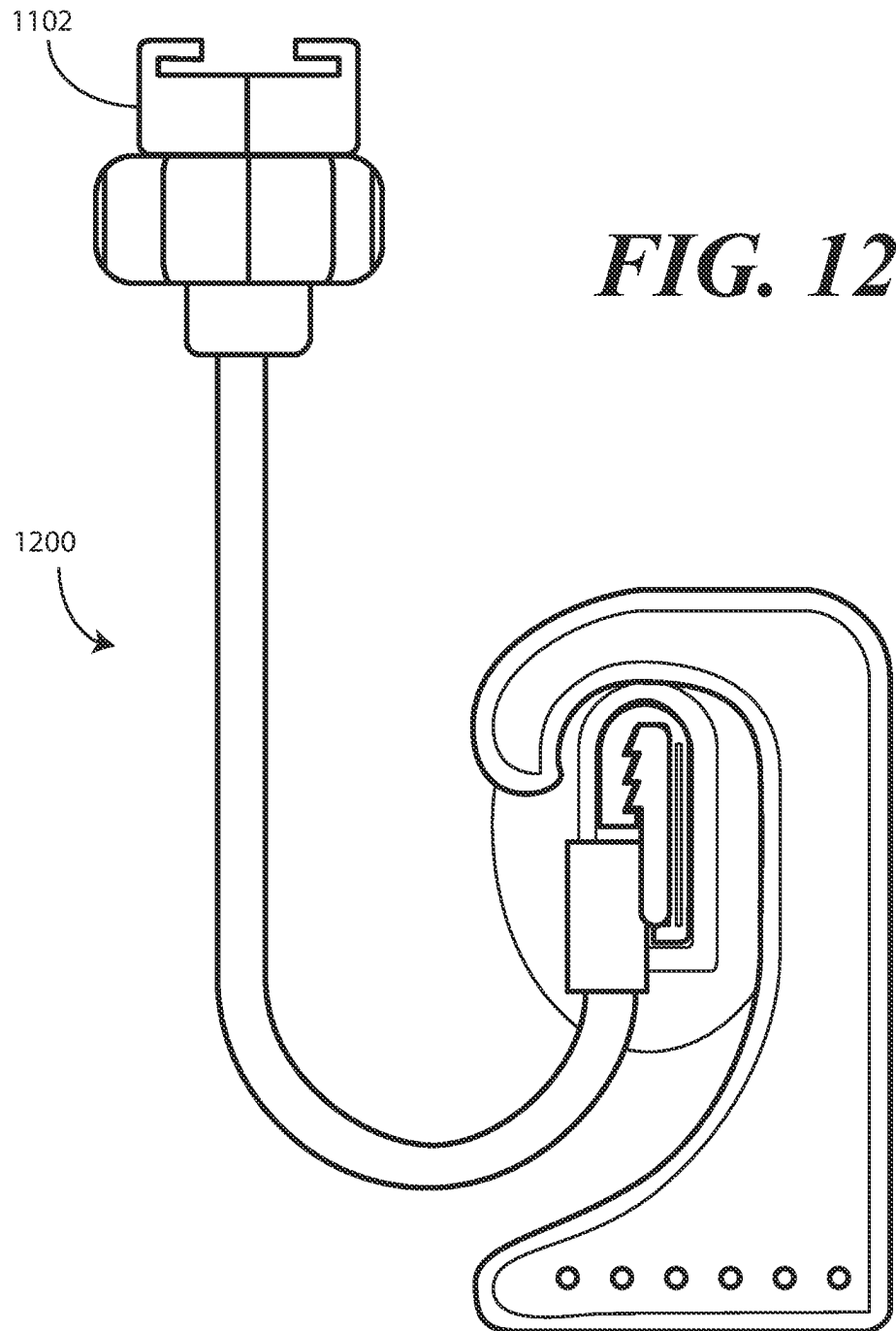


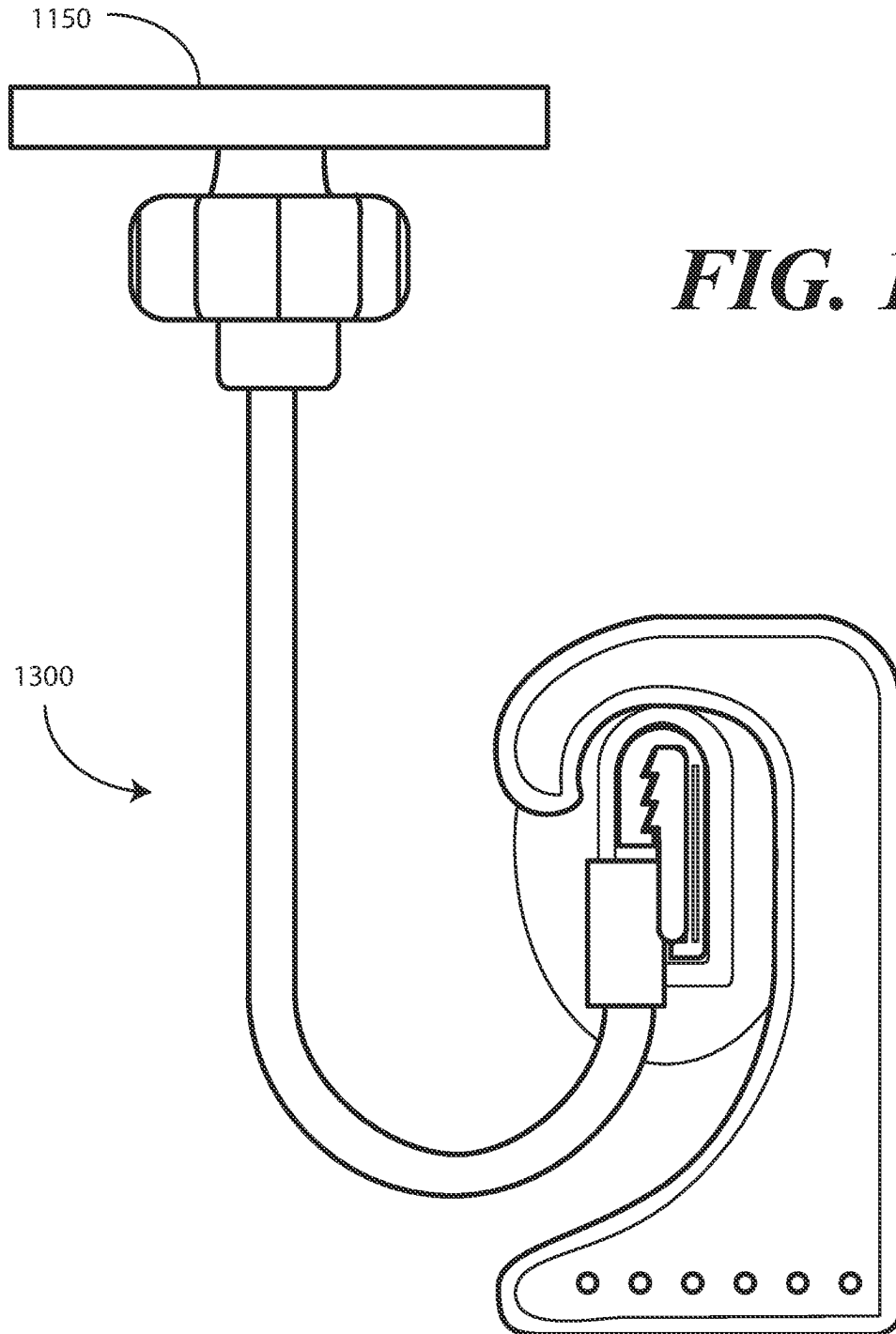
***FIG. 9***

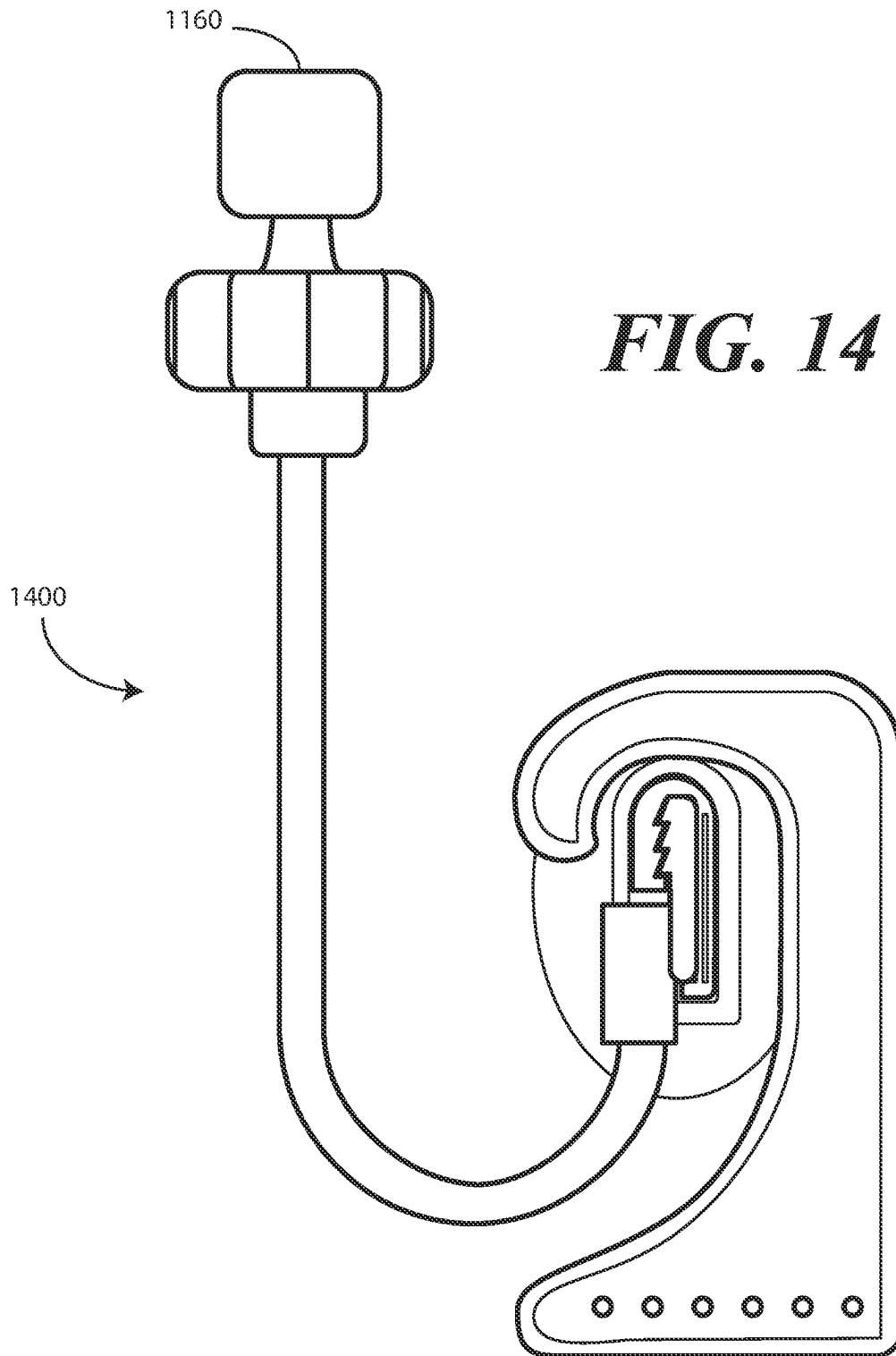


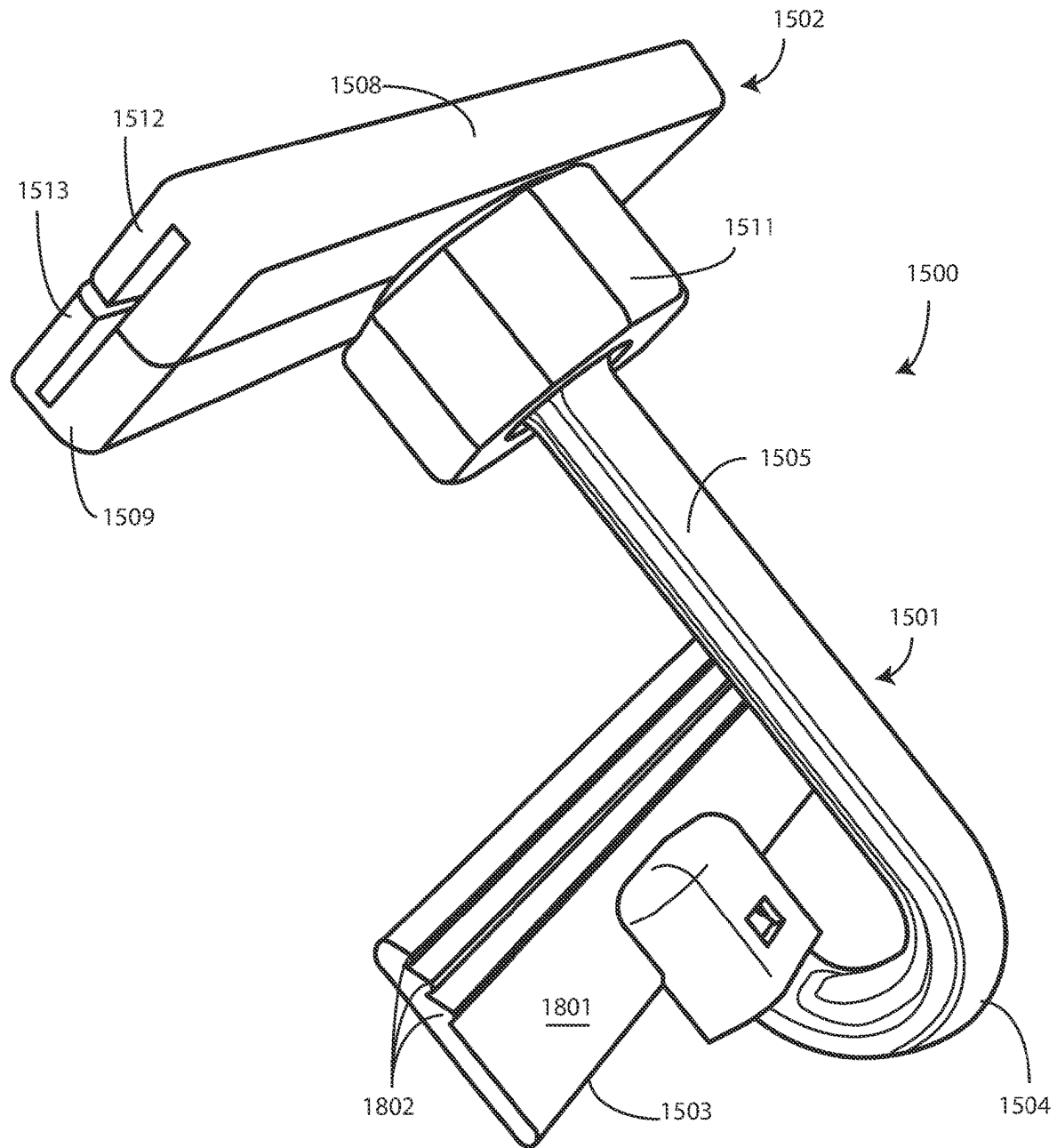
***FIG. 10***



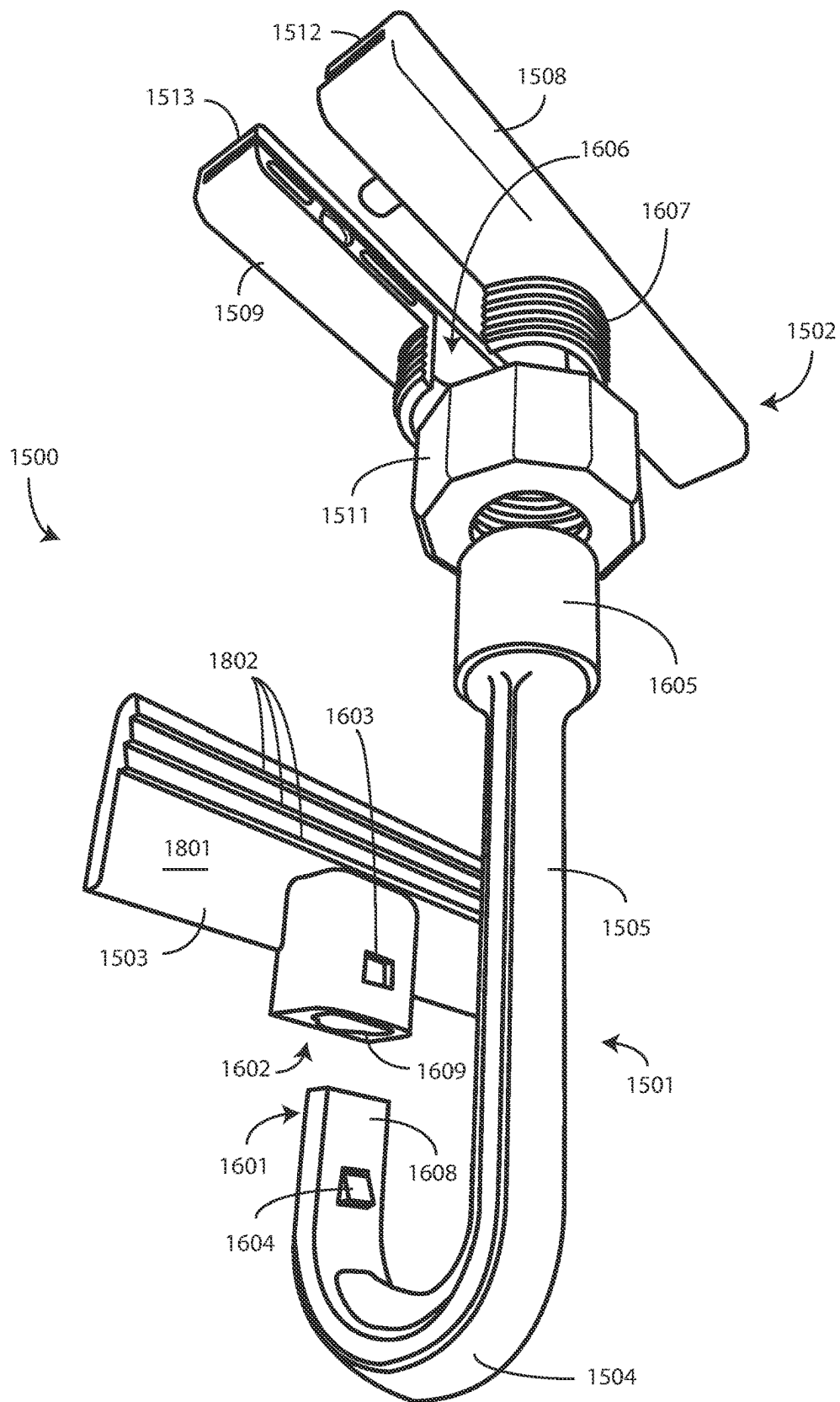






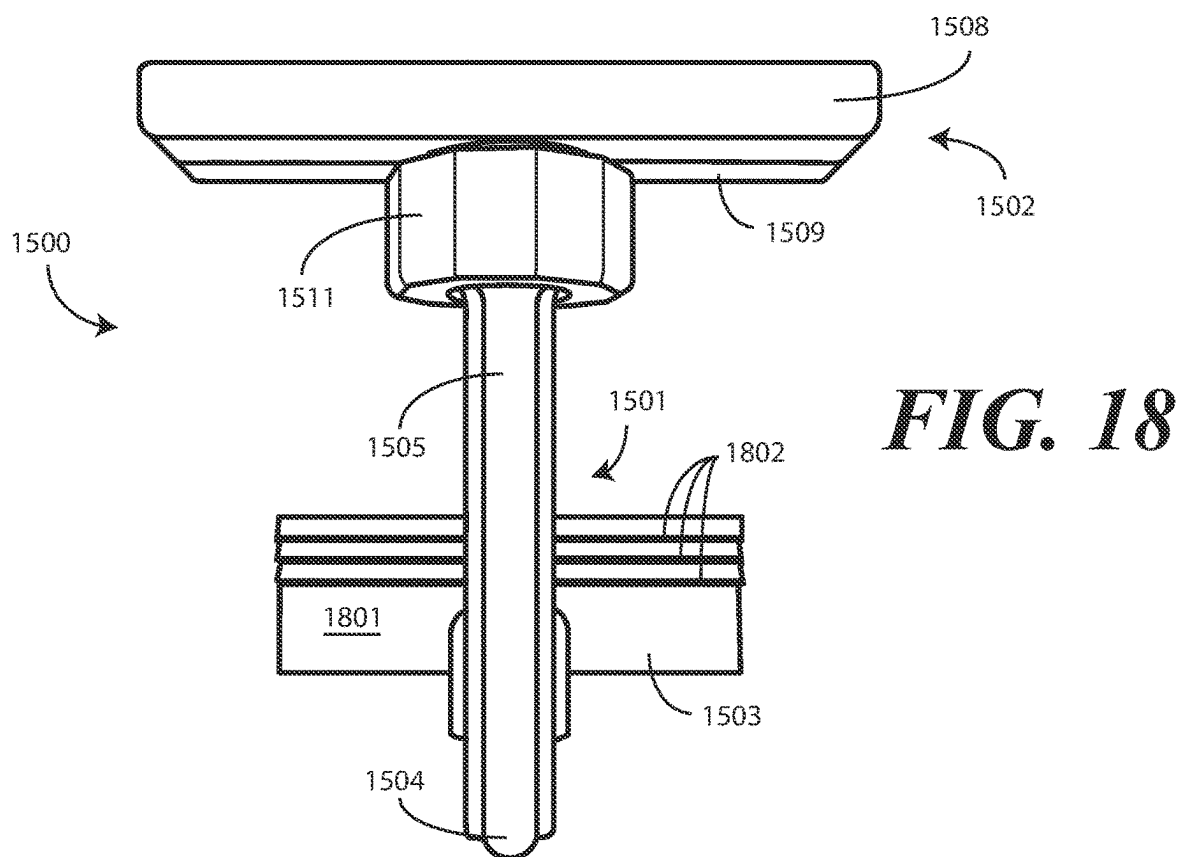
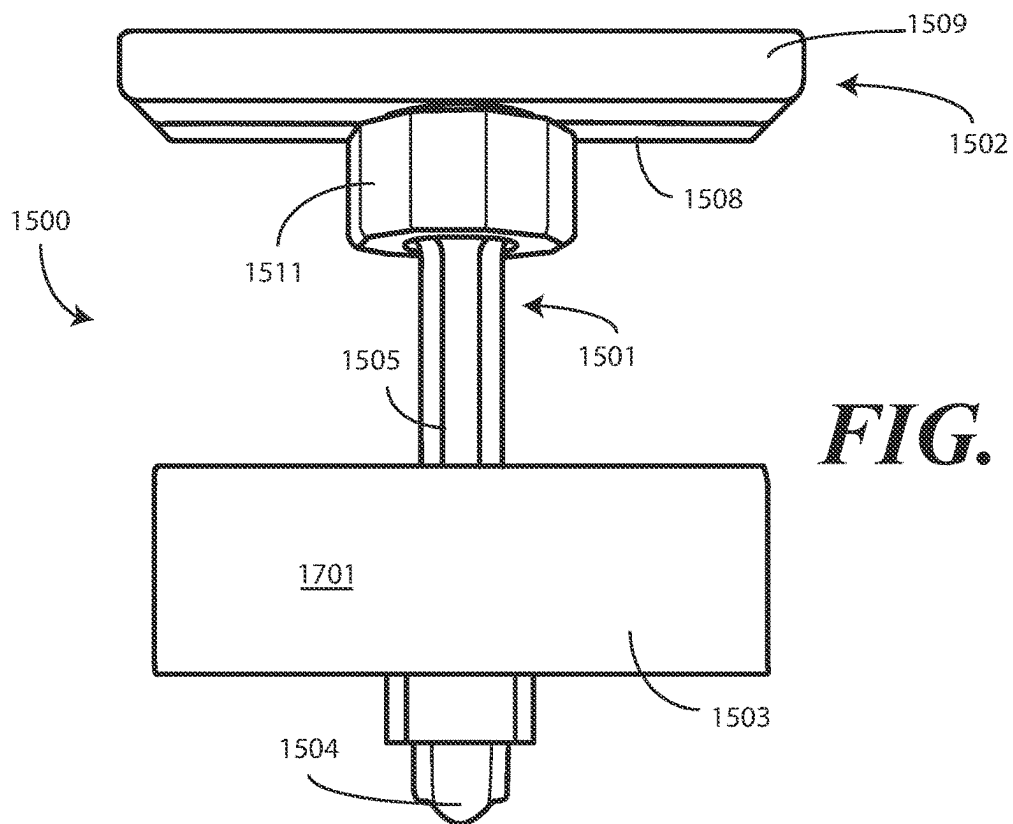


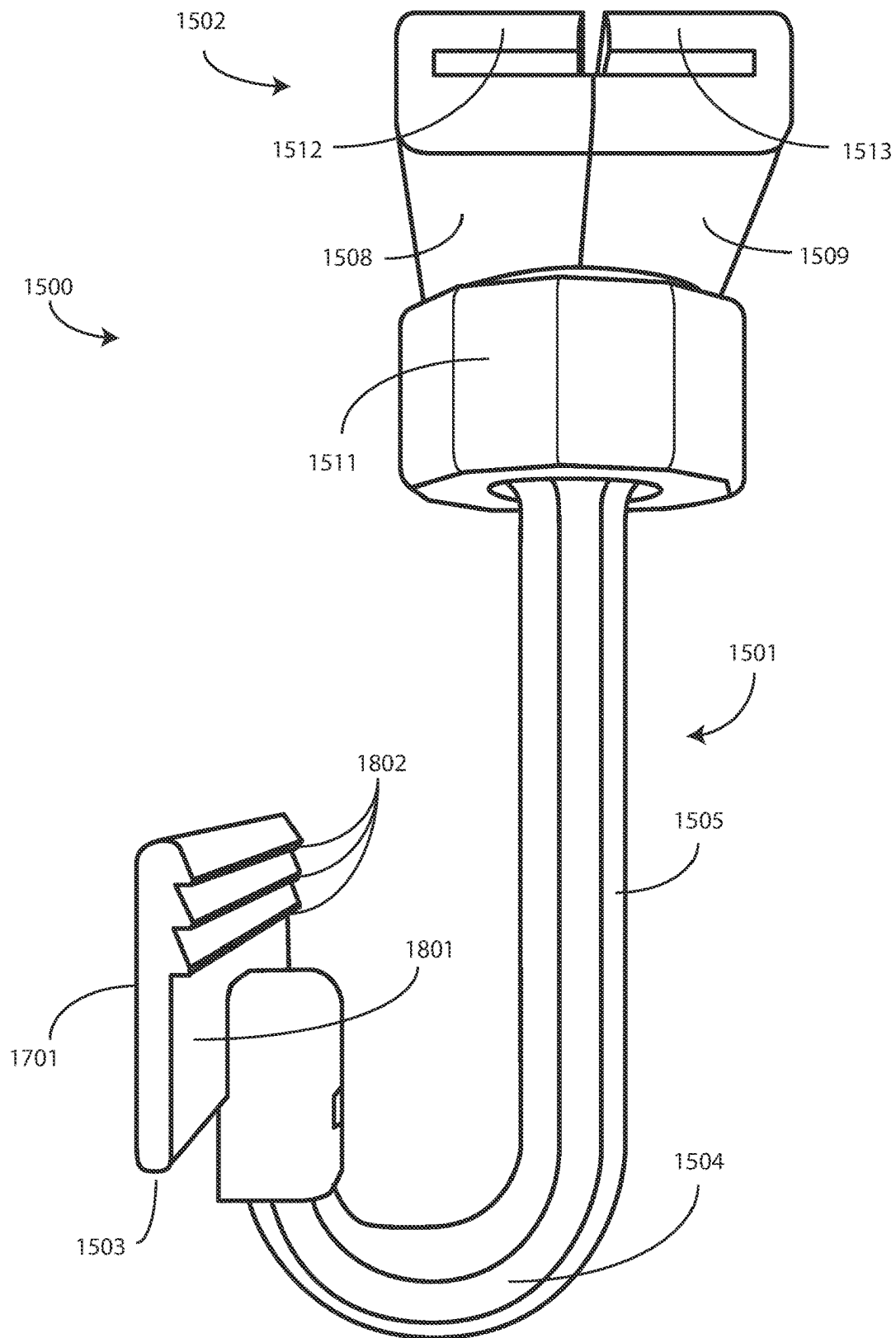
**FIG. 15**



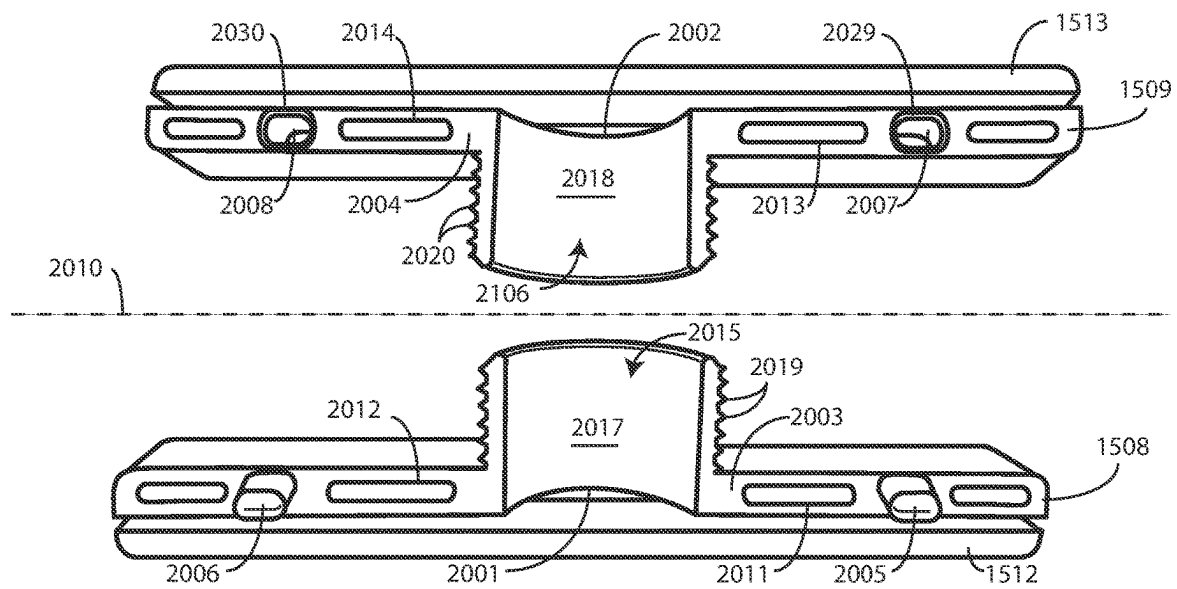
**FIG. 16**



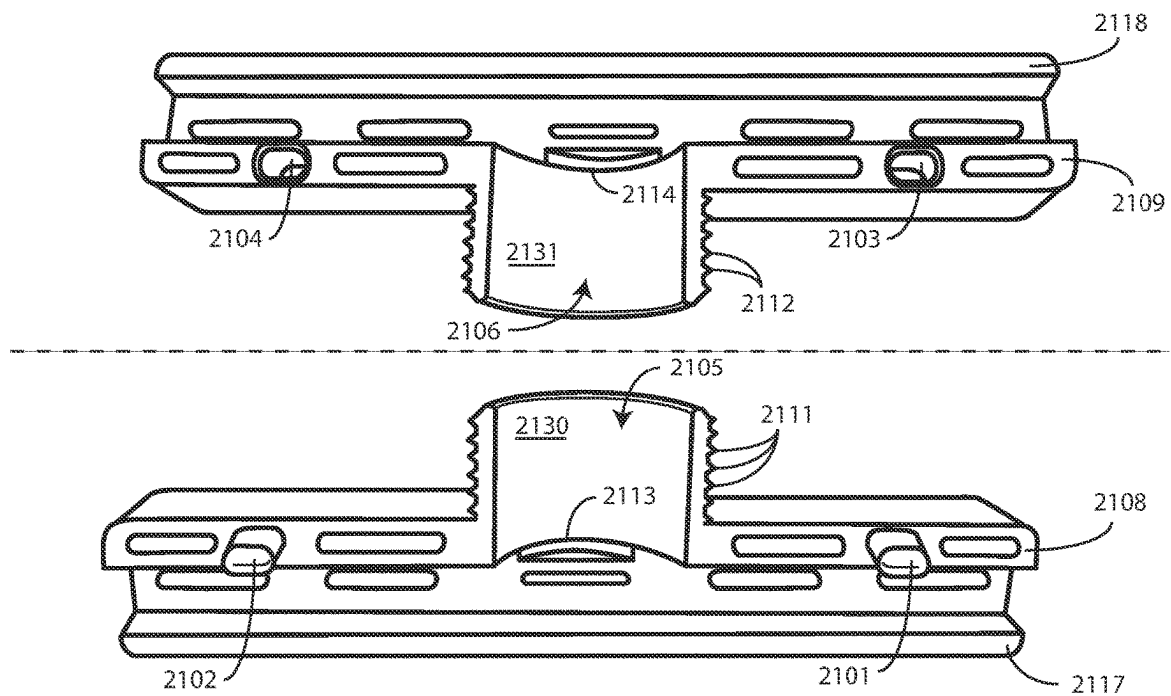




**FIG. 19**

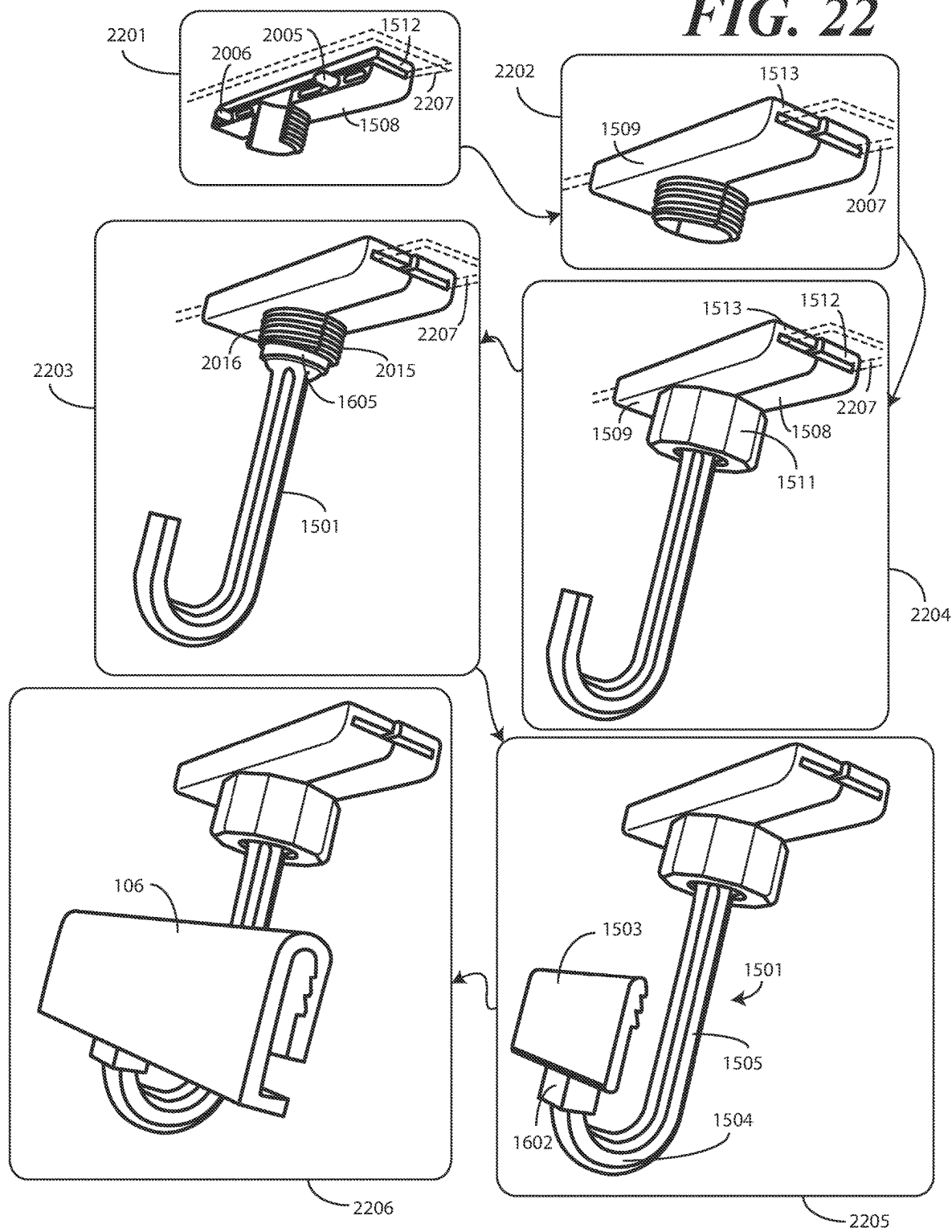


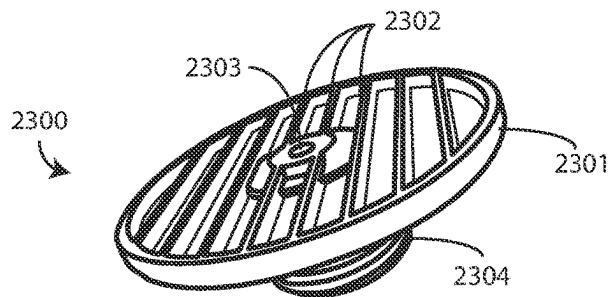
**FIG. 20**



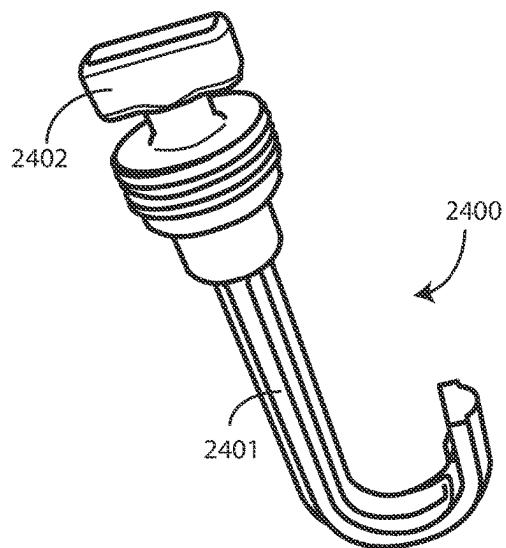
**FIG. 21**

**FIG. 22**

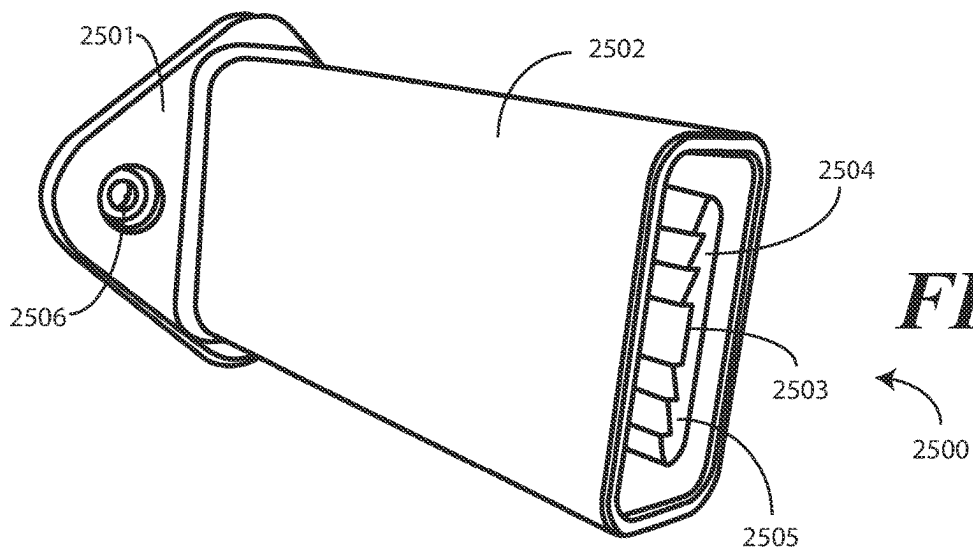




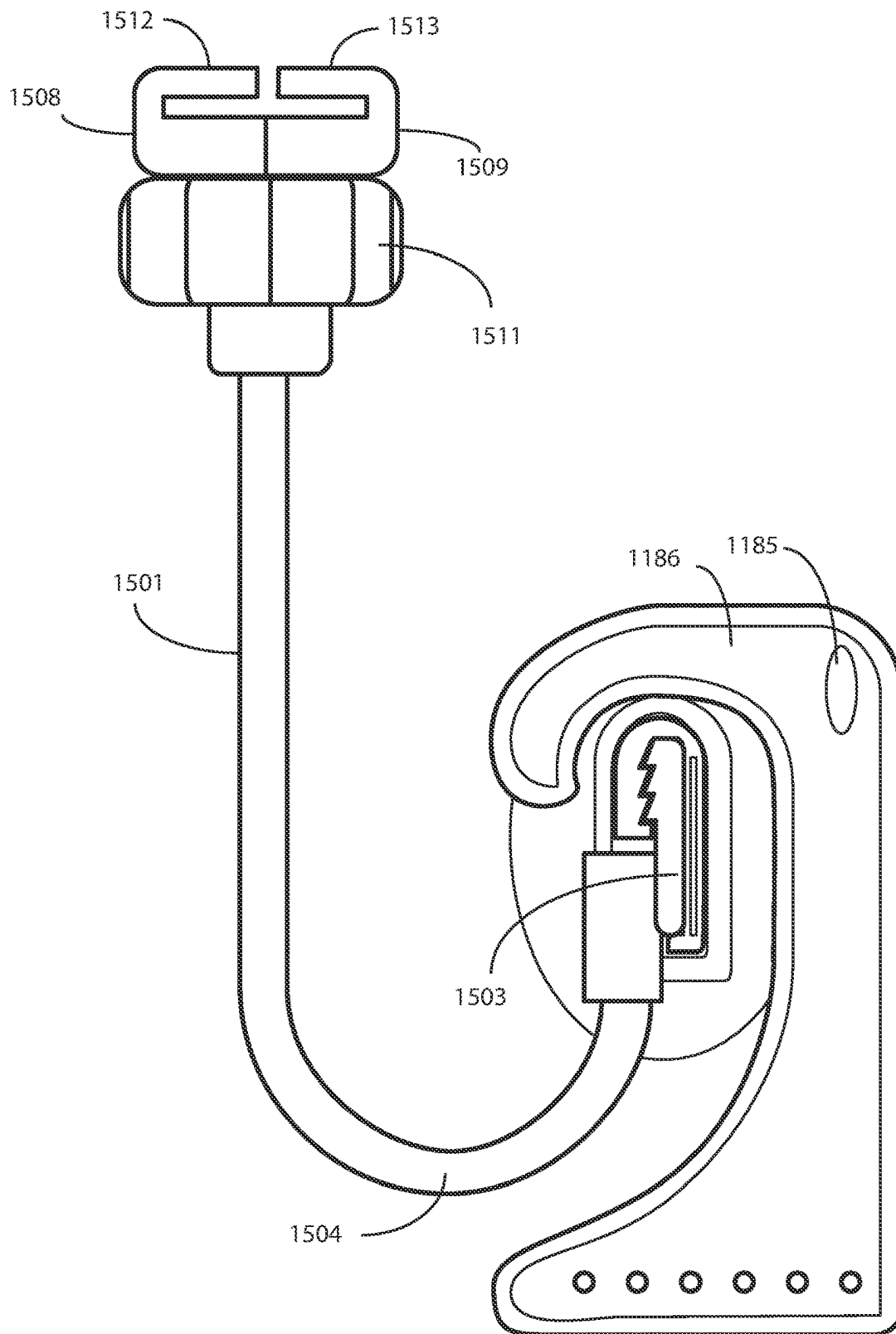
**FIG. 23**



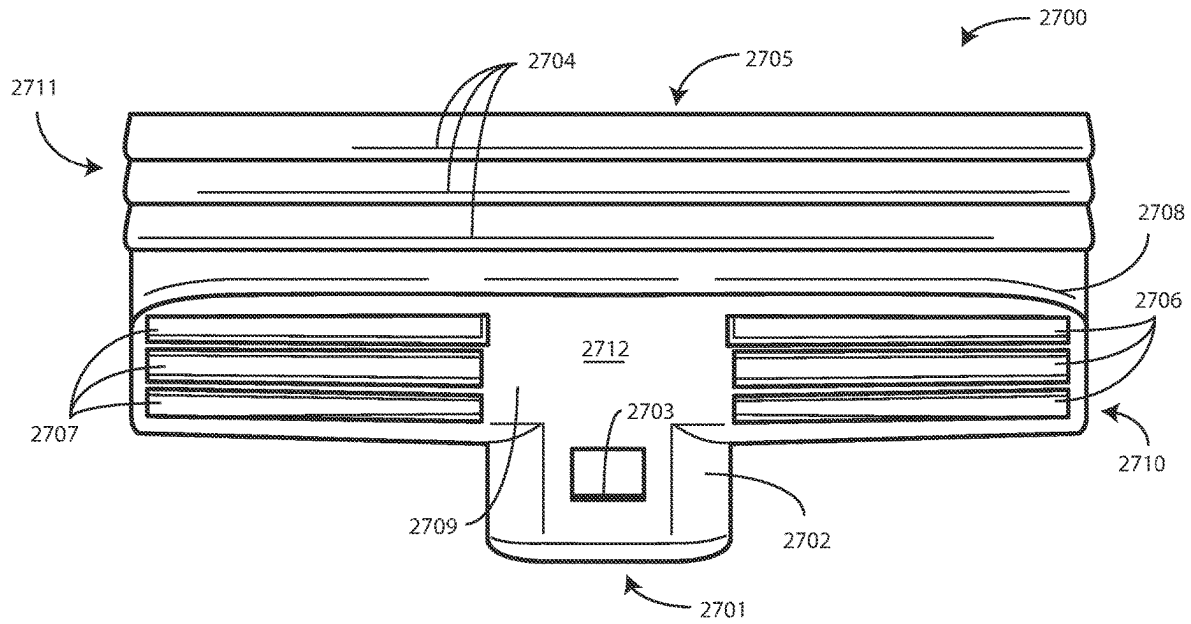
**FIG. 24**



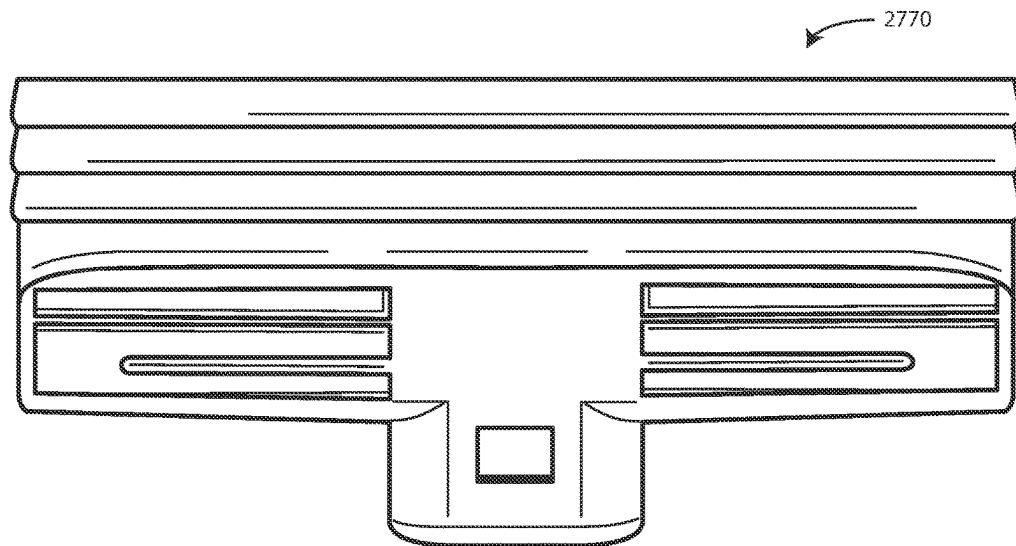
**FIG. 25**



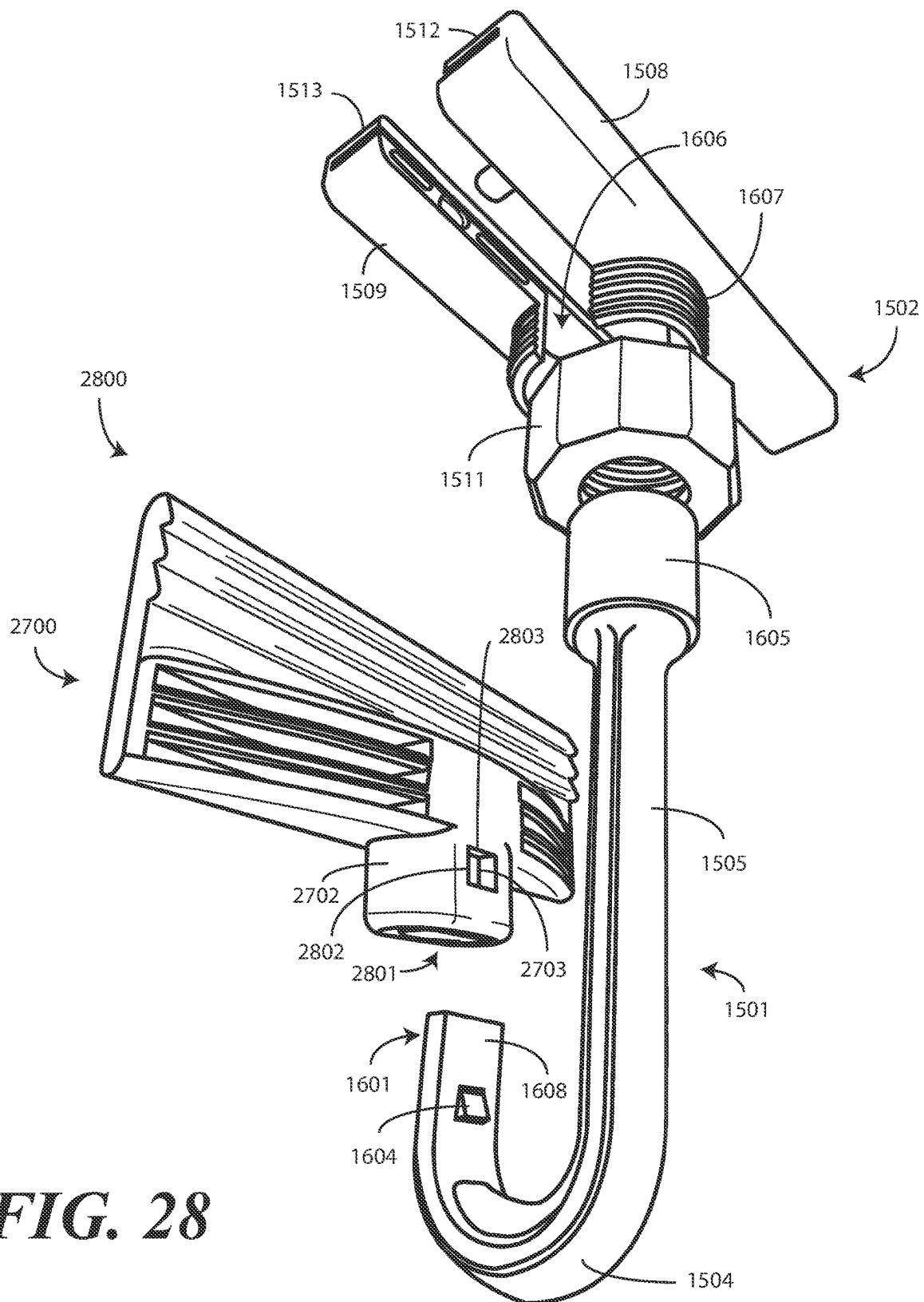
**FIG. 26**



**FIG. 27A**

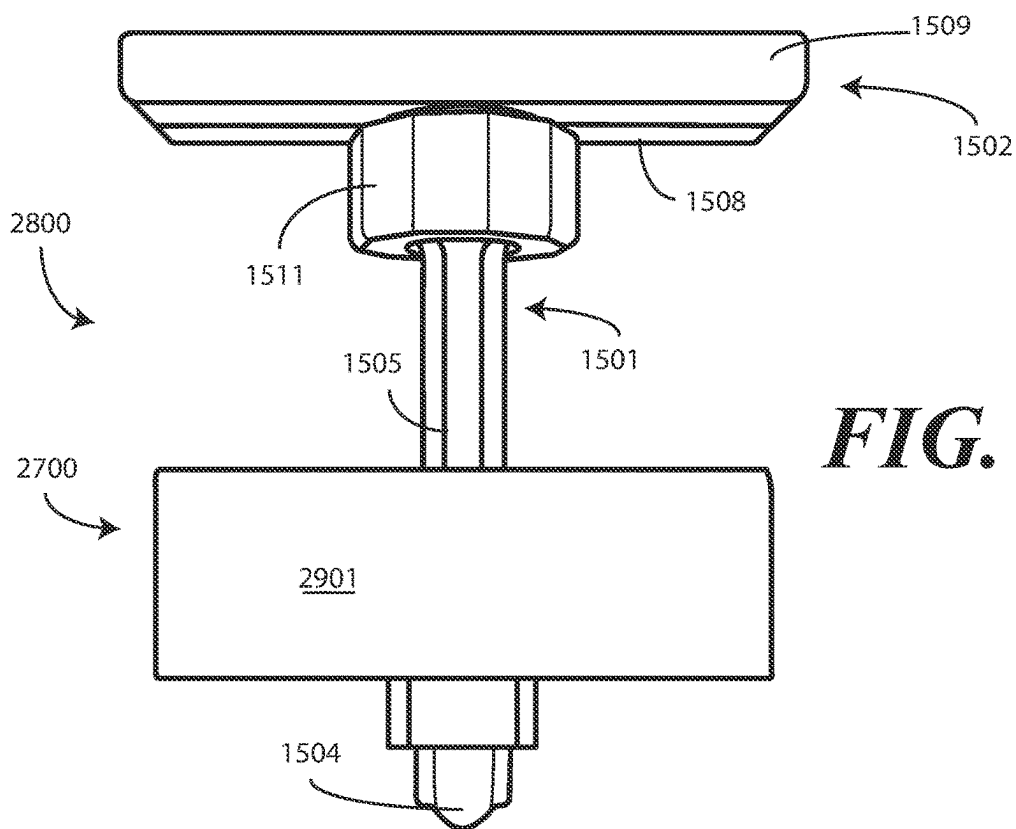


**FIG. 27B**

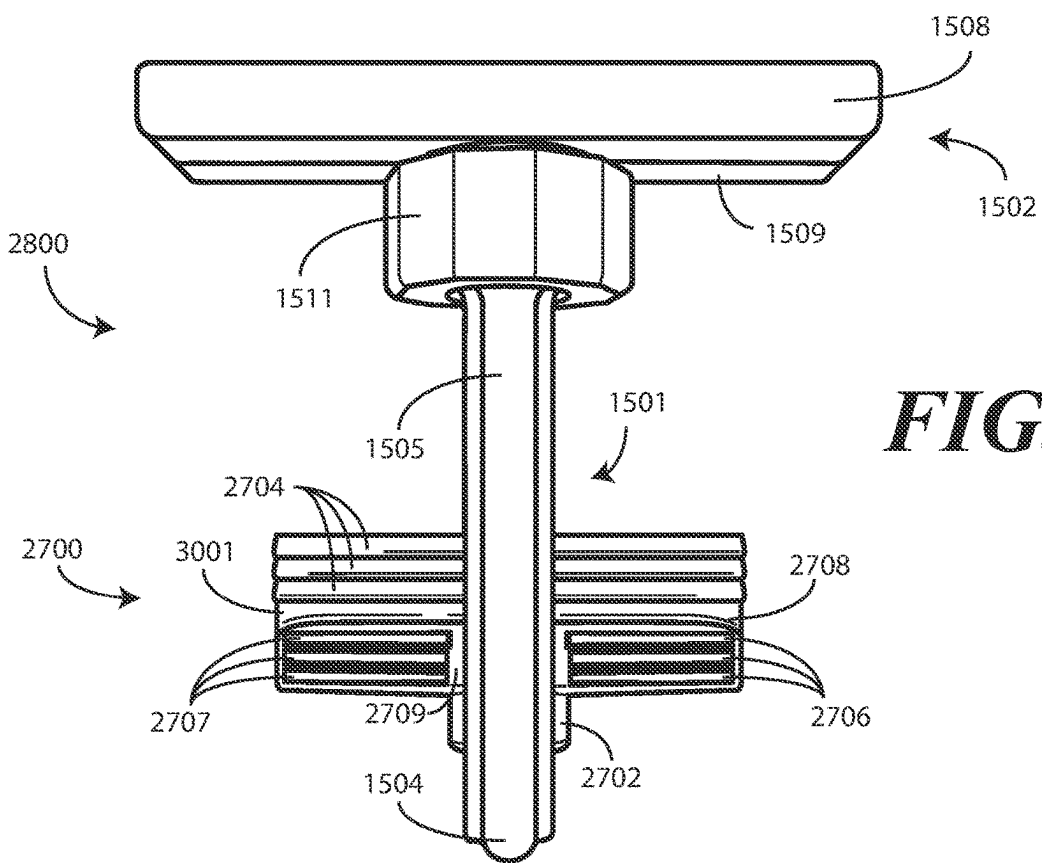


**FIG. 28**

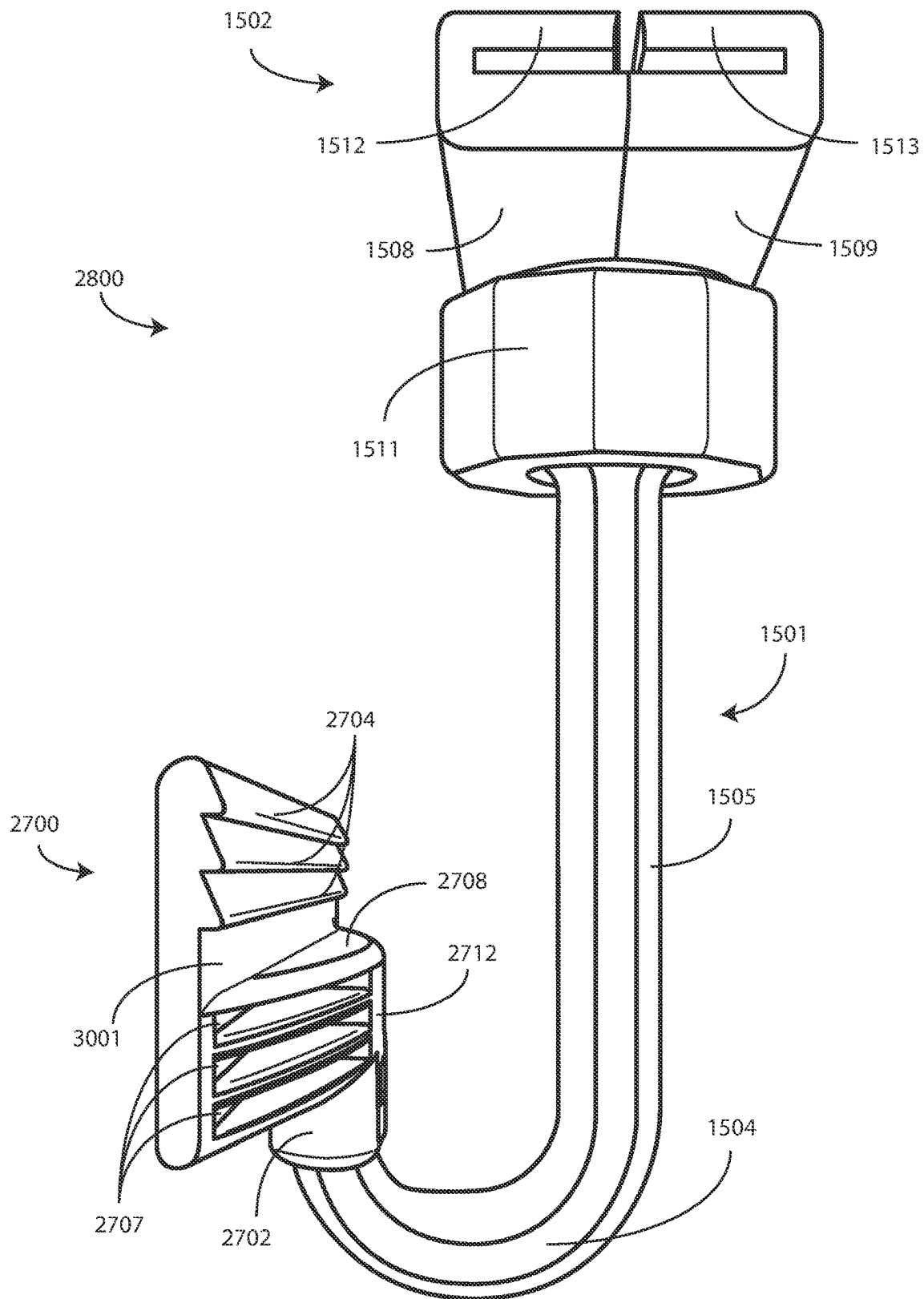




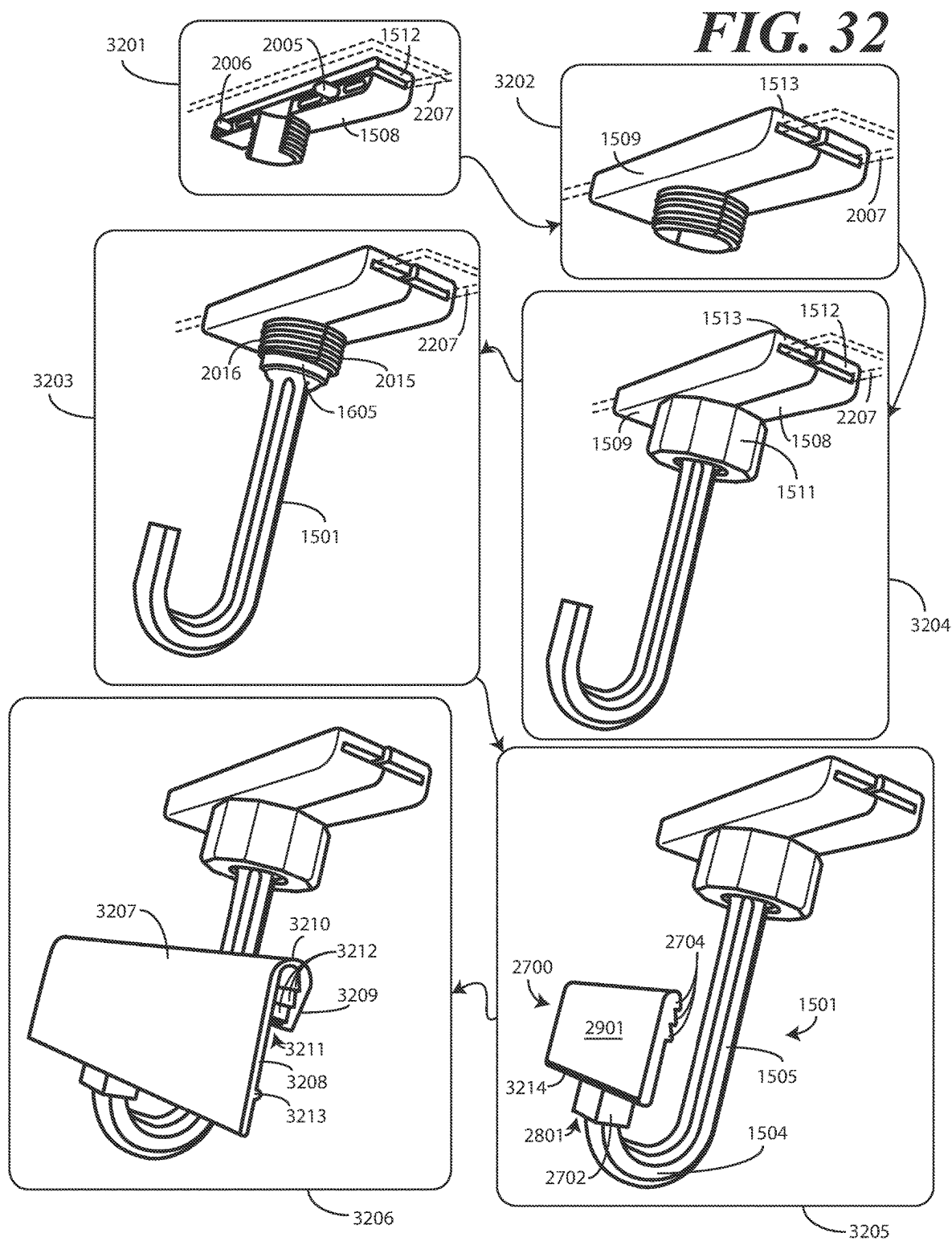
**FIG. 29**

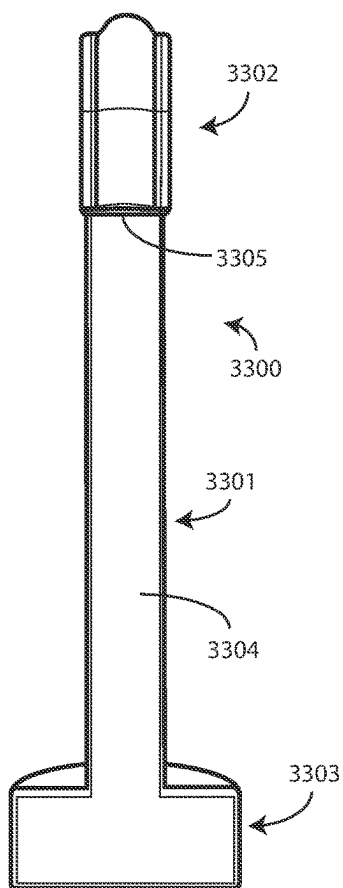


**FIG. 30**

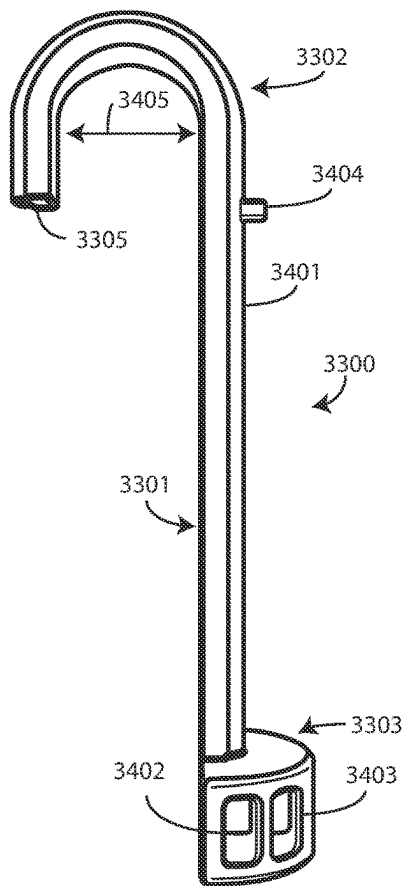


**FIG. 31**

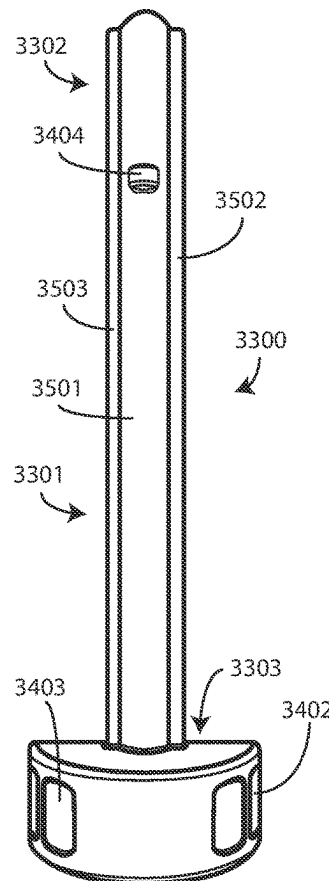




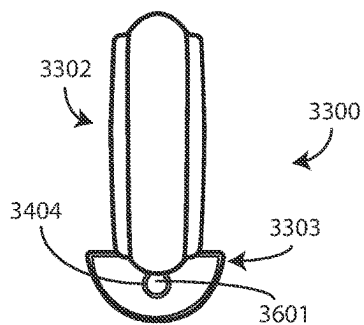
**FIG. 33**



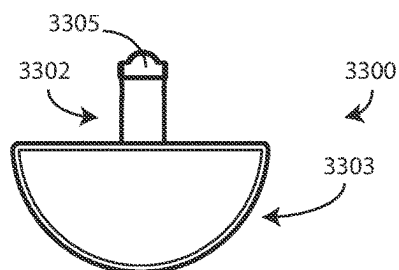
**FIG. 34**



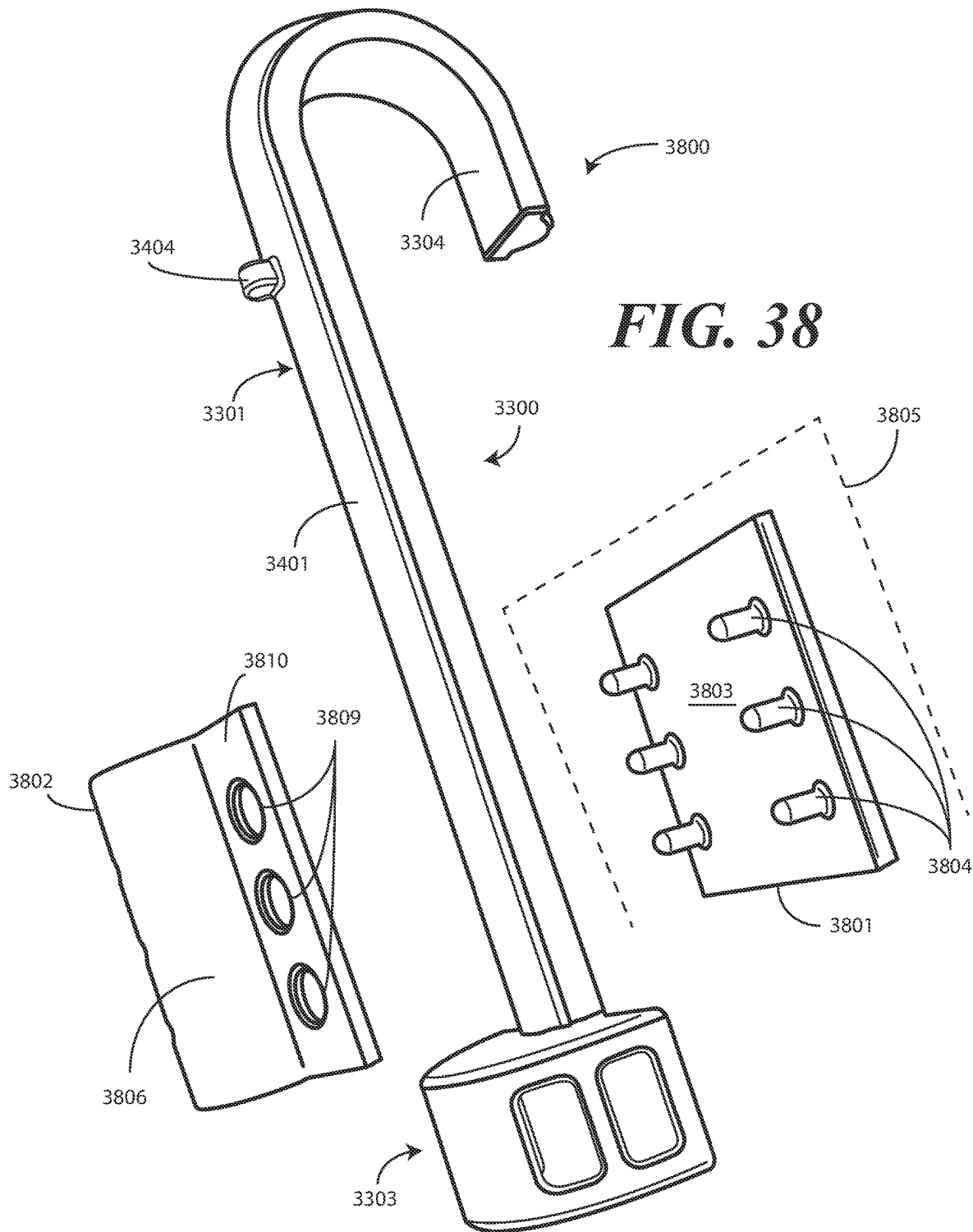
**FIG. 35**

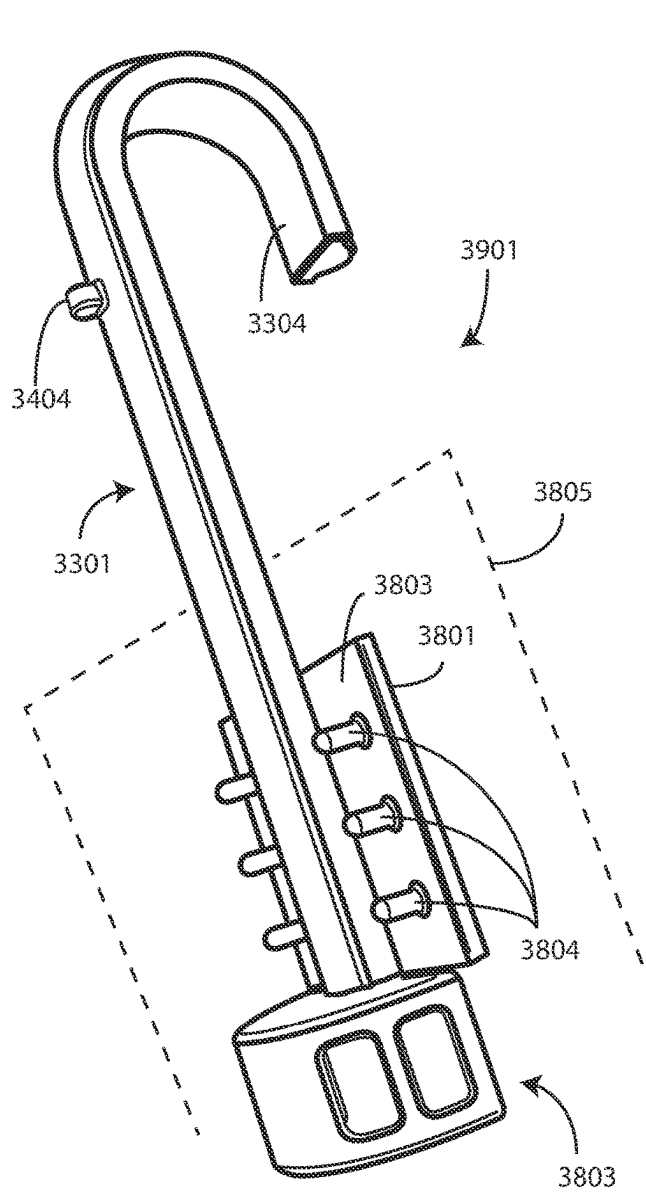


**FIG. 36**

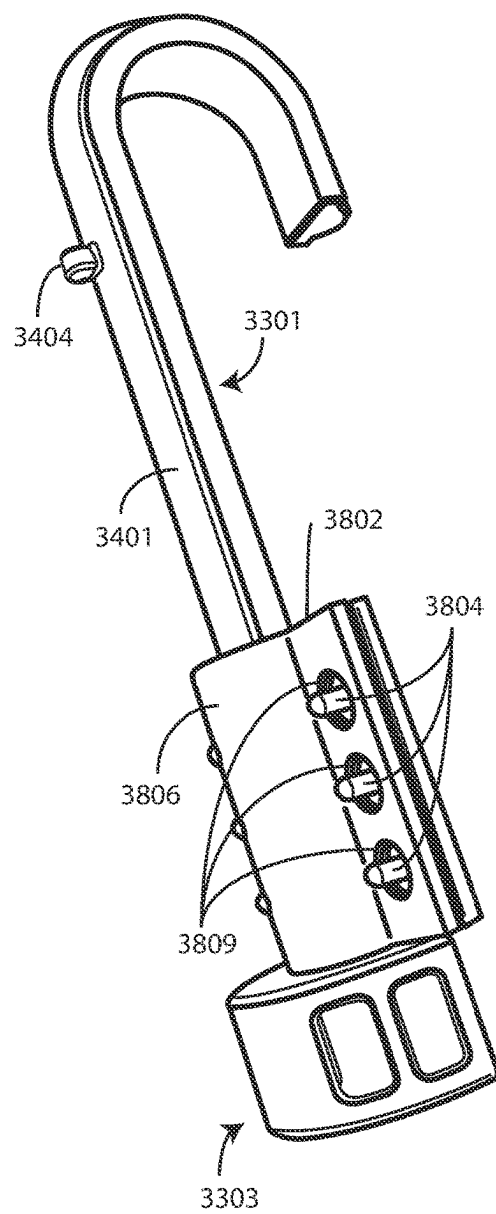


**FIG. 37**

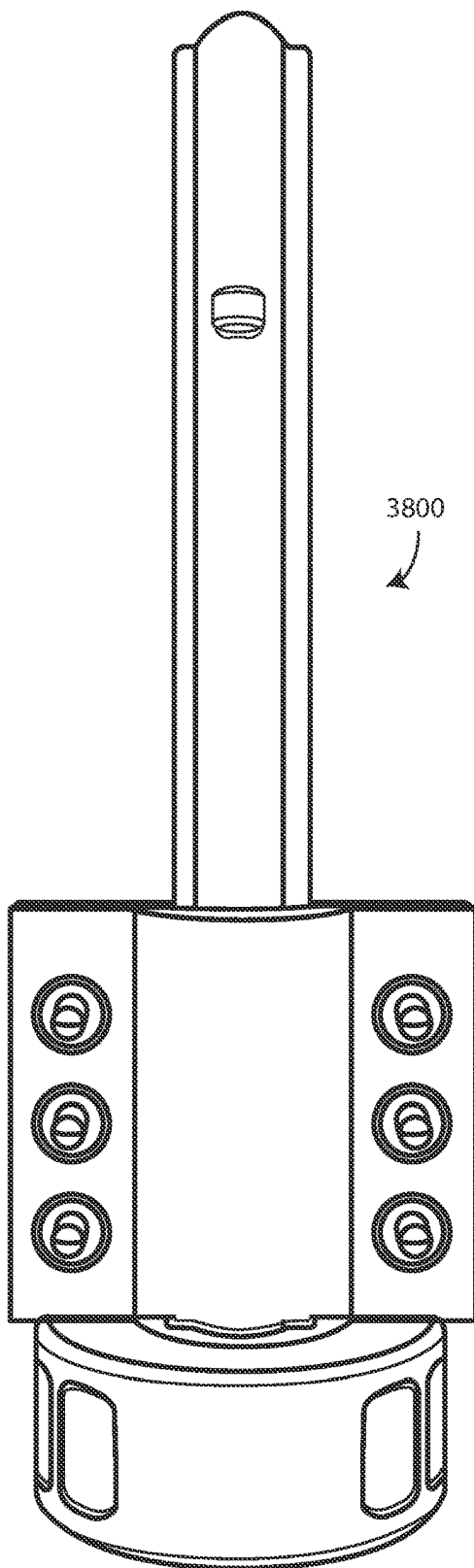




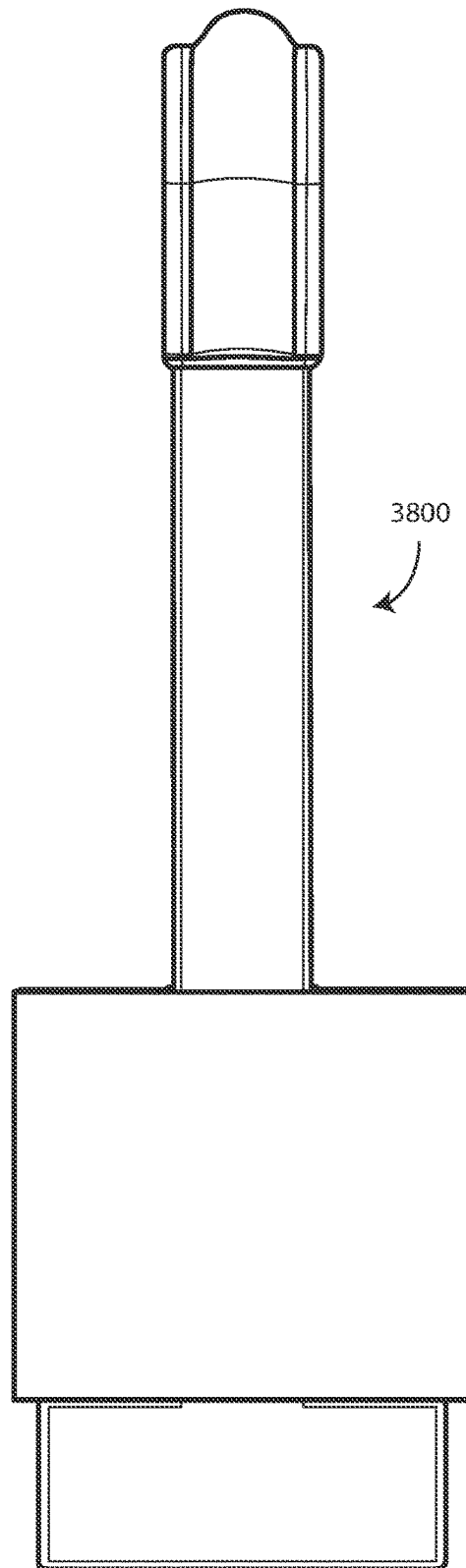
**FIG. 39**



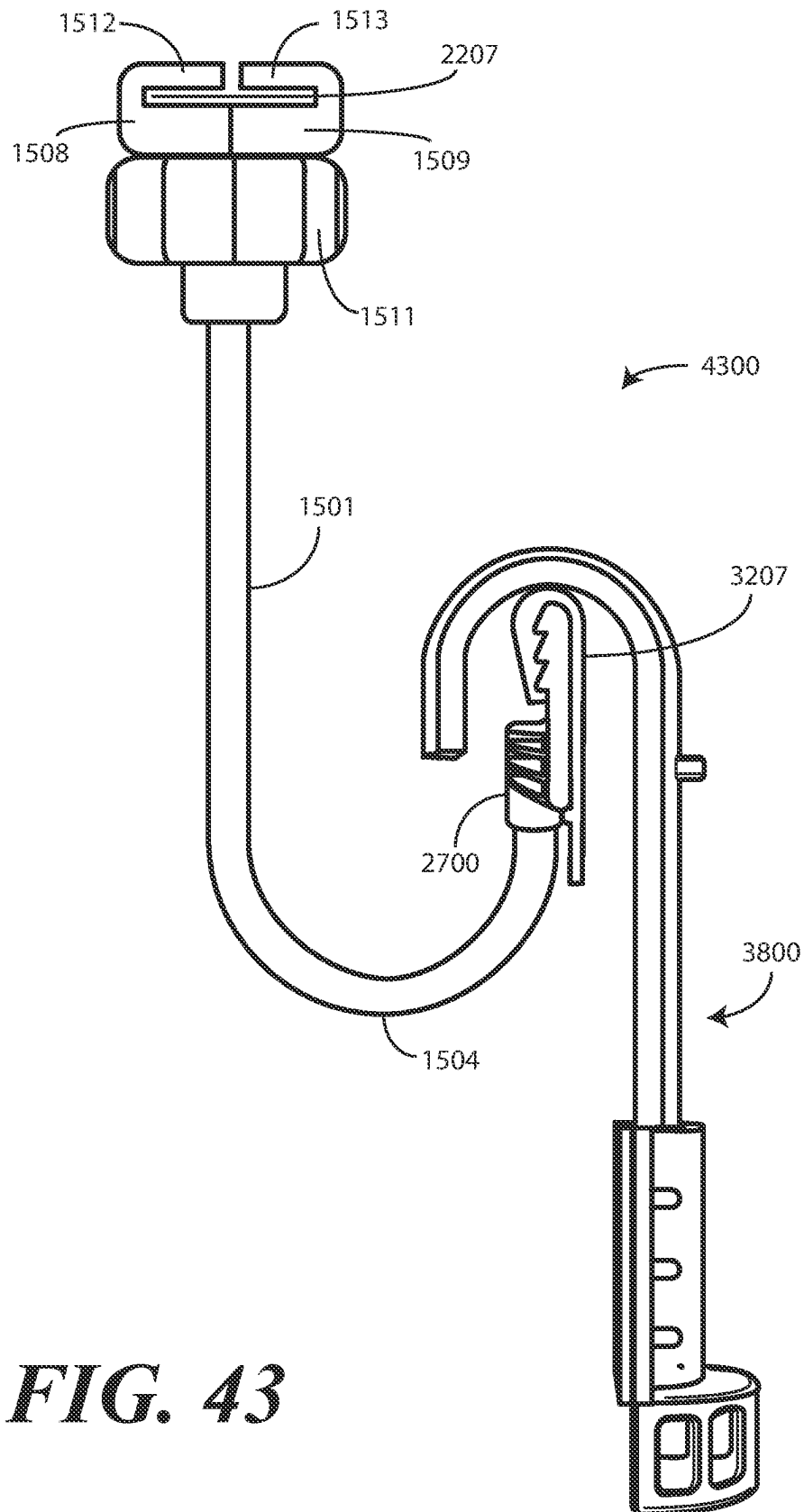
**FIG. 40**



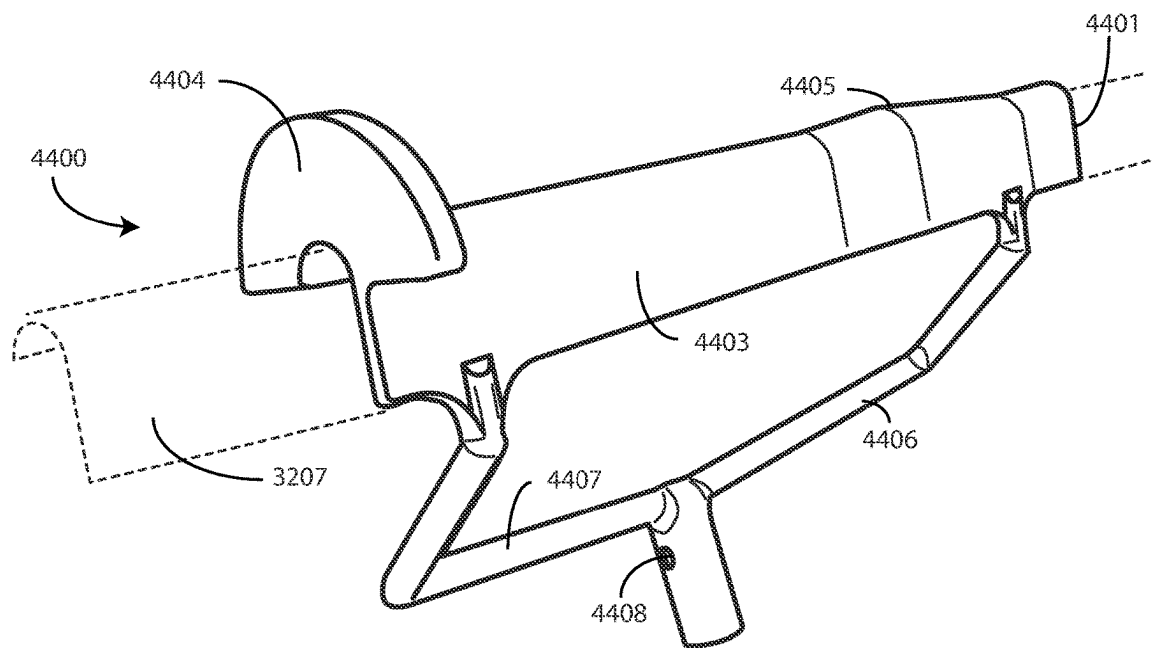
**FIG. 41**



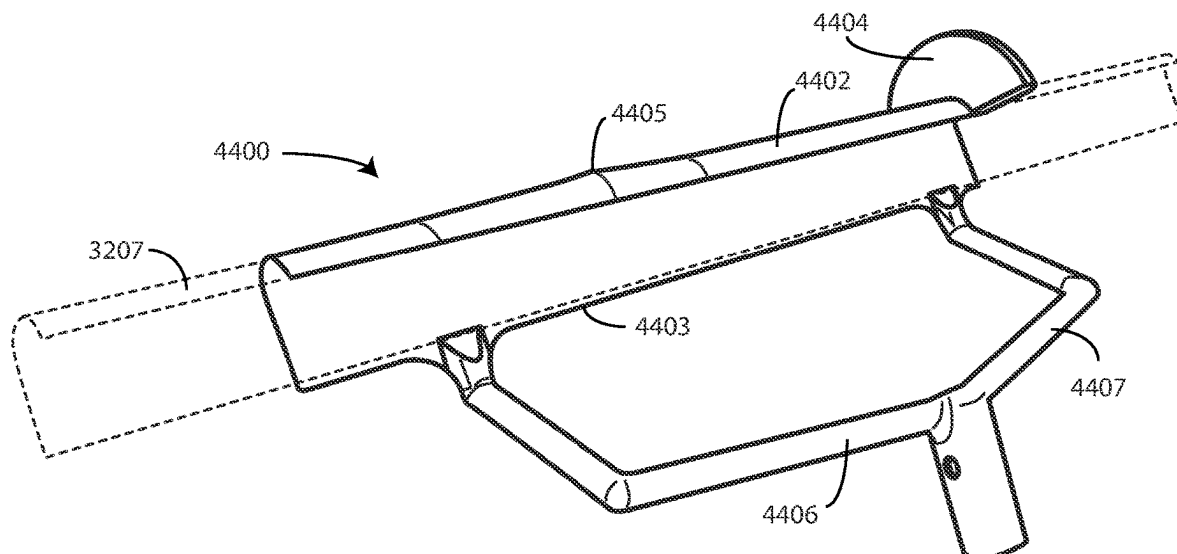
**FIG. 42**



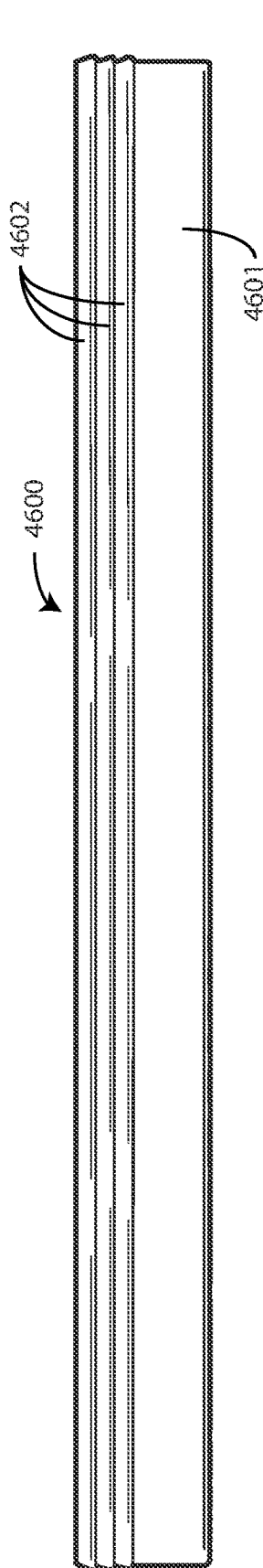




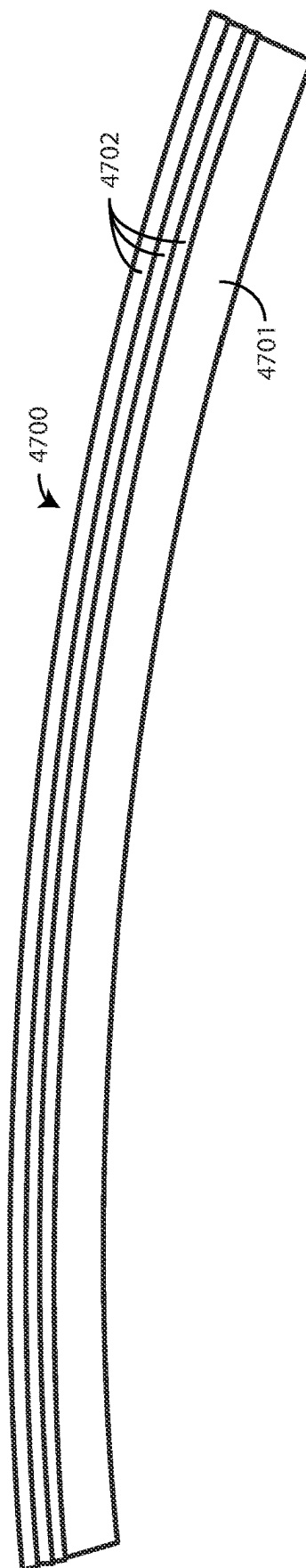
**FIG. 44**



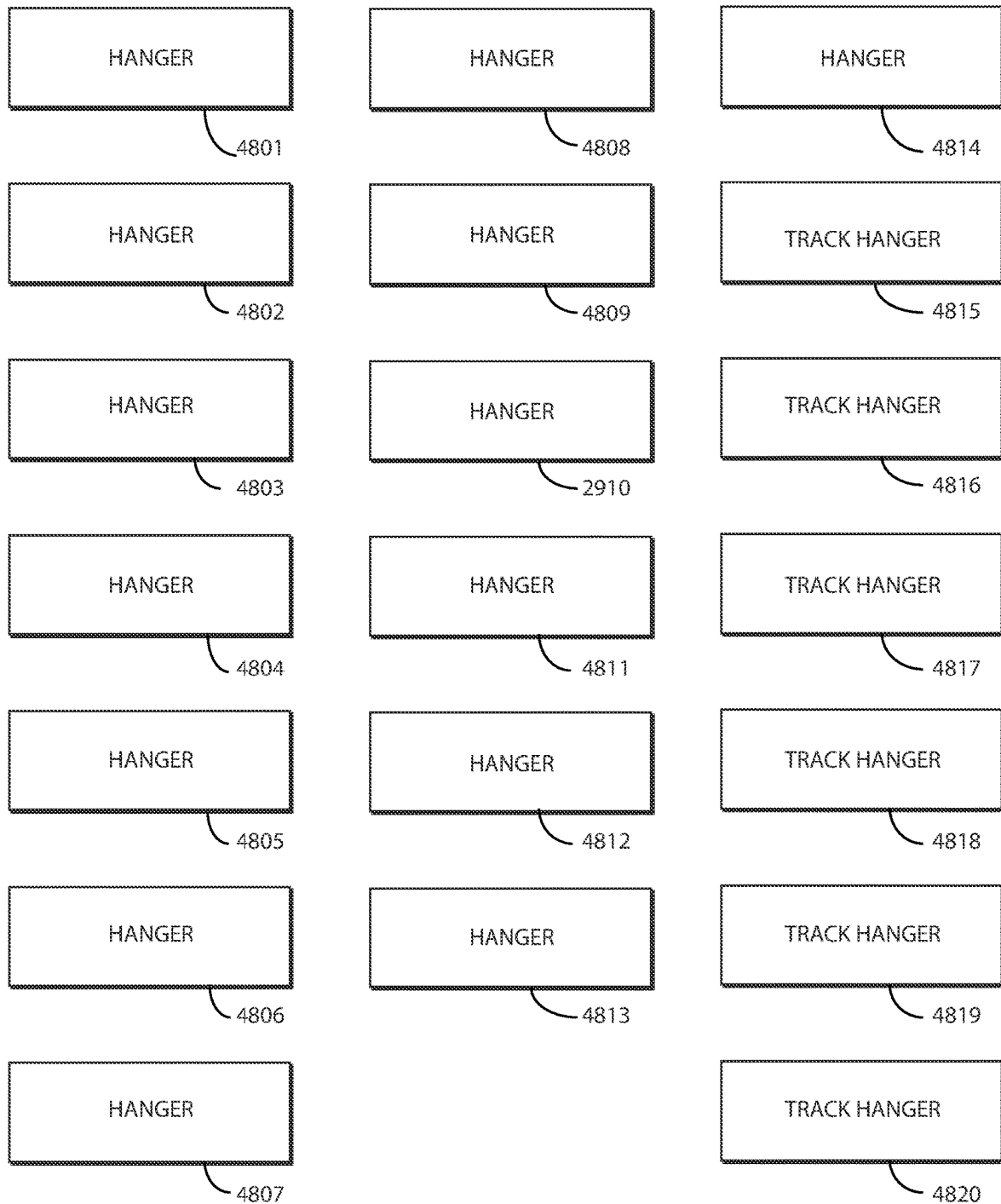
**FIG. 45**



**FIG. 46**



**FIG. 47**

**FIG. 48**

1

## HANGABLE APPARATUS AND SYSTEMS AND METHODS THEREFOR

### CROSS REFERENCE TO PRIOR APPLICATIONS

This application is a continuation application claiming and benefit under 35 U.S.C. § 120 from U.S. application Ser. No. 18/141,551, filed May 1, 2023, which is a continuation application claiming priority and benefit under 35 U.S.C. § 120 from U.S. application Ser. No. 17/684,282, filed Mar. 1, 2022, which is a continuation application claiming priority and benefit under 35 U.S.C. § 120 from U.S. application Ser. No. 16/913,815, filed Jun. 26, 2020, which is a continuation-in-part application claiming and benefit under 35 U.S.C. § 120 from U.S. application Ser. No. 16/588,771, filed Sep. 30, 2019, which is a continuation-in-part application claiming and benefit under 35 U.S.C. § 120 from U.S. application Ser. No. 16/045,391, filed Jul. 25, 2018, each of which is incorporated by reference for all purposes.

### BACKGROUND

#### Technical Field

This disclosure relates generally to hangable devices, and more particularly to an items comprising a hanger that hangs from a rail or track.

#### Background Art

Hanging items are popular in homes and businesses. Hanging items include curtains, drapes, plants, art, and so forth. Hanging items typically include a hanger, a mount, and something suspended from the hanger. Using a plant as one example, the hanger may comprise a metal hook, with the plant suspended beneath the hook. A user may couple the hook to a loop, perhaps mounted on the ceiling, to hang the plant.

While there are a variety of types of hangers and types of hanging items, curtains provide special challenges for designers. This is especially true in medical or hospital environments. It is frequently the case that medical service providers employ curtains to separate patients, conceal medical procedures from view, and to segregate areas of operating rooms and care centers. It is advantageous to launder such curtains to prevent the curtains from acting as a vector to transfer pathogens and bacteria from one patient to the next. Even where the curtains are disposable, they still need to be changed to prevent transfer of pathogens and bacteria from one patient to the next.

However, prior art curtains are difficult to take down and clean due to the fact that they are frequently attached to carriers that slide within a metal track. The dismounting process is costly and labor intensive. It would be advantageous to have an improved hangable apparatus, suitable for use in curtain and drape systems, which is easier and quieter to dismount.

### BRIEF DESCRIPTION OF THE DRAWINGS

FIG. 1 illustrates a perspective view of one explanatory track hanger system in accordance with one or more embodiments of the disclosure.

FIG. 2 illustrates a front elevation view of one explanatory track hanger system in accordance with one or more embodiments of the disclosure.

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FIG. 3 illustrates a rear elevation view of one explanatory track hanger system in accordance with one or more embodiments of the disclosure.

FIG. 4 illustrates a side elevation view of one explanatory track hanger in accordance with one or more embodiments of the disclosure.

FIG. 5 illustrates one explanatory key and hook tip in accordance with one or more embodiments of the disclosure.

FIG. 6 illustrates one explanatory base member in accordance with one or more embodiments of the disclosure.

FIG. 7 illustrates one explanatory track hanger prior to assembly in accordance with one or more embodiments of the disclosure.

FIG. 8 illustrates a partially assembled track hanger in accordance with one or more embodiments of the disclosure.

FIG. 9 illustrates an assembled track hanger in accordance with one or more embodiments of the disclosure.

FIG. 10 illustrates a partially assembled track hanger system in accordance with one or more embodiments of the disclosure.

FIG. 11 illustrates a perspective view of another explanatory track hanger system in accordance with one or more embodiments of the disclosure.

FIG. 12 illustrates another assembled track hanger in accordance with one or more embodiments of the disclosure.

FIG. 13 illustrates another assembled track hanger in accordance with one or more embodiments of the disclosure.

FIG. 14 illustrates yet another assembled track hanger in accordance with one or more embodiments of the disclosure.

FIG. 15 illustrates a perspective view of another explanatory track hanger system in accordance with one or more embodiments of the disclosure.

FIG. 16 illustrates another perspective view of another explanatory track hanger system in accordance with one or more embodiments of the disclosure, illustrating another explanatory key and hook tip in accordance with one or more embodiments of the disclosure.

FIG. 17 illustrates a front elevation view of another explanatory track hanger system in accordance with one or more embodiments of the disclosure.

FIG. 18 illustrates a rear elevation view of another explanatory track hanger system in accordance with one or more embodiments of the disclosure.

FIG. 19 illustrates a side elevation view of another explanatory track hanger in accordance with one or more embodiments of the disclosure.

FIG. 20 illustrates another explanatory base member in accordance with one or more embodiments of the disclosure.

FIG. 21 illustrates yet another explanatory base member in accordance with one or more embodiments of the disclosure.

FIG. 22 illustrates one or more method steps for assembling another explanatory track hanger configured in accordance with one or more embodiments of the disclosure.

FIG. 23 illustrates one explanatory accessory suitable for use with one or more embodiments of the disclosure.

FIG. 24 illustrates another explanatory accessory suitable for use with one or more embodiments of the disclosure.

FIG. 25 illustrates yet another explanatory accessory suitable for use with one or more embodiments of the disclosure.

FIG. 26 illustrates one explanatory system in accordance with one or more embodiments of the disclosure.

FIG. 27A illustrates yet another explanatory key in accordance with one or more embodiments of the disclosure.

FIG. 27B illustrates yet another explanatory key in accordance with one or more embodiments of the disclosure.

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FIG. 28 illustrates another perspective view of another explanatory track hanger system in accordance with one or more embodiments of the disclosure, illustrating another explanatory key and hook tip in accordance with one or more embodiments of the disclosure.

FIG. 29 illustrates a front elevation view of another explanatory track hanger system in accordance with one or more embodiments of the disclosure.

FIG. 30 illustrates a rear elevation view of another explanatory track hanger system in accordance with one or more embodiments of the disclosure.

FIG. 31 illustrates a side elevation view of another explanatory track hanger in accordance with one or more embodiments of the disclosure.

FIG. 32 illustrates one or more method steps for assembling another explanatory track hanger configured in accordance with one or more embodiments of the disclosure.

FIG. 33 illustrates a front elevation view of one explanatory hanger in accordance with one or more embodiments of the disclosure.

FIG. 34 illustrates a side elevation view of one explanatory hanger in accordance with one or more embodiments of the disclosure.

FIG. 35 illustrates a rear elevation view of one explanatory hanger in accordance with one or more embodiments of the disclosure.

FIG. 36 illustrates a top plan view of one explanatory hanger in accordance with one or more embodiments of the disclosure.

FIG. 37 illustrates a bottom plan view of one explanatory hanger in accordance with one or more embodiments of the disclosure.

FIG. 38 illustrates an exploded view of one explanatory hanger system in accordance with one or more embodiments of the disclosure.

FIG. 39 illustrates a partially assembled view of one explanatory hanger system in accordance with one or more embodiments of the disclosure.

FIG. 40 illustrates a perspective view of one explanatory assembled hanger system in accordance with one or more embodiments of the disclosure.

FIG. 41 illustrates a rear elevation view of one explanatory assembled hanger system in accordance with one or more embodiments of the disclosure.

FIG. 42 illustrates a front elevation view of one explanatory assembled hanger system in accordance with one or more embodiments of the disclosure.

FIG. 43 illustrates one explanatory system in accordance with one or more embodiments of the disclosure.

FIG. 44 illustrates a first perspective view of one explanatory hanger tool in accordance with one or more embodiments of the disclosure.

FIG. 45 illustrates another perspective view of one explanatory hanger tool in accordance with one or more embodiments of the disclosure.

FIG. 46 illustrates one explanatory track connector in accordance with one or more embodiments of the disclosure.

FIG. 47 illustrates another explanatory track connector in accordance with one or more embodiments of the disclosure.

FIG. 48 illustrates various embodiments of the disclosure.

Skilled artisans will appreciate that elements in the figures are illustrated for simplicity and clarity and have not necessarily been drawn to scale. For example, the dimensions of some of the elements in the figures may be exaggerated

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relative to other elements to help to improve understanding of embodiments of the present disclosure.

#### DETAILED DESCRIPTION OF THE DRAWINGS

Embodiments of the disclosure are now described in detail. Referring to the drawings, like numbers indicate like parts throughout the views. Apparatus components and method steps have been represented where appropriate by conventional symbols in the drawings, showing only those specific details that are pertinent to understanding the embodiments of the present disclosure so as not to obscure the disclosure with details that will be readily apparent to those of ordinary skill in the art having the benefit of the description herein.

As used in the description herein and throughout the claims, the following terms take the meanings explicitly associated herein, unless the context clearly dictates otherwise: the meaning of “a,” “an,” and “the” includes plural reference, the meaning of “in” includes “in” and “on.” Relational terms such as first and second, top and bottom, and the like may be used solely to distinguish one entity or action from another entity or action without necessarily requiring or implying any actual such relationship or order between such entities or actions. The terms “substantially” and “about” are used to refer to dimensions, orientations, or alignments inclusive of manufacturing tolerances. Thus, a “substantially orthogonal” angle with a manufacturing tolerance of plus or minus two degrees would include all angles between 88 and 92, inclusive. Also, reference designators shown herein in parenthesis indicate components shown in a figure other than the one in discussion. For example, talking about a device (10) while discussing figure A would refer to an element, 10, shown in figure other than figure A.

Embodiments of the disclosure provide a track hanger system for an item. In one embodiment, the item to be hung is a curtain. Illustrating by example, for a hospital setting where infection control is an area of high concern, track hangers configured in accordance with embodiments of the disclosure can be used to hang curtains between patients, procedures, areas, and so forth. Track hangers configured in accordance with embodiments of the disclosure are easier to mount and dismount than are prior art designs. Accordingly, embodiments of the disclosure advantageously make it easier for health care practitioners to change and launder the curtains to ensure that bacteria and other pathogens are not readily transferred from patient to patient or from patient to health care practitioner.

Embodiments of the disclosure contemplate that it is very difficult to change prior art curtain systems designed for health care environments. The hangers are difficult to dismount. Additionally, a technician, and sometimes multiple people, equipped with ladders or specialized equipment are required to dismount the curtains, making changing or laundering the curtains a labor-intensive and costly endeavor. For this reason, curtains are rarely changed or washed. In some situations, it is considered fortunate if the curtains were changed three times a year.

Advantageously, embodiments of the disclosure offer an improved track hanger that is easier to mount to a tiled ceiling, e.g., a dropped ceiling that includes a suspension grid of metal channels or rails, which are suspended to support ceiling panels, as are found in most hospitals, doctor's offices, and other health care offices. In one or more embodiments, the track hanger includes a hook, an extension, a head, a base member, and a key. Each component can be modular and detachable from the other. For example, the

key can be detachable from the hook. Similarly, the base member can be detachable from the extension, and so forth.

In one or more embodiments, the key includes a first major surface and a second major surface. One or both of the first major surface or the second major surface can define one or more barbs that facilitate a one-way insertion of the key into the track. This makes the track hangers and track extremely quick to mount.

Once mounted, curtains can be hung from the track. These curtains, such as those described in commonly assigned U.S. Ser. No. 15/651,774, filed Jul. 17, 2017, which is incorporated herein by reference, are quick and simple to mount and dismount. They require only a single person—working for a few minutes—to change the curtains. Accordingly, embodiments of the disclosure allow for more frequent changing and laundering of curtains, thereby promoting health and safety when used in hospitals or other health care settings.

In addition to potentially serving as vectors for bacteria and other pathogens, prior art curtain hanging systems have other problems as well. Most hangers require special mounting hardware and/or tools to mount to a wall or ceiling. Additionally, the mounting hardware leaves scars in the wall or ceiling should the hanger ever be taken down. Embodiments of the disclosure provide a solution to each of these problems by providing a base member that includes a first base member portion and a second base member portion. The first base member portion and the second base member portion are separable from each other along a medial major axis of the base member.

In one or more embodiments, the first base member portion comprises a first half rail clamp. Similarly, the second base member portion can include a second half rail clamp. In one or more embodiments, the first half rail clamp defines a first half head-receiving aperture, while the second half rail clamp defines a second half head-receiving aperture. In one or more embodiments, coupling the first base member portion to the second base member portion causes the first half head-receiving aperture to abut the second half head-receiving aperture, thereby resulting in the base member defining a head-receiving aperture into which the head of the hanger may situate.

In one or more embodiments, the hanger is then coupled to the base member. In one or more embodiments, the hanger includes a head, which is coupled to the extension. In one or more embodiments, the extension is situated between the head and the hook. In one or more embodiments, the head of the hanger is locked into the head-receiving aperture of the base member such that it is able to “clamp” onto a rail passing through the base member.

In one or more embodiments, separating the first base member portion from the second base member portion along the medial major axis therefore separates the first half rail clamp from the second half rail clamp and splits the head-receiving aperture. When used with a drop ceiling having a suspension grid of rails, one can simply place the first rail half clamp to the left of a rail and the second rail half clamp to the right of the rail, or vice versa.

The first base member portion can then be pressed against the second base member portion along the medial major axis, which causes the rail of the suspension grid to be caught between the first half rail clamp and the second half rail clamp. The head of the hanger can then be situated within the head-receiving aperture. A fastener can then be attached to the base member to both retain the first base member portion against the second base member portion and to clamp the head of the hanger against the rail. The fastener can be coupled and uncoupled as desired so as to allow quick

and easy coupling of the base member and head of the hanger to the rails of the suspension grid.

In one or more embodiments, the coupler engages one or more threads disposed on the exterior surface of a head receiver extending distally from the base member to retain the first base member portion against the second base member portion and the head of the hanger against the rail through the head-receiving aperture. As will be shown in more detail below, the coupler can also be used to bias and claim the head, which is attached to the extension and hook, against a major face of the rail within an interior socket supporting the threads. Thus, in one or more embodiments, the rail is clamped on three sides—two by the first base member portion and the second base member portion, and a third by the head of the hanger against the major surface of the rail. Advantageously, a track hanger for an item comprises a uniquely configured base member that facilitates simple mounting and dismounting of the track hanger to a rail of a conventional suspension grid of a drop ceiling while also biasing the head of a hanger against the same rail for additional coupling power.

Turning now to FIGS. 1-4, illustrated therein is one explanatory embodiment of a track hanger **100** configured in accordance with one or more embodiments of the disclosure. In one embodiment, the track hanger **100** is to suspend a track from a surface such as a wall or ceiling.

The track hanger **100**, in one or more embodiments, includes a hanger **101**, a base member **102**, and a key **403**. In one or more embodiments, each of the hanger **101**, the base member **102**, and the key **403** are separable from each other. For example, in one embodiment the hanger **101** can be detached from the base member **102**. Similarly, the key **403** can be detached from the hanger **101**. It should be noted that the track hanger **100** can be manufactured in different sizes and shapes so as to be compatible with, and fit appropriately, differently sized suspension grids of drop ceilings.

In one or more embodiments, the hanger **101** comprises a hook **103** and an extension **104**. In one or more embodiments, the key **403** attaches to a distal end **105** of the hook **103**. Accordingly, where the key **403** is detachable from the hanger **101**, the attachment location to attach or detach the key **403** from the hook **103** is the distal end **105**. Thus, in one or more embodiments the key **403** is detachable from the hook **103**.

In one embodiment, the hanger **101** is configured as a single, unitary element. Said differently, in one embodiment the hook **103** and the extension **104** are manufactured as a single, integral unit. As will be described in more detail below with reference to FIG. 7, the hanger **101** can also include a head. Where included, the hook **103**, the extension **104**, and the head can be manufactured as a single, integral unit.

Illustrating by example, in one embodiment the hook **103**, the extension **104**, and the head are manufactured from a thermoplastic material by way of an injection molding process. The hook **103**, the extension **104**, and the head can be manufactured from nylon, styrene, ABS, polycarbonate, or polycarbonate-ABS, PMMA, PVC, or other polyamide-based thermoplastics in one embodiment. Other materials suitable for manufacturing the hook **103**, the extension **104**, and the head will be obvious to those of ordinary skill in the art having the benefit of this disclosure. While the base member **102** and the key **403** are separable from the hanger **101**, in one or more embodiments the base member **102** and the key **403** can be manufactured from the same material as is the hanger **101**. In other embodiments, the hanger **101** is

manufactured from materials different from the base member 102 and/or key 403. The base member 102, hanger 101, and key 403 can each be manufactured from different materials as well.

As best seen in FIG. 4, in one embodiment the key 403 includes a first major surface 401 and a second major surface 402. In this illustrative embodiment, the first major surface 401 defines one or more barbs 404. Each barb comprises a hemi-triangular protrusion with the base of each hemi-triangular protrusion being disposed beneath the side of the hemi-triangular protrusion. In this illustrative embodiment, the second major surface 402 is substantially planar.

In one or more embodiments, a track 106 attaches to the key 403. In this illustrative embodiment, the track 106 has a first flat side 407 and a second flat side 408. An arch 409 spans between ends of the first flat side 407 and the second flat side 408 in this embodiment.

In one embodiment, the track 106 is flexible so that it can be shaped into different contours when attached to a track hanger 100. For example, in one embodiment the track 106 is manufactured from plastic. One suitable plastic for the track 106 is polypropylene, although other flexible materials, such as polyethylene, will be obvious to those of ordinary skill in the art having the benefit of this disclosure. Additionally, other material such as ABS plastic can be used in other embodiments. In one embodiment, the track 106 is manufactured from an extrusion process.

In one embodiment, the track 106 is a continuous piece. In other embodiments, segments of different tracks can be aligned end-to-end to form a composite track. In one embodiment, the track 106 is malleable. While the cross section of the track 106 shown in FIG. 4 is generally flat along each of the first flat side 407 and the second flat side 408, it should be noted that the cross section could take other shapes as well, such as ovalar or flat.

In one embodiment the track 106 is configured so as to be easily cleanable. In one embodiment, the track 106 is manufactured so as to be light beige in color. In another embodiment, the track 106 is manufactured so as to be white in color. Other colors for the track 106 will be obvious to those of ordinary skill in the art having the benefit of this disclosure. In one embodiment, the track 106 is cut to predefined lengths, such as twenty-foot lengths. In one or more embodiments, the ends of the track 106 can be contoured for smooth interconnection to adjacent track segments.

In one embodiment, the track 106 comprises a coating 107. For example, in one embodiment the track 106 is coated with a silicon-based coating to allow curtains or other hangers to more smoothly slide along the track 106. It should be noted that one primary advantage offered by embodiments of the disclosure is that hanging systems configured in accordance with embodiments of the disclosure are very, very quiet when in operation. For example, where the track 106 is manufactured from polyethylene and coated with silicon, and a curtain having a hanger such as those described in commonly assigned U.S. Ser. No. 15/651,774, filed Jul. 17, 2017, which is incorporated herein by reference, which is made from a woven polyester mesh, moving the hanger along the track 106 is nearly a silent procedure. This is advantageous in hospitals and other medical environments where noise is problematic. Prior art hanging systems, which primarily include metal, are loud and intrusive. In one embodiment of the present disclosure, each of the track 106, the hanger, and any item attached

thereto is made without any metal. This greatly reduces—if not eliminates—noise when the hangers are moved on the track 106.

A second advantage of not including metal in either the track 106 or items hanging therefrom is that components of systems configured in accordance with various embodiments of the disclosure can be extremely light in weight. This enables the track 106 to easily be mounted on the key 403. The process can be accomplished by anyone, regardless of size or strength.

The inclusion of the first flat side 407, the second flat side 408, and the arch 409 define a peninsular indentation 410 into which the key 403 may be inserted. In this illustrative embodiment, the track 106 comprises one or more complementary barbs 405. Each of the one or more complementary barbs 405 is complementary in shape to the one or more barbs 404 of the key 403. Here, each complementary barb comprises an inverted hemi-triangular protrusion with a base of each inverted hemi-triangular protrusion being disposed above a side of the inverted hemi-triangular protrusion.

Using this configuration, the one or more barbs 404 of the key 403 facilitate one-way penetration of the key 403 into the track 106. In one or more embodiments, the track 106 is manufactured from a pliable material, such as a thermoplastic. When the key 403 is inserted into the peninsular indentation 410, the first flat side 407 of the track 106 flexes so that the one or more complementary barbs 405 of the track 106 pass over the one or more barbs 404 of the key 403. Once the key 403 is fully inserted into the peninsular indentation 410 of the track 106, first flat side 407 of the track 106 flex back toward the key 403, thereby causing the one or more complementary barbs 405 to engage the one or more barbs 404 of the key 403. This results in the key 403 being frictionally retained within the track 106.

In one or more embodiments, the second flat side 408 of the track 106 terminates at an end opposite the arch 409 in an L-shaped latch 411. In one or more embodiments, the second major surface 402 of the key includes a complementary L-shaped indentation 412 into which the base of the “L” of the L-shaped latch 411 seats when the key 403 is fully inserted into the track 106.

In operation, when the key 403 is inserted into the peninsular indentation 410, the second flat side 408 of the track 106 flexes so the base of the “L” of the L-shaped latch 411 of the track 106 can pass over the second flat side 408 of the key 403. Once the key 403 is fully inserted into the peninsular indentation 410 of the track 106, second flat side 408 of the track 106 flexes back toward the key 403, thereby causing the base of the “L” of the L-shaped latch 411 to seat within the complementary L-shaped indentation 412 of the key 403. This assists in retaining the key 403 within the track 106.

In one or more embodiments, the base member 102 comprises a first base member portion 108 and a second base member portion 109. In one or more embodiments, the first base member portion 108 and the second base member portion 109 are separable along a medial major axis 110. In the illustrative embodiment of FIGS. 1-4, the first base member portion 108 and the second base member portion 109 have been placed together such that their inner surfaces abut along the medial major axis 110. A coupler 111, which in this embodiment is a plastic hex nut, couples the first base member portion 108 and the second base member portion 109 together to retain the inner surfaces abutting at the medial major axis 110.

In one or more embodiments, the first base member portion 108 defines a first half rail clamp 112. Similarly, the

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second base member portion 109 defines a second half rail clamp 113. Each of the first half rail clamp 112 and the second half rail clamp 113 includes a vertical member extending distally from the first base member portion 108 and the second base member portion 109, respectively, and a horizontal member extending distally from its respective vertical member only a portion of the width of the first base member portion 108 and the second base member portion 109. This leaves a gap between each horizontal member, as shown in FIGS. 2-3.

Separating the first base member portion 108 from the second base member portion 109 along the medial major axis 110 therefore separates the first half rail clamp 112 from the second half rail clamp 113. When used with a drop ceiling having a suspension grid of rails, one can simply place the first half rail clamp 112 to the left of a rail and the second half rail clamp 113 to the right of the rail, or vice versa. The first base member portion 108 can then be pressed against the second base member portion 109 along the medial major axis 110. This causes the rail of the suspension grid to be caught between the first half rail clamp 112 and the second half rail clamp 113.

When the coupler 111 is then be attached to the base member 102 to retain the first base member portion 108 against the second base member portion 109, this causes the first half rail clamp 112 from the second half rail clamp 113 to couple the track hanger 100 to the rail of the suspension grid. In one or more embodiments, as will be described below with reference to FIGS. 6-9, the coupler 111 engages one or more threads disposed on the exterior surface of a head receiver extending distally from the base member 102 to retain the first base member portion 108 against the second base member portion 109. As will be also shown in more detail below in these figures, the coupler 111 can also be used to retain a head, which is attached to the extension 104 and hook 103, within an interior socket supporting the threads. Thus, in one or more embodiments, the track hanger 100 comprises a uniquely configured base member 102 that facilitates simple mounting and dismounting of the track hanger 100 to a track of a conventional suspension grid of a drop ceiling.

In the illustrative embodiment of FIGS. 1-4, hanger 101 of the track hanger 100 includes one or more bends 114, 115, 116. In this illustrative embodiment, the hanger 101 includes three bends 114, 115, 116. Bend 114 and bend 115 are obtuse, while bend 116 is substantially orthogonal.

As noted above, in one or more embodiments each of the hanger 101, the base member 102, and the key 403 are separable from each other. For example, in one embodiment the hanger 101 can be detached from the base member 102. Similarly, the key 403 can be detached from the hanger 101. Turning now to FIG. 5, illustrated therein is the key 403 detached from the end 501 of the hook 103. In this illustrative embodiment, the key 403 is selectively attachable to the end 501 of the hook 103. To attach the key 403 to the end 501 of the hook 103, the end 501 of the hook 103 opposite the extension (104) of the hanger 101 is inserted into an aperture 502 defined within a connector 504 of the key. In one or more embodiments, the aperture 502 is configured to frictionally retain the key 403 to the end 501 of the hook 103. However, in other embodiments, a latch, snap, one-way latch, or other mating feature can be incorporated into either the aperture 502 or the end 501 of the hook 103 to frictionally retain the key 403 to the end 501 of the hook 103.

In this illustrative embodiment, the aperture 502 and the end 501 of the hook 103 are geometrically configured to prevent rotation of the key 403 about the end 501 of the

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hook. To wit, here the aperture 502 defines a flat side 505 and two arched protrusions 506, 507. The end 501 of the hook 103 have a flat side 509 and two arched indentations 508 that are complementary in shape to the two arched protrusions 506, 507. When the end 501 of the hook 103 opposite the extension (104) of the hanger 101 is inserted into an aperture 502 defined within a connector 504 of the key 403, engagement of the two arched protrusions 506, 507 with the two arched indentations 508, opposite engagement of the flat side 505 of the aperture 502 with the flat side 505 of the end 501 of the hook 103 prevents rotation of the key 403 about the end 501 of the hook 103.

Turning now to FIG. 6, illustrated therein are the first base member portion 108 and the second base member portion 109 after having been separated along the medial major axis 110. This separation exposes the inner surfaces 601, 602 of the first base member portion 108 and the second base member portion 109, respectively.

In one or more embodiments, the inner surface 601 of the first base member portion 108 comprises one or more bosses 603, 604. Here, two bosses 603, 604 are shown. However, more or fewer bosses can be included in other embodiments.

In this illustrative embodiment, the inner surface 602 of the second base member portion 109 comprises one or more boss receivers 605, 606. Here, two boss receivers 605, 606 are shown. However, more or fewer boss receivers can be included in other embodiments. In one or more embodiments, the boss receivers 605, 606 comprise apertures that extend into the inner surface 602 of the second base member portion 109, and which have shapes that are complementary to the one or more bosses 603, 604 of the inner surface 601 of the first base member portion 108. The one or more boss receivers 605 can optionally include wider mouths 607, 608 at their openings to facilitate easier insertion of the one or more bosses 603, 604 into the one or more boss receivers 605, 606 when the first base member portion 108 and the second base member portion 109 are pressed together such that their inner surfaces 601, 602 abut at the medial major axis 110.

In one or more embodiments, when the first base member portion 108 and the second base member portion 109 are pressed together such that their inner surfaces 601, 602 abut at the medial major axis 110, the one or more bosses 603, 604 insert into the one or more boss receivers 605, 606. This prevents the inner surfaces 601, 602 of the first base member portion 108 and the second base member portion 109 from sliding about.

In the illustrative embodiment of FIG. 6, the base member 102 includes a head receiver. As will be described in more detail below with reference to FIG. 7, the hanger (101) can also include a head. In this illustrative embodiment, the first base member portion 108 has a first half head receiver 609 extending distally from a side of the first base member portion 108 disposed opposite the first half rail clamp 112. Similarly, the second base member portion 109 has a second half head receiver 610 extending distally from a side of the second base member portion 109 disposed opposite the second half rail clamp 113.

In this illustrative embodiment, the first half head receiver 609 and the second half head receiver 610 are both hemi-cylindrical. When the first base member portion 108 and the second base member portion 109 are pressed together such that their inner surfaces 601, 602 abut at the medial major axis 110, the edges of the first half head receiver 609 and the second half head receiver 610 also abut to define a cylindrical head receiver. While a cylinder is one suitable shape for a head receiver formed by abutment of the edges of the



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first half head receiver 609 and the second half head receiver 610, other shapes can be used as well. In other embodiments, the head receiver is rectangular, triangular, polygonal, or free form shapes. Still other shapes for the head receiver will be obvious to those of ordinary skill in the art having the benefit of this disclosure.

In one or more embodiments, the inner surfaces 611,612 of the first half head receiver 609 and the second half head receiver 610 define an interior socket to receive the head of a hanger (101) configured in accordance with one or more embodiments of the disclosure. In one or more embodiments, the exterior surfaces 613,614 of the first half head receiver 609 and the second half head receiver 610 define one or more threads 615,616. When the first base member portion 108 and the second base member portion 109 are pressed together such that their inner surfaces 601,602 abut at the medial major axis 110, the edges of the first half head receiver 609 and the second half head receiver 610 also abut to define a cylindrical head receiver having an interior socket defined by the inner surfaces 611,612 of the first half head receiver 609 and the second half head receiver 610, and having threads defined by the exterior surfaces 613,614 of the first half head receiver 609 and the second half head receiver 610. The coupler (111) can then be threaded onto the threads defined by the exterior surfaces 613,614 of the first half head receiver 609 and the second half head receiver 610 to couple the first base member portion 108 and the second base member portion 109 together.

Turning now to FIG. 7, illustrated therein is the hanger 101 separated from the base member 102. As can be seen in this separated state, in one or more embodiments the hanger 101 includes a head 701, which is coupled to the extension 104. In this illustrative embodiment, the extension 104 is situated between the head 701 and the hook 103.

In this illustrative embodiment, the head 701 is cylindrical. While a cylinder is one suitable shape for the head 701, other shapes can be used as well. In other embodiments, the head 701 is rectangular, triangular, polygonal, or free form shapes. Still other shapes for the head 701 will be obvious to those of ordinary skill in the art having the benefit of this disclosure.

In one or more embodiments, the head 701 and the head receiver defined by the first half head receiver 609 and the second half head receiver 610 are complementary in shape. In this illustrative embodiment, the head 701 and the head receiver defined by the first half head receiver 609 and the second half head receiver 610, when abutting, are cylindrical.

The head receiver defined by the first half head receiver 609 and the second half head receiver 610, or more particularly, the interior socket defined by the inner surfaces (611,612) of the first half head receiver 609 and the second half head receiver 610, is configured to receive the head 701 when the head 701 is inserted into the head receiver defined by the first half head receiver 609 and the second half head receiver 610.

In one or more embodiments, the head 701 can include one or more teeth 702. Where included, the one or more teeth 702 can engage an interior surface of the interior socket defined by the inner surfaces (611,612) of the first half head receiver 609 and the second half head receiver 610, thereby preventing rotation of the head 701 when inserted into the head receiver defined by the first half head receiver 609 and the second half head receiver 610. In one or more embodiments, the interior surface of the interior socket defined by the inner surfaces (611,612) of the first half head receiver 609 and the second half head receiver 610 can include

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complementary teeth to engage the one or more teeth 702 of the head 701, thereby further preventing rotation of the head 701 when inserted into the head receiver defined by the first half head receiver 609 and the second half head receiver 610. Where rotation of the head 701 within head receiver defined by the first half head receiver 609 and the second half head receiver 610 is desired, the head 701 and the interior surface of the interior socket defined by the inner surfaces (611,612) of the first half head receiver 609 and the second half head receiver 610 can be smooth so as to facilitate movement. Lubricants can be placed between the head 701 and the interior surface of the interior socket defined by the inner surfaces (611,612) of the first half head receiver 609 and the second half head receiver 610 as well.

As shown in FIG. 7, prior to insertion of the head 701 into the interior socket defined by the inner surfaces (611,612) of the first half head receiver 609 and the second half head receiver 610, the coupler 111 is placed about the hanger 101. The head 701 is then inserted into the interior socket defined by the inner surfaces (611,612) of the first half head receiver 609 and the second half head receiver 610. This is shown in FIG. 8.

Turning now to FIG. 8, once the head (701) is then inserted into the interior socket defined by the inner surfaces (611,612) of the first half head receiver 609 and the second half head receiver 610. The coupler 111, which includes interior threads, can then engage the one or more threads 801 disposed along the exterior surface of the head receiver defined by the first half head receiver 609 and the second half head receiver 610 to retain the first base member portion 108 and the second base member portion 109 together. This engagement of the coupler 111 to the one or more threads 801 disposed along the exterior surface of the head receiver defined by the first half head receiver 609 and the second half head receiver 610 also retains the head (701) of the hanger 101 within the interior socket defined by the inner surfaces (611,612) of the first half head receiver 609 and the second half head receiver 610. Said differently, in one or more embodiments, when the head (701) is inserted into the head receiver defined by the first half head receiver 609 and the second half head receiver 610, the coupler 111 is operable to engage the exterior surface of the head receiver defined by the first half head receiver 609 and the second half head receiver 610 to retain the head (701) within the head receiver defined by the first half head receiver 609 and the second half head receiver 610.

The completed hanger assembly 900 is shown in FIG. 9. As shown in FIG. 10, the track 106 can then be attached to the key 403 as described above with reference to FIG. 4, thereby yielding the embodiment shown in FIGS. 1-3.

Turning now FIG. 11, illustrated therein is another explanatory embodiment of a track hanger 1100 configured in accordance with one or more embodiments of the disclosure. In one embodiment, the track hanger 1100 is to suspend a track from a surface such as a wall or ceiling.

The track hanger 1100, in one or more embodiments, includes a hanger 1101, a base member 1102, and a key 1143. In this illustrative embodiment, the hanger 1101 has a circular cross section 1181. However, the hanger 1101 can have other cross sections as well. For example, in another embodiment the hanger 1101 has a cross section 1180 configured as a cross or an "X." In still another embodiment, the hanger 1101 has a cross section 1182 that is a square or diamond. Other cross sections, such as ovals, free form shapes, polygons, and the like, will be obvious to those of ordinary skill in the art having the benefit of this disclosure.

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In one or more embodiments, each of the hanger **1101**, the base member **1102**, and the key **1143** are separable from each other. For example, in one embodiment the hanger **1101** can be detached from the base member **1102**. Similarly, the key **1143** can be detached from the hanger **1101**. However, in other embodiments, one or more of the hanger **1101**, base member **1102**, or the key **1143** can be integrated with, or perdurably connected to, another of the hanger **1101**, base member **1102**, or the key **1143**. Illustrating by example, in one or more embodiments the base member **1102** and the hanger **1101** can be constructed as a single, integrated, unitary component. In one embodiment, base member **1102** and hanger **1101** are constructed as a single, integrated, unitary component. Similarly, base member **1150** and hanger **1101** can be constructed as a single, integrated, unitary component. Moreover, base member **1160** and hanger **1101** can be constructed as a single, integrated, unitary component, and so forth.

In this illustration, three different and interchangeable base members **1102**, **1150**, **1160** are shown. The first base member **1102** is a track clip base member, while the second base member **1150** is a screw adaptor base member. The third base member **1160** is a track insertion base member. The first base member **1102** is similar to that shown above with reference to FIGS. 1-4.

The second base member **1150** includes a quadrilateral attachment support **1151** that defines a recess **1152** along its upper surface. Two apertures **1153**, **1154**, through which screws can be inserted from the lower surface of the quadrilateral attachment support **1151**, through the apertures **1153**, **1154**, and out the upper surface of the quadrilateral attachment support **1151** to attach the second base member **1150** to a ceiling. Each aperture **1153**, **1154** is surrounded by a boss **1155**, **1156** in the recess **1152**. The recess **1152** is bounded by a perimeter wall as shown in this illustrative embodiment. A head receiver **1158** includes one or more threads **1159** as previously described. A neck **1157**, which has a diameter that is smaller than that of the head receiver **1158**, and is also narrower than a minor dimension of the quadrilateral of the quadrilateral attachment support **1151**, separates the quadrilateral attachment support **1151** and the head receiver **1158**.

The third base member **1160** includes a rectangular track insertion head **1161** that can be inserted into a track with the major axis of the rectangular track insertion head **1161** parallel to the major axis of the track. A head receiver **1163** includes one or more threads **1164** as previously described. A neck **1162**, which has a diameter that is smaller than that of the head receiver **1163**, and is also narrower than a minor dimension of the rectangular track insertion head **1161**, separates the rectangular track insertion head **1161** and the head receiver **1163**.

In one or more embodiments, the hanger **1101** comprises a hook **1103** and an extension **1104**. In one or more embodiments, the key **1143** attaches to a distal end **1105** of the hook **1103**. Accordingly, where the key **1143** is detachable from the hanger **1101**, the attachment location to attach or detach the key **1143** from the hook **1103** is the distal end **1105**. Thus, in one or more embodiments the key **1143** is detachable from the hook **1103**.

In one embodiment, the distal end **1105** of the hook **1103** simply inserts into the key **1143**, with a friction fit holding the two components together. In another embodiment, a protrusion **1183** disposed along the distal end **1105** of the hook **1103** inserts into a recess **1184** configured in the key **1143** for a snap fit. Other engagements for the distal end

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**1105** of the hook **1103** and the key **1143** will be obvious to those of ordinary skill in the art having the benefit of this disclosure.

In one embodiment, the hanger **1101** is configured as a single, unitary element. Said differently, in one embodiment the hook **1103** and the extension **1104** are manufactured as a single, integral unit. In this illustrative embodiment, the hanger **1101** also includes a head **1170**. Where included, the hook **1103**, the extension **1104**, and the head **1170** can be manufactured as a single, integral unit as previously described.

In one embodiment the key **1143** includes a first major surface **1144** and a second major surface **1145**. In this illustrative embodiment, the first major surface **1144** defines one or more barbs **1146**. Each barb comprises a hemi-triangular protrusion with the base of each hemi-triangular protrusion being disposed beneath the side of the hemi-triangular protrusion. In this illustrative embodiment, the second major surface **1145** is substantially planar.

In one or more embodiments, a track **1106** attaches to the key **1143**. In this illustrative embodiment, the track **1106** has a first flat side **1147** and a second flat side **1148**. An arch spans between ends of the first flat side **1147** and the second flat side **1148** in this embodiment. In one embodiment, the track **1106** is flexible so that it can be shaped into different contours when attached to a track hanger **1100**.

In one embodiment, the track **1106** is a continuous piece. In other embodiments, segments of different tracks can be aligned end-to-end to form a composite track. In one embodiment, the track **1106** is malleable. While the cross section of the track **1106** shown in FIG. 10 is generally flat along each of the first flat side **1147** and the second flat side **1148**, it should be noted that the cross section could take other shapes as well, such as ovular or flat.

In one embodiment the track **1106** is configured so as to be easily cleanable. In one embodiment, the track **1106** is manufactured so as to be light beige in color. In one embodiment, the track **1106** is cut to predefined lengths, such as twenty-foot lengths. In one or more embodiments, the ends of the track **1106** can be contoured for smooth interconnection to adjacent track segments.

The inclusion of the first flat side **1147**, the second flat side **1148**, and the arch define a peninsular indentation into which the key **1143** may be inserted. In this illustrative embodiment, the track **1106** comprises one or more complementary barbs **1149**. Each of the one or more complementary barbs **1149** is complementary in shape to the one or more barbs **1146** of the key **1143**. Here, each complementary barb comprises an inverted hemi-triangular protrusion with a base of each inverted hemi-triangular protrusion being disposed above a side of the inverted hemi-triangular protrusion.

Using this configuration, the one or more barbs **1146** of the key **1143** facilitate one-way penetration of the key **1143** into the track **1106**. In one or more embodiments, the track **1106** is manufactured from a pliable material, such as a thermoplastic. When the key **1143** is inserted into the peninsular indentation, the first flat side **1147** of the track **1106** flexes so that the one or more complementary barbs **1149** of the track **1106** pass over the one or more barbs **1146** of the key **1143**. Once the key **1143** is fully inserted into the peninsular indentation of the track **1106**, first flat side **1147** of the track **1106** flex back toward the key **1143**, thereby causing the one or more complementary barbs **1149** to engage the one or more barbs **1146** of the key **1143**. This results in the key **1143** being frictionally retained within the track **1106**.

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In one or more embodiments, the second flat side **1148** of the track **1106** terminates at an end opposite the arch in an L-shaped latch. In one or more embodiments, the second major surface **1145** of the key includes a complementary L-shaped indentation into which the base of the “L” of the L-shaped latch seats when the key **1143** is fully inserted into the track **1106**.

In operation, when the key **1143** is inserted into the peninsular indentation, the second flat side **1148** of the track **1106** flexes so the base of the “L” of the L-shaped latch of the track **1106** can pass over the second flat side **1148** of the key **1143**. Once the key **1143** is fully inserted into the peninsular indentation of the track **1106**, second flat side **1148** of the track **1106** flexes back toward the key **1143**, thereby causing the base of the “L” of the L-shaped latch to seat within the complementary L-shaped indentation of the key **1143**. This assists in retaining the key **1143** within the track **1106**.

In one or more embodiments, the base member **1102** comprises a first base member portion **1108** and a second base member portion **1109**. In one or more embodiments, the first base member portion **1108** and the second base member portion **1109** are separable along a medial major axis **1110**. In the illustrative embodiment of FIG. 10, the first base member portion **1108** and the second base member portion **1109** have been placed together such that their inner surfaces abut along the medial major axis **1110**. A coupler **1111**, which in this embodiment is a plastic hex nut, couples the first base member portion **1108** and the second base member portion **1109** together to retain the inner surfaces abutting at the medial major axis **1110**.

In one or more embodiments, the first base member portion **1108** defines a first half rail clamp **1112**. Similarly, the second base member portion **1109** defines a second half rail clamp **1113**. Each of the first half rail clamp **1112** and the second half rail clamp **1113** includes a vertical member extending distally from the first base member portion **1108** and the second base member portion **1109**, respectively, and a horizontal member extending distally from its respective vertical member only a portion of the width of the first base member portion **1108** and the second base member portion **1109**. This leaves a gap between each horizontal member, as shown.

Separating the first base member portion **1108** from the second base member portion **1109** along the medial major axis **1110** therefore separates the first half rail clamp **1112** from the second half rail clamp **1113**. When used with a drop ceiling having a suspension grid of rails, one can simply place the first half rail clamp **1112** to the left of a rail and the second half rail clamp **1113** to the right of the rail, or vice versa. The first base member portion **1108** can then be pressed against the second base member portion **1109** along the medial major axis **1110**. This causes the rail of the suspension grid to be caught between the first half rail clamp **1112** and the second half rail clamp **1113**.

When the coupler **1111** is then be attached to the base member **1102** to retain the first base member portion **1108** against the second base member portion **1109**, this causes the first half rail clamp **1112** from the second half rail clamp **1113** to couple the track hanger **1100** to the rail of the suspension grid. In one or more embodiments, the coupler **1111** engages one or more threads **1171** disposed on the exterior surface of a head receiver **1172** extending distally from the base member **1102** to retain the first base member portion **1108** against the second base member portion **1109**. The coupler **1111** can also be used to retain a head **1170**, which is attached to the extension **1104** and hook **1103**,

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within an interior socket supporting the threads. Thus, in one or more embodiments, the track hanger **1100** comprises a uniquely configured base member **1102** that facilitates simple mounting and dismounting of the track hanger **1100** to a track of a conventional suspension grid of a drop ceiling. The hanger **1186** can optionally include an aperture **1185** in its top corner. The track hanger **1100** can be assembled as previously described above with reference to FIGS. 7-9.

The completed hanger assembly **1200** using the first base member **1102** is shown in FIG. 12. The completed hanger assembly **1300** using the second base member **1150** is shown in FIG. 13. The completed hanger assembly **1400** using the first base member **1102** is shown in FIG. 14.

Turning now to FIGS. 15-20, illustrated therein is another explanatory embodiment of a track hanger **1500** configured in accordance with one or more embodiments of the disclosure. In one embodiment, the track hanger **1500** is to suspend a track from a surface such as a wall or ceiling.

The track hanger **1500**, in one or more embodiments, includes a hanger **1501**, a base member **1502**, and a key **1503**. In one or more embodiments, each of the hanger **1501**, the base member **1502**, and the key **1503** are separable from each other. For example, in one embodiment the hanger **1501** can be detached from the base member **1502**. Similarly, as shown in FIG. 16, the key **1503** can be detached from the hanger **1501**. It should be noted that the track hanger **1500** can be manufactured in different sizes and shapes so as to be compatible with, and fit appropriately, differently sized suspension grids of drop ceilings.

In one or more embodiments, the hanger **101** comprises a hook **1504** and an extension **1505**. In one or more embodiments, the key **1503** attaches to a distal end **1601** of the hook **1504**. Accordingly, where the key **1503** is detachable from the hanger **1501**, the attachment location to attach or detach the key **1503** from the hook **1504** is the distal end **1601**. In one or more embodiments, the key **1503** includes a distal end receiver **1602** into which the distal end **1601** of the hook **1504** inserts. The distal end receiver **1602** can define one or more engagement recesses **1603** to receive one or more engagement protrusions **1604** projecting from the distal end **1601** of the hook **1504** to retain the key **1503** to the distal end **1601** of the hook **1504** in one or more embodiments. In one or more embodiments the key **1503** is detachable from the hook **1504**.

In one or more embodiments, the hanger **1501** comprises a head **1605** that is attached to the extension **1505**. In one embodiment shown in FIG. 16, the hanger **1501** is configured as a single, unitary component. Said differently, in one embodiment the hook **1504**, the extension **1505**, and the head **1605** coupled to the extension **1505** are manufactured as a single, integral unit.

Illustrating by example, in one embodiment the hook **1504**, the extension **1505**, and the head **1605** are manufactured from a thermoplastic material by way of an injection molding process. The hook **1504**, the extension **1505**, and the head **1605** can be manufactured from nylon, styrene, ABS, polycarbonate, or polycarbonate-ABS, PMMA, PVC, or other polyamide-based thermoplastics in one embodiment. Other materials suitable for manufacturing the hook **1504**, the extension **1505**, and the head **1605** will be obvious to those of ordinary skill in the art having the benefit of this disclosure.

While the base member **1502** and the key **1503** can be separable from the hanger **101**, in one or more embodiments the base member **1502** and the key **1503** can be manufactured from the same material as is the hanger **1501**. In other embodiments, the hanger **1501** is manufactured from mate-

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rials different from the base member **1502** and/or key **1503**. The base member **1502**, hanger **1501**, and key **1503** can each be manufactured from different materials as well.

As best seen by comparing FIGS. 17-18, in one embodiment the key **1503** includes a first major surface **1801** and a second major surface **1701**. In this illustrative embodiment, the first major surface **1801** defines one or more barbs **1802**. As best shown in FIG. 19, in one or more embodiments each barb **1802** comprises a hemi-triangular protrusion with the base of each hemi-triangular protrusion being disposed beneath the side of the hemi-triangular protrusion. In this illustrative embodiment, the second major surface **1701** is substantially planar.

In one or more embodiments, a track (**106**) attaches to the key **1503** as previously described. For example, in one or more embodiments the track (**106**) has a first flat side (**407**) and a second flat side (**408**), with an arch (**409**) spanning between ends of the first flat side (**407**) and the second flat side (**408**). The track (**106**) can be flexible so that it can be shaped into different contours when attached to a track hanger **1500**.

In one or more embodiments, the inclusion of the first flat side (**407**), the second flat side (**408**), and the arch (**409**) define a peninsular indentation (**410**) into which the key **1503** may be inserted. The track (**106**) can comprise one or more complementary barbs (**405**). Each of the one or more complementary barbs (**405**) is, in one or more embodiments, complementary in shape to the one or more barbs **1802** of the key **1503**. For example, each complementary barb can comprise an inverted hemi-triangular protrusion with a base of each inverted hemi-triangular protrusion being disposed above a side of the inverted hemi-triangular protrusion.

Using this configuration, the one or more barbs **1802** of the key **1503** facilitate one-way penetration of the key **1503** into the track (**106**). In one or more embodiments, when the key **1503** is inserted into the peninsular indentation (**410**), the first flat side (**407**) of the track (**106**) flexes so that the one or more complementary barbs (**405**) of the track (**106**) pass over the one or more barbs **1802** of the key **1503**. Once the key **1503** is fully inserted into the peninsular indentation (**410**) of the track (**106**), first flat side (**407**) of the track (**106**) flex back toward the key **1503**, thereby causing the one or more complementary barbs (**405**) to engage the one or more barbs **1802** of the key **1503**. This results in the key **1503** being frictionally retained within the track (**106**) as previously described.

In one or more embodiments, the base member **1502** comprises a first base member portion **1508** and a second base member portion **1509**. In one or more embodiments, the first base member portion **1508** and the second base member portion **1509** are separable along a medial major axis **2010**. In the illustrative embodiment of FIGS. 15-20, the first base member portion **1508** and the second base member portion **1509** can be placed together such that their inner surfaces abut along the medial major axis **2010** to define the base member **1502**. A coupler **1511**, which in this embodiment is a plastic hex nut, couples the first base member portion **1508** and the second base member portion **1509** together to retain the inner surfaces abutting at the medial major axis **2010**.

In one or more embodiments, the first base member portion **1508** defines a first half rail clamp **1512**. Similarly, the second base member portion **1509** defines a second half rail clamp **1513**. Each of the first half rail clamp **1512** and the second half rail clamp **1513** includes a vertical member extending distally from the first base member portion **1508** and the second base member portion **1509**, respectively, and

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a horizontal member extending distally from its respective vertical member only a portion of the width of the first base member portion **1508** and the second base member portion **1509**. This leaves a gap between each horizontal member, as shown in FIGS. 15, 16, 19, and 20.

Separating the first base member portion **1508** from the second base member portion **1509** along the medial major axis **2010** therefore separates the first half rail clamp **1512** from the second half rail clamp **1513**. When used with a drop ceiling having a suspension grid of rails, one can simply place the first half rail clamp **1512** to the left of a rail and the second half rail clamp **1513** to the right of the rail, or vice versa. The first base member portion **1508** can then be pressed against the second base member portion **1509** along the medial major axis **2010**. This causes the rail of the suspension grid to be caught between the first half rail clamp **1512** and the second half rail clamp **1513**.

In one or more embodiments, the first base member portion **1508** defines a first half head-receiving aperture **2001**, while the second base member portion **1509** defines a second half head-receiving aperture **2002**. In one or more embodiments, coupling the first base member portion **1508** to the second base member portion **1509** causes the first half head-receiving aperture **2001** to abut the second half head-receiving aperture **2002**, thereby resulting in the base member **1502** defining a head-receiving aperture **1606** into which the head **1605** of the hanger **1501** may situate.

As shown in FIG. 20, the head-receiving aperture **1606** is defined in a planar surface of the base member formed by coupling the first base member portion **1508** to the second base member portion **1509** that is situated opposite the first half rail clamp **1512** and the second half rail clamp **1513**. Effectively, this allows the head **1605** of the hanger **1501** to be exposed to the first half rail clamp **1512** and the second half rail clamp **1513**. That the head **1605** of the hanger **1501** is exposed to the first half rail clamp **1512** and the second half rail clamp **1513** means that the head **1605** can be inserted into the head receiver defined by the first half head receiver **2015** and the second half head receiver **2016** until it touches the first half rail clamp **1512** and the second half rail clamp **1513**. When a rail is positioned or situated within the first half rail clamp **1512** and the second half rail clamp **1513**, as shown below in FIG. 22, that the head **1605** of the hanger **1501** is exposed to the first half rail clamp **1512** and the second half rail clamp **1513** means that the head **1605** can be inserted into the head receiver defined by the first half head receiver **2015** and the second half head receiver **2016** until it abuts or intersects with the rail.

That the head **1605** of the hanger **1501** is exposed to the first half rail clamp **1512** and the second half rail clamp **1513** additionally means that threading the coupler **1511** onto the first base member portion **1508** and the second base member portion **1509** moves the head **1605** of the hanger **1501** toward the first half rail clamp **1512** and the second half rail clamp **1513** until at least a portion of the head **1605** is exposed within the gap defined by the first half rail clamp **1512** and the second half rail clamp **1513**. It is this exposure that provides the unique clamping function of the head **1605** of the hanger **1501** against the rail.

In one or more embodiments, the hanger **1501** is then coupled to the base member **1502**. The coupler **1511** couples the first base member portion **1508** and the second base member portion **1509**. Where the base member **1502** defines the head-receiving aperture **1606**, with the head **1605** of the hanger **1501** situated therein, the coupler **1511** not only retains the inner surfaces abutting at the medial major axis **2010**, but also moves the head **1605** of the hanger **1501**

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toward the first half rail clamp **1512** and the second half rail clamp **1513**. Where a rail is situated in the first half rail clamp **1512** and the second half rail clamp **1513**, this causes the head **1605** of the hanger **1501** to lock into the head-receiving aperture **1606** of the base member **1502** such that the head **1605** is able to “clamp” onto the rail passing through the base member **1502**.

Thus, in one or more embodiments a first base member portion **1508** comprises a first half rail clamp **1512** and defines a first half head-receiving aperture **2001**. A second base member portion **1509** is coupled to the first base member portion **1508**. In one or more embodiments, the second base member portion **1509** comprises a second half rail clamp **1513** and defines a second half head-receiving aperture **2002**. A head **1605** of a hanger **1501** situates within the head receiving aperture defined by the first half head-receiving aperture **2001** and the second half head-receiving aperture **2002** abutting. In one or more embodiments, an upper surface of the head **1605** is exposed to the first half rail clamp **1512** and the second half rail clamp **1513** through the head-receiving aperture. A coupler **1511** couples the first base member portion **1508** to the second base member portion **1509**. In one or more embodiments, threading the coupler **1511** onto the first base member portion **1508** and the second base member portion **1509** moves the head **1605** of the hanger **1501** toward the first half rail clamp **1512** and the second half rail clamp **1513** due to the head’s exposure through the head-receiving aperture. Continuing to thread the coupler **1511** onto the first base member portion **1508** and the second base member portion **1509**, when a rail is situated between the first half rail clamp **1512** and the second half rail clamp **1513**, causes the head **1605** of the hanger **1501** to clamp onto the rail.

In one or more embodiments, separating the first base member portion **1508** from the second base member portion **1509** along the medial major axis **2010** then separates the first half rail clamp **1512** from the second half rail clamp **1513** while also separating the head-receiving aperture **1606**. When used with a drop ceiling having a suspension grid of rails, one can simply place the first half rail clamp **1512** to the left of a rail and the second half rail clamp **1513** to the right of the rail, or vice versa.

The first base member portion **1508** can then be pressed against the second base member portion **1509** along the medial major axis **2010**. In one or more embodiments, this causes the rail of the suspension grid to be caught between the first half rail clamp **1512** and the second half rail clamp **1513**. The head **1605** of the hanger **1501** can then be situated within the head-receiving aperture **1606**. The coupler **1511** can then be attached to the base member **1502** to both retain the first base member portion **1508** against the second base member portion **1509** and to clamp the head **1605** of the hanger **1501** against the rail. The coupler **1511** can be coupled and uncoupled as desired so as to allow quick and easy coupling of the base member **1502** and head **1605** of the hanger **1501** to the rails of the suspension grid.

In one or more embodiments, the coupler **1511** engages one or more threads **1607** disposed on the exterior surface of the head-receiving aperture **1606**, which extend distally from the base member **1502** to retain the first base member portion **1508** against the second base member portion **1509** and the head **1605** of the hanger **1501** against the rail through the head-receiving aperture **1606**.

The coupler **1511**, moving toward the first half rail clamp **1512** and the second half rail clamp **1513** translates the head **1605** of the hanger **1501** toward the rail, thereby biasing and clamping the head **1605** against a major face of the rail.

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Thus, in one or more embodiments, the rail is clamped on three sides—two by the first base member portion **1508** and the second base member portion **1509**, and a third by the head **1605** of the hanger **1501** against the major surface of the rail. Advantageously, a track hanger **1500** of FIGS. **15-20** comprises a uniquely configured base member **1502** that facilitates simple mounting and dismounting of the track hanger **1500** to a rail of a conventional suspension grid of a drop ceiling while also biasing the head **1605** of the hanger **1501** against the same rail for additional coupling power. Said differently, in one or more embodiments the head **1605** is insertable into the head receiver and the coupler **1511** is operable to engage an exterior surface of the head receiver to move the head **1605** through the head receiving aperture toward the first half rail clamp **1512** and the second half rail clamp **1513**.

As noted above, in one or more embodiments each of the hanger **1501**, the base member **1502**, and the key **1503** are separable from each other. For example, in one embodiment the hanger **1501** can be detached from the base member **1502**. Similarly, the key **1503** can be detached from the hanger **1501**. As best shown in FIG. **16**, in one or more embodiments the key **1503** can be detached from the distal end **1601** of the hook **1504**. In this illustrative embodiment, the key **1503** is selectively attachable to the distal end **1601** of the hook **1504**.

In one or more embodiments, to attach the key **1503** to the distal end **1601** of the hook **1504**, the distal end **1601** of the hook **1504** opposite the extension **1505** of the hanger **1501** is inserted into a distal end receiver **1602** defined within a connector of the key **1503**. In one or more embodiments, the distal end receiver **1602** is configured to frictionally retain the key **1503** to the distal end **1601** of the hook **1504**. However, in other embodiments, a latch, snap, one-way latch, or other mating feature, such as engagement protrusion **1604**, can be incorporated into either the distal end receiver **1602** or the distal end **1601** of the hook **1504** to engage an engagement recess **1603** or otherwise frictionally retain the key **1503** to the distal end **1601** of the hook **1504**.

Thus, in one or more embodiments the distal end receiver **1602** defines at least one engagement recess **1603**. In one or more embodiments, the distal end **1601** of the hook **1504** comprises at least one engagement protrusion **1604** projecting from the distal end **1601**. In one or more embodiments, the at least one engagement protrusion **1604** engages the engagement recess **1603** when the distal end **1601** is inserted into the distal end receiver **1602** to retain the key **1503** to the distal end **1601**.

In this illustrative embodiment, a flat surface **1608** and the distal end **1601** of the hook **1504** mates with a corresponding flat surface **1609** of the distal end receiver **1602** to prevent rotation of the key **1503** about the distal end **1601** of the hook **1504**. To wit, here the distal end receiver **1602** defines a corresponding flat surface **1609** and two arched protrusions. The distal end **1601** of the hook **1504** has a flat surface **1608** and two arched indentations that are complementary in shape to the two arched protrusions.

In one or more embodiments, when the distal end **1601** of the hook **1504** opposite the extension **1505** of the hanger **1501** is inserted into the distal end receiver **1602** of the key **1503**, engagement of the two arched protrusions with the two arched indentations, opposite engagement of the complementary flat surface **1609** of the distal end receiver **1602** with the flat surface **1608** of the distal end **1601** of the hook **1504** prevents rotation of the key **1503** about the distal end **1601** of the hook **1504**.

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Turning now to FIG. 20, illustrated therein are the first base member portion 1508 and the second base member portion 1509 after having been separated along the medial major axis 2010. This separation exposes the inner surfaces 2003, 2004 of the first base member portion 1508 and the second base member portion 1509, respectively.

In one or more embodiments, the inner surface 2003 of the first base member portion 1508 comprises one or more bosses 2005, 2006. Here, two bosses 2005, 2006 are shown. However, more or fewer bosses can be included in other embodiments.

In this illustrative embodiment, the inner surface 2004 of the second base member portion 1509 comprises one or more boss receivers 2007, 2008. Here, two boss receivers 2007, 2008 are shown. However, more or fewer boss receivers can be included in other embodiments.

In one or more embodiments, the boss receivers 2007, 2008 comprise apertures that extend into the inner surface 2004 of the second base member portion 1509, and which have shapes that are complementary to the one or more bosses 2005, 2006 of the inner surface 2003 of the first base member portion 1508. The one or more boss receivers 2007, 2008 can optionally include wider mouths 2029, 2030 at their openings to facilitate easier insertion of the one or more bosses 2005, 2006 into the one or more boss receivers 2007, 2008 when the first base member portion 1508 and the second base member portion 1509 are pressed together such that their inner surfaces 2003, 2004 abut at the medial major axis 2010.

In one or more embodiments, when the first base member portion 1508 and the second base member portion 1509 are pressed together such that their inner surfaces 2003, 2004 abut at the medial major axis 2010, the one or more bosses 2005, 2006 insert into the one or more boss receivers 2007, 2008. This prevents the inner surfaces 2003, 2004 of the first base member portion 1508 and the second base member portion 1509 from sliding about. Additional apertures 2011, 2012, 2013, 2014 can be tooled into the first base member portion 1508 and second base member portion 1509, respectively, to make each part lighter and to conserve material.

In the illustrative embodiment of FIG. 20, the base member (1502) includes a head receiver. In this illustrative embodiment, the first base member portion 1508 has a first half head receiver 2015 extending distally from a side of the first base member portion 1508 disposed opposite the first half rail clamp 1512. Similarly, the second base member portion 1509 has a second half head receiver 2016 extending distally from a side of the second base member portion 1509 disposed opposite the second half rail clamp 1513.

In this illustrative embodiment, the first half head receiver 2015 and the second half head receiver 2016 are both hemi-cylindrical. When the first base member portion 1508 and the second base member portion 1509 are pressed together such that their inner surfaces 2003, 2004 abut at the medial major axis 2010, the edges of the first half head receiver 2015 and the second half head receiver 2016 also abut to define a cylindrical head receiver in this illustrative embodiment.

While a cylinder is one suitable shape for a head receiver formed by abutment of the edges of the first half head receiver 2015 and the second half head receiver 2016, other shapes can be used as well. In other embodiments, the head receiver is rectangular, triangular, polygonal, or free form shapes. Still other shapes for the head receiver will be obvious to those of ordinary skill in the art having the benefit of this disclosure.

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In one or more embodiments, the inner surfaces 2017, 2018 of the first half head receiver 2015 and the second half head receiver 2016 define an interior socket to receive the head (1605) of a hanger (1501) configured in accordance with one or more embodiments of the disclosure. In one or more embodiments, the exterior surfaces 2019, 2020 of the first half head receiver 2015 and the second half head receiver 2016 define one or more threads. When the first base member portion 1508 and the second base member portion 1509 are pressed together such that their inner surfaces 2003, 2004 abut at the medial major axis 2010, the edges of the first half head receiver 2015 and the second half head receiver 2016 also abut to define a cylindrical head receiver having an interior socket defined by the inner surfaces 2017, 2018 of the first half head receiver 2015 and the second half head receiver 2016, and having threads defined by the exterior surfaces 2019, 2020 of the first half head receiver 2015 and the second half head receiver 2016. The coupler (1511) can then be threaded onto the threads defined by the exterior surfaces 2019, 2020 of the first half head receiver 2015 and the second half head receiver 2016 to couple the first base member portion 1508 and the second base member portion 1509 together.

In this illustrative embodiment, the first half head receiver 2015 defines a first half head-receiving aperture 2001. Similarly, the second half head receiver 2016 defines a second half head-receiving aperture 2002. In one or more embodiments, when coupling the first base member portion 1508 to the second base member portion 1509 causes the first half head receiver 2015 to abut the second half head receiver 2016, a head-receiving aperture (1606) is created into which the head (1605) of the hanger (1501) may situate.

When the head (1605) of the hanger (1501) is situated within the head-receiving aperture (1606), and the coupler (1511) is threaded onto the threads defined by the exterior surfaces 2019, 2020 of the first half head receiver 2015 and the second half head receiver 2016 to couple the first base member portion 1508 and the second base member portion 1509 together, this causes the head (1605) of the hanger (1501) to advance toward the first half rail clamp 1512 and second half rail clamp 1513, respectively. When a rail is situated within the first half rail clamp 1512 and second half rail clamp 1513, this advancement of the head (1605) of the hanger (1501) toward the first half rail clamp 1512 and second half rail clamp 1513 causes the head (1605) of the hanger (1501) to “pinch” or “clamp” or “grab” the rail between the top of the head (1605) and the underside of the first half rail clamp 1512 and second half rail clamp 1513.

It should be noted that the first base member portion 1508 and the second base member portion 1509 can be configured in a variety of ways. Turning now to FIG. 21, illustrated therein is an alternate first base member portion 2108 and an alternate second base member portion 2109 configured in accordance with one or more embodiments of the disclosure.

As with the first base member portion (1508) and the second base member portion (1509) of FIG. 20, the alternate first base member portion 2108 and the alternate second base member portion 2109 of FIG. 21 include one or more bosses 2101, 2102 and one or more boss receivers 2103, 2104. The boss receivers 2103, 2104 comprise apertures that extend into the inner surface of the second base member portion 2109 have shapes that are complementary to the one or more bosses 2101, 2102 of the inner surface of the first base member portion 2108.

The base member of FIG. 21 also includes a head receiver. The first base member portion 2108 has a first half head receiver 2105 extending distally from a side of the first base

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member portion 2108 disposed opposite the first half rail clamp 2117, which is substantially wider than was the first half rail clamp (1512) of FIG. 20. Similarly, the second base member portion 2109 has a second half head receiver 2106 extending distally from a side of the second base member portion 2109 disposed opposite the second half rail clamp 2118, which is also substantially wider than the second half rail clamp (1513) of FIG. 20.

The inner surfaces 2130,2131 of the first half head receiver 2105 and the second half head receiver 2106 once again define an interior socket to receive the head (1605) of a hanger (1501). The exterior surfaces 2111,2112 of the first half head receiver 2105 and the second half head receiver 2106 again define one or more threads.

The first half head receiver 2105 defines a first half head-receiving aperture 2113. Similarly, the second half head receiver 2106 defines a second half head-receiving aperture 2114. In one or more embodiments, when coupling the first base member portion 2108 to the second base member portion 2109 causes the first half head receiver 2105 to abut the second half head receiver 2106, a head-receiving aperture (1606) is created into which the head (1605) of the hanger (1501) may situate.

Threading the coupler (1511) onto the threads defined by the exterior surfaces 2111,2112 of the first half head receiver 2105 and the second half head receiver 2106 then couples the first base member portion 2108 and the second base member portion 2109 together. It also causes the head (1605) of the hanger (1501) to advance toward the first half rail clamp 2117 and second half rail clamp 2118, respectively. When a rail is situated within the first half rail clamp 2117 and second half rail clamp 2118, this advancement of the head (1605) of the hanger (1501) toward the first half rail clamp 2117 and second half rail clamp 2118 causes the head (1605) of the hanger (1501) to clamp the rail between the top of the head (1605) and the underside of the first half rail clamp 2117 and second half rail clamp 2118.

Turning now to FIG. 22, illustrated therein are one or more method steps in accordance with one or more embodiments of the disclosure. Beginning at step 2201, the first base member portion 1508 is positioned such that its first half rail clamp 1512 situates about a rail 2207. At step 2202, the second base member portion 1509 is positioned such that its second half rail clamp 1513 situates about the rail 2207. In one or more embodiments, this results in the one or more bosses 2005,2006 of the first base member portion 1508 inserting into the one or more boss receivers (2007,2008) of the second base member portion 1509 when the first base member portion 1508 and the second base member portion 1509 are pressed together such that their inner surfaces (2003,2004) abut at the medial major axis (2010). It also captures the rail 2207 between the first half rail clamp 1512 and the second half rail clamp 1513.

At step 2203, the head 1605 of the hanger 1501 is inserted into the head receiver formed by the first half head receiver 2015 extending distally from the bottom of the first base member portion 1508 and the second half head receiver 2016 extending distally from the bottom of the second base member portion 1509. In one or more embodiments, the head 1605 of the hanger 1501 is inserted into the head receiver until its upper surface abuts the rail 2207.

At step 2204, the coupler 1511 is threaded onto the threads defined by the exterior surfaces (2019,2020) of the first half head receiver 2015 and the second half head receiver 2016 to couple the first base member portion 1508 and the second base member portion 1509 together.

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In this illustrative embodiment, the first half head receiver 2015 defines a first half head-receiving aperture (2001). Similarly, the second half head receiver 2016 defines a second half head-receiving aperture (2002). When the head 1605 of the hanger 1501 is situated within the head-receiving aperture 1606 created by the first half head-receiving aperture (2001) and second half head-receiving aperture (2002) abutting, threading the coupler 1511 is threaded onto the threads defined by the exterior surfaces (2019,2020) of the first half head receiver 2015 and the second half head receiver 2016 causes the head 1605 of the hanger (1501) to advance toward the rail 2207. This advancement of the head 1605 toward the rail 2207 causes the head 1605 to clamp the rail 2207 between the top of the head 1605 and the underside of the first half rail clamp 1512 and second half rail clamp 1513.

At step 2205, the key 1503 can be attached to the distal end (1601) of the hook 1504. In one or more embodiments, to attach the key 1503 to the distal end (1601) of the hook 1504, the distal end (1601) of the hook 1504 opposite the extension 1505 of the hanger 1501 is inserted into a distal end receiver 1602 defined within a connector of the key 1503. In one or more embodiments, the distal end receiver 1602 is configured with an engagement recess (1603) that receives an engagement protrusion (1604) projecting from a surface of the hook 1504. A flat surface (1608) and the distal end (1601) of the hook (1504) can mate with a corresponding flat surface (1609) of the distal end receiver 1602 to prevent rotation of the key 1503 about the distal end (1601) of the hook 1504. As step 2206, the track 106 can then be attached to the key 1503 as described above with reference to FIG. 10.

Turning now FIG. 23, illustrated therein is one explanatory accessory 2300 that can be used in place of the base member of FIGS. 15-22. In FIG. 23, the accessory 2300 is a screw adaptor base member.

In one or more embodiments, the accessory 2300 includes a circular attachment support 2301 that defines one or more recesses 2302 along its upper surface. An aperture 2303 is centrally disposed along the circular attachment support 2301. The aperture 2303 allows a screw or other fastener to be inserted from the lower surface of the circular attachment support 2301, through the aperture 2303, and out the upper surface of the circular attachment support 2301 to attach the accessory 2300 to a ceiling.

In one or more embodiments, a boss surrounds the aperture 2303 in a recess. A perimeter wall bounds the recesses 2302 in this illustrative embodiment. A head receiver 2304 includes one or more threads as previously described.

Turning now FIG. 24, illustrated therein is another explanatory accessory 2400 that can be used in place of the combined base member and hanger of FIGS. 15-22. In FIG. 23, the accessory 2400 is a combined hanger 2401 and rectangular track insertion head 2402. The rectangular track insertion head 2402 can be inserted into a track with the major axis of the rectangular track insertion head 2402 parallel to the major axis of the track. The accessory 2400 includes one or more threads as previously described.

Turning now to FIG. 25, illustrated therein is another explanatory accessory 2500 configured in accordance with one or more embodiments of the disclosure. In FIG. 25, the accessory 2400 is an end cap for a track (106). In one or more embodiments, the end cap can be attached to a track (106), as shown above in FIGS. 1-4 and 11.

In this illustrative embodiment, the end cap includes a wall abutment 2501 and an extension 2502 that situates around the exterior of an end of the track (106). Situated

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within an interior of the extension 2502, along a center axis of the extension 2502, is a track insertion key 2503. In this illustrative embodiment, the track insertion key 2503 includes an upper track insertion key 2504 and a lower track insertion key 2505. In this embodiment, the lower track insertion key 2505 is a mirror image of the upper track insertion key 2504. This allows the accessory 2500 to be ambidextrous in that it can insert into the track (106) of FIG. 4 with either minor surface of the accessory 2500 situated on the top of the track (106). Thus, where the track (106) comprising one or more complementary barbs (405), as shown above in FIG. 4, the end cap can include a track insertion key 2505 that inserts into the one or more complementary barbs (405) regardless of whether the end cap is right side up or upside down.

In this illustrative embodiment, the wall abutment 2501 includes a quadrilateral attachment support that includes two apertures 2506 (with another being a mirror image of aperture 2506 similar to apertures 1153, 1154 of FIG. 11). Screws can be inserted through the apertures 2506 to attach the quadrilateral attachment support to a wall. The track (106) can then be inserted into the extension 2502 around the track insertion key 2503 to securely adhere the track (106).

Turning now to FIG. 26, illustrated therein is one explanatory assembled system 2600 in accordance with one or more embodiments of the disclosure. As shown in FIG. 26, the first base member portion 1508 is positioned such that its first half rail clamp 1512 situates about a rail 2207. The second base member portion 1509 is positioned such that its second half rail clamp 1513 situates about the rail 2207. The head (1605) of the hanger 1501 is inserted into the head receiver formed by the first half head receiver (2015) extending distally from the bottom of the first base member portion 1508 and the second half head receiver (2016) extending distally from the bottom of the second base member portion 1509 until its upper surface abuts the rail 2207.

The coupler 1511 is threaded onto the threads defined by the exterior surfaces (2019, 2020) of the first half head receiver (2015) and the second half head receiver (2016) to couple the first base member portion 1508 and the second base member portion 1509 together. The key 1503 is attached to the distal end (1601) of the hook 1504. The track 106 is attached to the key 1503.

One or more hangers 1186 are then placed upon the track 106. The one or more hangers 1186 can be coupled to a curtain or other substrate, and can optionally include an aperture 1185 in its top corner.

Turning now to FIG. 27A, illustrated therein is yet another key 2700 configured in accordance with one or more embodiments of the disclosure. As with prior embodiments, the key 2700 of FIG. 27A is selectively detachable from the end of a hook, such as the distal end (1601) of hook (1504). The key 2700 is shown in as being completely detached from the end of a hook in FIG. 27A.

To attach the key 2700 to the distal end (1601) of a hook (1504), the distal end (1601) of the hook (1504) opposite the extension (1505) of the hanger (1501) is inserted into an aperture 2701 situated at the base of a connector 2702 of the key 2700. In one or more embodiments, the aperture 2701 is configured to frictionally retain the key 2700 to the distal end (1601) of the hook (1504). However, in other embodiments, such as that described above with reference to FIG. 16, a latch, snap, one-way latch, or other mating feature can be incorporated into either the aperture 2701 or the distal end (1601) of the hook (1504) to mechanically retain the key 2700 coupled to the distal end (1601) of the hook (1504).

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As previously described, in the key 2700 of FIG. 27A, the aperture 2701 and the distal end (1601) of the hook (1504) are geometrically configured to prevent rotation of the key 2700 about the distal end (1601) of the hook (1504). Illustrating by example, in one or more embodiments the aperture 2701 defines a flat surface 2703 and two arched protrusions (identical to the arched protrusions shown in FIG. 16). The distal end (1601) of the hook (1504) have a flat surface (1608) and two arched indentations that are complementary in shape to the two arched protrusions. When the distal end (1601) of the hook (1504) opposite the extension (1505) of the hanger (1501) is inserted into an aperture 2701 defined within the connector 2702 of the key 2700, engagement of the two arched protrusions with the two arched indentations, opposite engagement of the flat surface 2703 of the aperture 2701 with the flat surface (1608) of the distal end (1601) of the hook (1504) prevents rotation of the key 2700 about the distal end (1601) of the hook (1504).

The key 2700 of FIG. 27A comprises a first major surface (shown in FIG. 27A) and a second major surface (shown in FIG. 29). In this illustrative embodiment, the first major surface defines one or more barbs 2704. In one or more embodiments each barb of the one or more barbs 2704 comprises a hemi-triangular protrusion with the base of each hemi-triangular protrusion being disposed beneath the side of the hemi-triangular protrusion. In this illustrative embodiment, the second major surface is substantially planar.

In one or more embodiments, a track, one example of which is track (106), is selectively attachable to the key 2700 when the one or more barbs 2704 insert between a first flat side (407) and a second flat side (408) of the track (106). For example, in one or more embodiments the track (106) has a first flat side (407) and a second flat side (408), with an arch (409) spanning between ends of the first flat side (407) and the second flat side (408), the top 2705 of the key 2700 can be inserted between the first flat side (407) and the second flat side (408) until it abuts the interior surface of the arch (409) spanning the first flat side (407) and the second flat side (408).

In this illustrative embodiment, a plurality of ledges is situated beneath the one or more barbs 2704. For example, in FIG. 27A a first plurality of ledges 2706 is situated to the right of the connector 2702, while a second plurality of ledges 2707 is situated to the left of the connector. In this illustrative embodiment, the first plurality of ledges 2706 comprises three ledges, as does the second plurality of ledge 2707. While three ledges are shown in this embodiment, it should be noted that other numbers of ledges, such as fewer than three ledges, or more than three ledges, can be included in other embodiments. For example, in another embodiment, the first plurality of ledges 2706 and the second plurality of ledges 2707 each comprise two ledges. In another embodiment, the first plurality of ledges 2706 and the second plurality of ledges 2707 each comprise four ledges. In another embodiment, the first plurality of ledges 2706 and the second plurality of ledges 2707 each comprise five ledges, and so forth.

Additionally, while the first plurality of ledges 2706 and the second plurality of ledges 2707 each comprise the same number of ledges in the illustrative embodiment of FIG. 27A, in other embodiments the first plurality of ledges 2706 and the second plurality of ledges 2707 will each comprise different numbers of ledges. The first plurality of ledges 2706 may comprise three ledges, while the second plurality of ledges 2707 comprises five ledges, and so forth. Moreover, while the first plurality of ledges 2706 and the second plurality of ledges 2707 each have a common width in the



illustrative embodiment of FIG. 27A, in other embodiments, such as is the case with the key 2770 of FIG. 27B, ledges in the first plurality of ledges and the second plurality of ledges can have differing lengths.

In this illustrative embodiment, the first plurality of ledges 2706 and the second plurality of ledges 2707 are situated in a partially cylindrical protuberance 2708 extending distally outward (out from the page as viewed in FIG. 27) from the first major surface of the key 2700. Since the partially cylindrical protuberance 2708 is partially cylindrical, the depth of each ledge of the first plurality of ledges 2706 and the second plurality of ledges 2707 gets deeper closer to the connector 2702 than they are toward either the right edge 2710 or the left edge 2711 of the key 2700, respectively. In this illustrative embodiment, the first plurality of ledges 2706 and the second plurality of ledges 2707 include ledges situated atop each other with widths that run from the right edge 2710 and the left edge 2711 of the key 2700, respectively, in an inward direction toward the connector 2702. Each ledge of the first plurality of ledges 2706 and the second plurality of ledges 2707 then terminates at a central mass 2709 having an arched central face 2712.

In one or more embodiments, the key 2700 is manufactured as a unitary component from a thermoplastic material by way of an injection molding process. The key 2700 can be manufactured from nylon, styrene, ABS, polycarbonate, or polycarbonate-ABS, PMMA, PVC, or other polyamide-based thermoplastics in one embodiment. Other materials suitable for manufacturing the key 2700 will be obvious to those of ordinary skill in the art having the benefit of this disclosure.

Turning now to FIGS. 28-31, illustrated therein is another explanatory embodiment of a track hanger 2800 configured in accordance with one or more embodiments of the disclosure. In one embodiment, the track hanger 2800 is to suspend a track from a surface such as a wall, track, ceiling, or other surface or object.

The track hanger 2800, in one or more embodiments, includes a hanger 1501, a base member 1502, and a key 2700. In one or more embodiments, each of the hanger 1501, the base member 1502, and the key 2700 are separable from each other. For example, in one embodiment the hanger 1501 can be detached from the base member 1502. Similarly, as shown in FIG. 28, the key 2700 can be detached from the hanger 1501. It should be noted that the track hanger 2800 can be manufactured in different sizes and shapes so as to be compatible with, and fit appropriately, differently sized suspension grids of drop ceilings.

In one or more embodiments, the hanger 1501 comprises a hook 1504 and an extension 1505. In one or more embodiments, the key 2700 attaches to a distal end 1601 of the hook 1504. Accordingly, where the key 2700 is detachable from the hanger 1501, the attachment location to attach or detach the key 2700 from the hook 1504 is the distal end 1601. In one or more embodiments, the key 2700 includes a distal end receiver 2801 into which the distal end 1601 of the hook 1504 inserts. The distal end receiver 2801 can define one or more engagement recesses 2802 to receive one or more engagement protrusions 1604 projecting from the distal end 1601 of the hook 1504 to retain the key 2700 to the distal end 1601 of the hook 1504 in one or more embodiments. In one or more embodiments the key 2700 is detachable from the hook 1504.

In one or more embodiments, the hanger 1501 comprises a head 1605 that is attached to the extension 1505. In one embodiment shown in FIG. 28, the hanger 1501 is configured as a single, unitary component. Said differently, in one

embodiment the hook 1504, the extension 1505, and the head 1605 coupled to the extension 1505 are manufactured as a single, integral unit.

Illustrating by example, in one embodiment the hook 1504, the extension 1505, and the head 1605 are manufactured from a thermoplastic material by way of an injection molding process. The hook 1504, the extension 1505, and the head 1605 can be manufactured from nylon, styrene, ABS, polycarbonate, or polycarbonate-ABS, PMMA, PVC, or other polyamide-based thermoplastics in one embodiment. Other materials suitable for manufacturing the hook 1504, the extension 1505, and the head 1605 will be obvious to those of ordinary skill in the art having the benefit of this disclosure.

While the base member 1502 and the key 2700 can be separable from the hanger 1501, in one or more embodiments the base member 1502 and the key 2700 can be manufactured from the same material as is the hanger 1501. In other embodiments, the hanger 1501 is manufactured from materials different from the base member 1502 and/or key 2700. The base member 1502, hanger 1501, and key 2700 can each be manufactured from different materials as well.

As best seen by comparing FIGS. 29-30, in one embodiment the key 2700 includes a first major surface 3001 and a second major surface 2901. In this illustrative embodiment, the first major surface 3001 defines one or more barbs 2704 and a plurality of ledges situated beneath the one or more barbs 2704. As described above with reference to FIG. 27, a first plurality of ledges 2706 is situated to the right of the connector 2702, while a second plurality of ledges 2707 is situated to the left of the connector.

In this illustrative embodiment, the first plurality of ledges 2706 and the second plurality of ledges 2707 are situated in a partially cylindrical protuberance 2708 extending distally from the first major surface 3001 of the key 2700. Since the partially cylindrical protuberance 2708 is partially cylindrical, the depth of each ledge of the first plurality of ledges 2706 and the second plurality of ledges 2707 gets deeper closer to the connector 2702 than they are toward either the right edge 2710 or the left edge 2711 of the key 2700, respectively. This is shown in FIG. 31. As also shown in FIG. 31, in one or more embodiments each barb of the one or more barbs 2704 comprises a hemi-triangular protrusion with the base of each hemi-triangular protrusion being disposed beneath the side of the hemi-triangular protrusion. In this illustrative embodiment, the second major surface 2901 is substantially planar between the one or more barbs 2704 and the partially cylindrical protuberance 2708, with its arched central face 2712 situated between the first plurality of ledges 2706 and the second plurality of ledges 2707.

When the key 2700 is inserted into a track, as will be described in more detail below with reference to FIG. 32, in addition to being aesthetically pleasing, the first plurality of ledges 2706 and the second plurality of ledges 2707 can provide a utilitarian function as well. Illustrating by example, when a curtain hook slides along the track and engages the portion of the track sitting atop the key 2700, the first plurality of ledges 2706 and the second plurality of ledges 2707, being situated within the partially cylindrical protuberance 2708, can function to keep the hook on the track while sliding across the portion attached to the key 2700 by serving as a ramp that gently moves the hook toward the extension 1505 as the hook rides the first plurality of ledges 2706 and the second plurality of ledges 2707 across the partially cylindrical protuberance 2708. In one or more embodiments, the first plurality of ledges 2706 and the

second plurality of ledges **2707** allow hooks to smoothly slide over the portion of track attached to the key **2700** rather than abruptly bumping into the connector **2702**. Gaps in between the first plurality of ledges **2706** and the second plurality of ledges **2707** also facilitate removal of the key **2700** from a mold when the key **2700** is manufactured using an injection molding process.

In one or more embodiments, a track, which could be track **(106)** described above, track **(3207)** described below, or another track, attaches to the key **2700** as previously described. For example, in one or more embodiments the track **(106)** has a first flat side **(407)** and a second flat side **(408)**, with an arch **(409)** spanning between ends of the first flat side **(407)** and the second flat side **(408)**. The track **(106)** can be flexible so that it can be shaped into different contours when attached to a track hanger **2800**.

In one or more embodiments, the inclusion of the first flat side **(407)**, the second flat side **(408)**, and the arch **(409)** define a peninsular indentation **(410)** into which the key **2700** may be inserted. The track **(106)** can comprise one or more complementary barbs **(405)**. Each of the one or more complementary barbs **(405)** is, in one or more embodiments, complementary in shape to the one or more barbs **2704** of the key **2700**. For example, each complementary barb can comprise an inverted hemi-triangular protrusion with a base of each inverted hemi-triangular protrusion being disposed above a side of the inverted hemi-triangular protrusion.

Using this configuration, the one or more barbs **2704** of the key **2700** facilitate one-way penetration of the key **2700** into the track **(106)**. In one or more embodiments, when the key **2700** is inserted into the peninsular indentation **(410)**, the first flat side **(407)** of the track **(106)** flexes so that the one or more complementary barbs **(405)** of the track **(106)** pass over the one or more barbs **2704** of the key **2700**. Once the key **2700** is fully inserted into the peninsular indentation **(410)** of the track **(106)**, first flat side **(407)** of the track **(106)** flex back toward the key **1503**, thereby causing the one or more complementary barbs **(405)** to engage the one or more barbs **2704** of the key **2700**. This results in the key **2700** being frictionally retained within the track **(106)** in one or more embodiments.

In one or more embodiments, the base member **1502** comprises a first base member portion **1508** and a second base member portion **1509**. In one or more embodiments, the first base member portion **1508** and the second base member portion **1509** are separable along a medial major axis **2010**. In the illustrative embodiment of FIGS. **28-31**, the first base member portion **1508** and the second base member portion **1509** can be placed together such that their inner surfaces abut along the medial major axis **2010** to define the base member **1502**. A coupler **1511**, which in this embodiment is a plastic hex nut, couples the first base member portion **1508** and the second base member portion **1509** together to retain the inner surfaces abutting at the medial major axis **2010**.

In one or more embodiments, the first base member portion **1508** defines a first half rail clamp **1512**. Similarly, the second base member portion **1509** defines a second half rail clamp **1513**. Each of the first half rail clamp **1512** and the second half rail clamp **1513** includes a vertical member extending distally from the first base member portion **1508** and the second base member portion **1509**, respectively, and a horizontal member extending distally from its respective vertical member only a portion of the width of the first base member portion **1508** and the second base member portion **1509**. This leaves a gap between each horizontal member, as shown in FIG. **31**.

Separating the first base member portion **1508** from the second base member portion **1509** along the medial major axis **2010** therefore separates the first half rail clamp **1512** from the second half rail clamp **1513**. When used with a drop ceiling having a suspension grid of rails, one can simply place the first half rail clamp **1512** to the left of a rail and the second half rail clamp **1513** to the right of the rail, or vice versa. The first base member portion **1508** can then be pressed against the second base member portion **1509** along the medial major axis **2010**. This causes the rail of the suspension grid to be caught between the first half rail clamp **1512** and the second half rail clamp **1513**.

In one or more embodiments, the first base member portion **1508** defines a first half head-receiving aperture **(2001)**, while the second base member portion **1509** defines a second half head-receiving aperture **(2002)**. In one or more embodiments, coupling the first base member portion **1508** to the second base member portion **1509** causes the first half head-receiving aperture **(2001)** to abut the second half head-receiving aperture **(2002)**, thereby resulting in the base member **1502** defining a head-receiving aperture **1606** into which the head **1605** of the hanger **1501** may situate.

A head **1605** of a hanger **1501** situates within the head receiving aperture defined by the first half head-receiving aperture **(2001)** and the second half head-receiving aperture **(2002)** abutting. In one or more embodiments, an upper surface of the head **1605** can be exposed to the first half rail clamp **1512** and the second half rail clamp **1513** through the head-receiving aperture in one or more embodiments. A coupler **1511** couples the first base member portion **1508** to the second base member portion **1509**. In one or more embodiments, threading the coupler **1511** onto the first base member portion **1508** and the second base member portion **1509** moves the head **1605** of the hanger **1501** toward the first half rail clamp **1512** and the second half rail clamp **1513** due to the head's exposure through the head-receiving aperture. Continuing to thread the coupler **1511** onto the first base member portion **1508** and the second base member portion **1509**, when a rail is situated between the first half rail clamp **1512** and the second half rail clamp **1513**, causes the head **1605** of the hanger **1501** to clamp onto the rail.

In one or more embodiments, the coupler **1511** engages one or more threads **1607** disposed on the exterior surface of the head-receiving aperture **1606**, which extend distally from the base member **1502** to retain the first base member portion **1508** against the second base member portion **1509** and the head **1605** of the hanger **1501** against the rail through the head-receiving aperture **1606**.

In one or more embodiments, to attach the key **2700** to the distal end **1601** of the hook **1504**, the distal end **1601** of the hook **1504** opposite the extension **1505** of the hanger **1501** is inserted into a distal end receiver **2801** defined within a connector **2702** of the key **2700**. In one or more embodiments, the distal end receiver **2801** is configured to frictionally retain the key **2700** to the distal end **1601** of the hook **1504**. However, in other embodiments, a latch, snap, one-way latch, or other mating feature, such as engagement protrusion **1604**, can be incorporated into either the distal end receiver **2801** or the distal end **1601** of the hook **1504** to engage an engagement recess **2803** or otherwise frictionally retain the key **2700** to the distal end **1601** of the hook **103**.

Thus, in one or more embodiments the distal end receiver **2801** defines at least one engagement recess **2803**. In one or more embodiments, the distal end **1601** of the hook **1504** comprises at least one engagement protrusion **1604** projecting from the distal end **1601**. In one or more embodiments,

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the at least one engagement protrusion **1604** engages the engagement recess **2803** when the distal end **1601** is inserted into the distal end receiver **2801** to retain the key **2700** to the distal end **1601**.

In this illustrative embodiment, a flat surface **1608** and the distal end **1601** of the hook **1504** mates with a corresponding flat surface (**2703**) of the distal end receiver **2801** to prevent rotation of the key **2700** about the distal end **1601** of the hook **1504**. To wit, here the distal end receiver **2801** defines a corresponding flat surface **2703** and two arched protrusions. The distal end **1601** of the hook **1504** has a flat surface **1608** and two arched indentations that are complementary in shape to the two arched protrusions.

In one or more embodiments, when the distal end **1601** of the hook **1504** opposite the extension **1505** of the hanger **1501** is inserted into the distal end receiver **2801** of the key **2700**, engagement of the two arched protrusions with the two arched indentations, opposite engagement of the complementary flat surface **2703** of the distal end receiver **2801** with the flat surface **1608** of the distal end **1601** of the hook **1504** prevents rotation of the key **2700** about the distal end **1601** of the hook **103**.

Turning now to FIG. **32**, illustrated therein are one or more method steps in accordance with one or more embodiments of the disclosure. Beginning at step **3201**, the first base member portion **1508** is positioned such that its first half rail clamp **1512** situates about a rail **2207**. At step **3202**, the second base member portion **1509** is positioned such that its second half rail clamp **1513** situates about the rail **2207**. In one or more embodiments, this results in the one or more bosses **2005,2006** of the first base member portion **1508** inserting into the one or more boss receivers (**2007,2008**) of the second base member portion **1509** when the first base member portion **1508** and the second base member portion **1509** are pressed together such that their inner surfaces (**2003,2004**) abut at the medial major axis (**2010**). It also captures the rail **2207** between the first half rail clamp **1512** and the second half rail clamp **1513**.

At step **3203**, the head **1605** of the hanger **1501** is inserted into the head receiver formed by the first half head receiver **2015** extending distally from the bottom of the first base member portion **1508** and the second half head receiver **2016** extending distally from the bottom of the second base member portion **1509**. In one or more embodiments, the head **1605** of the hanger **1501** is inserted into the head receiver until its upper surface abuts the rail **2207**.

At step **3204**, the coupler **1511** is threaded onto the threads defined by the exterior surfaces (**2019,2020**) of the first half head receiver **2015** and the second half head receiver **2016** to couple the first base member portion **1508** and the second base member portion **1509** together.

In this illustrative embodiment, the first half head receiver **2015** defines a first half head-receiving aperture (**2001**). Similarly, the second half head receiver **2016** defines a second half head-receiving aperture (**2002**). When the head **1605** of the hanger **1501** is situated within the head-receiving aperture **1606** created by the first half head-receiving aperture (**2001**) and second half head-receiving aperture (**2002**) abutting, threading the coupler **1511** is threaded onto the threads defined by the exterior surfaces (**2019,2020**) of the first half head receiver **2015** and the second half head receiver **2016** causes the head **1605** of the hanger (**1501**) to advance toward the rail **2207**. This advancement of the head **1605** toward the rail **2207** causes the head **1605** to clamp the rail **2207** between the top of the head **1605** and the underside of the first half rail clamp **1512** and second half rail clamp **1513**.

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At step **3205**, the key **2700** can be attached to the distal end (**1601**) of the hook **1504**. In one or more embodiments, to attach the key **2700** to the distal end (**1601**) of the hook **1504**, the distal end (**1601**) of the hook **1504** opposite the extension **1505** of the hanger **1501** is inserted into a distal end receiver **2801** defined within the connector **2702** of the key **2700**. In one or more embodiments, the distal end receiver **2801** is configured with an engagement recess (**2803**) that receives an engagement protrusion (**1604**) projecting from a surface of the hook **1504**. A flat surface (**1608**) and the distal end (**1601**) of the hook (**1504**) can mate with a corresponding flat surface (**2703**) of the distal end receiver **2801** to prevent rotation of the key **2700** about the distal end (**1601**) of the hook **1504**.

As step **3206**, a track **3207** can then be attached to the key **2700**. In this illustrative embodiment, the track **3207** includes a first flat side **3208** and a second flat side **3209**. An arch **3210** spans the first flat side **3208** and the second flat side **3209** in this embodiment. In one embodiment, the track **3207** is flexible so that it can be shaped into different contours when attached to the key **2700**.

In one embodiment, the track **3207** is a continuous piece. In other embodiments, segments of different tracks can be aligned end-to-end to form a composite track. For example, in some embodiments sections of the track **3207** can be coupled together in a straight line using a track connector illustrated in FIG. **46**, or in curved liens using a track connector illustrated in FIG. **47**. Other techniques for coupling sections of the track **3207** together will be obvious to those of ordinary skill in the art having the benefit of this disclosure.

In one embodiment, the track **3207** is malleable. In one embodiment the track **3207** is configured so as to be easily cleanable. In one embodiment, the track **3207** is manufactured so as to be light beige in color. In another embodiment, the track **3207** is manufactured so as to be white in color. Other colors for the track **3207** will be obvious to those of ordinary skill in the art having the benefit of this disclosure.

In one embodiment, the track **3207** is cut to predefined lengths, such as twenty-foot lengths. In one or more embodiments, the ends of the track **3207** can be contoured for smooth interconnection to adjacent track segments.

The inclusion of the first flat side **3208**, the second flat side **3209**, and the arch **3210** define a peninsular indentation **3211** into which the key **2700** may be inserted. In this illustrative embodiment, the track **3207** comprises one or more complementary barbs **3212**. Each of the one or more complementary barbs **3212** is complementary in shape to the one or more barbs **2704** of the key **2700**. Here, each complementary barb **3212** comprises an inverted hemi-triangular protrusion with a base of each inverted hemi-triangular protrusion being disposed above a side of the inverted hemi-triangular protrusion.

Using this configuration, the one or more barbs **2704** of the key **2700** facilitate one-way penetration of the key **2700** into the track **3207**. In one or more embodiments, the track **3207** is manufactured from a pliable material, such as a thermoplastic. When the key **2700** is inserted into the peninsular indentation **3211**, the second flat side **3209** of the track **3207** flexes so that the one or more complementary barbs **3212** of the track **3207** pass over the one or more barbs **2704** of the key **2700**. Once the key **2700** is fully inserted into the peninsular indentation of the track **3207**, second flat side **3209** of the track **3207** flexes back toward the key **2700**, thereby causing the one or more complementary barbs **3212**

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to engage the one or more barbs **2704** of the key **2700**. This results in the key **2700** being frictionally retained within the track **3207**.

In one or more embodiments, the first flat side **3208** of the track **3207** comprises a hemi-cylindrical latching protrusion **3213** that wraps about the bottom edge **3214** of the second major surface **2901** of the key **2700**. In one or more embodiments, the bottom edge **3214** of the second major surface **2901** of the key **2700** is concave so as to provide a complementary indentation into which the hemi-cylindrical latching protrusion **3213** seats when the key **2700** is fully inserted into the track **3207**. This configuration advantageously covers the bottom portion of the key **2700** like a hood, which prevents curtain hooks sliding along the track **3207** from bumping into the key **2700** or the hanger **1501** when passing over portions of the track **3207** coupled to the key **2700**.

In operation, when the key **2700** is inserted into the peninsular indentation **3211**, the first flat side **3208** of the track **3207** flexes so the hemi-cylindrical latching protrusion **3213** of the track **3207** can pass over the second major surface **2901** of the key **2700**. Once the key **2700** is fully inserted into the peninsular indentation **3211** of the track **3207**, first flat side **3208** of the track **3207** flexes back toward the key **2700**. In one or more embodiments, this cause the hemi-cylindrical latching protrusion **3213** to seat around, or where convex within, the bottom edge **3214** of the second major surface **2901** of the key **2700**. This engagement assists in retaining the key **2700** within the track **3207** in one or more embodiments.

Turning now to FIGS. **33-37**, illustrated therein is one explanatory hanger **3300** configured in accordance with one or more embodiments of the disclosure. In one or more embodiments, the hanger **3300** can be suspended from a track (**3207**) to hang an object coupled thereto, such as a curtain.

In one or more embodiments, the hanger **3300** comprises a hanger arm **3301**, a hanger crook **3302**, and a hanger footer **3303**. In one or more embodiments, each of the hanger arm **3301**, the hanger crook **3302**, and the hanger footer **3303** are manufactured as a singular, unitary component. For example, the hanger **3300** can be manufactured as a singular, unitary component comprising a thermoplastic material manufactured by way of an injection molding process. The hanger **3300** can be manufactured from nylon, polypropylene, high-density polyethylene, low-density polyethylene, styrene, ABS, polycarbonate, or polycarbonate-ABS, PMMA, PVC, or other polyamide-based thermoplastics in one embodiment. Other materials suitable for manufacturing the hanger **3300** will be obvious to those of ordinary skill in the art having the benefit of this disclosure.

In this illustrative embodiment, the hanger arm **3301** includes a flat surface **3304** and a contoured surface **3401**. In this illustrative embodiment, the contoured surface **3401** comprises a central arch **3501** that extends distally outward from the rear side of the hanger arm **3301**. In one or more embodiments, the central arch **3501** sits between a first minor convex contour **3502** and a second minor convex contour **3503**. As will be described in more detail below with reference to FIGS. **39-40**, in one or more embodiments the flat surface **3304** and contoured surface **3401** can be used with a pair of backer plates to prevent objects to which the backer plates are coupled, e.g., a curtain, from twisting about the hanger arm **3301**.

In one or more embodiments, the hanger arm **3301** is situated between the hanger crook **3302** and the hanger footer **3303**. In this illustrative embodiment, the hanger

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footer **3303** is a hemi-cylindrical member having a radius that is greater than its height, as can best be seen in FIG. **33**. This shape for the hanger footer **3303** is illustrative only, as the hanger footer **3303** can be configured with other shapes as well. Illustrating by example, in another embodiment the hanger footer **3303** is configured with a rectangular shape. Other shapes for the hanger footer **3303** will be obvious to those of ordinary skill in the art having the benefit of this disclosure.

In one or more embodiments, the hanger footer **3303** is wider than the width of the hanger arm **3301**. The hanger footer **3303** can include one or more apertures **3402,3403**. In one embodiment, the one or more apertures **3402,3403** pass through the hanger footer **3303** along an axis that is situated parallel to the diameter of the hanger footer **3303**. In other embodiments, the one or more apertures **3402,3403** partially penetrate into the hanger footer **3303**, but do not pass all the way through. Where included, the one or more apertures **3402,3403** can be used as insertion points for a tool passing through the one or more apertures **3402,3403** to mount the hanger **3300** to a track (**3207**), move the hanger **3300** along the track (**3207**), or otherwise manipulate the hanger **3300** as desired. Mechanical stop **3404** can be used in a similar manner.

However, the one or more apertures **3402,3403** can offer other advantages as well, such as reducing the overall thickness of sections of the hanger footer **3303** to prevent sinks or other issues when the hanger **3300** is manufactured. Illustrating by example, as shown in FIGS. **33-37**, the hanger footer **3303** has a thickness and size that provides weight to the bottom of the hanger **3300**, which helps to keep the hanger **3300** on a track while sliding.

In one or more embodiments, the contoured surface **3401** of the hanger **3300** comprises a mechanical stop **3404** situated between the hanger arm **3301** and the hanger crook **3302**. As will be described in more detail below with reference to FIGS. **39-40**, in one or more embodiments a pair of backer plates can be coupled about the hanger arm **3301** between the mechanical stop **3404** and the hanger footer **3303**. In one or more embodiments, this configuration allows the backer plates, and anything coupled thereto, such as a curtain, to translate vertically along the hanger arm **3301** between the mechanical stop **3404** and the hanger footer **3303**. As best shown in FIG. **36**, in one or more embodiments the mechanical stop **3404** has at least one circular major face **3601**. In this illustration, the mechanical stop **3404** is cylindrical in contour and thus includes two circular major faces.

In one or more embodiments, the hanger crook **3302** extends from the hanger arm **3301** where the mechanical stop **3404** is attached toward a distal end **3305**. In one or more embodiments, the hanger crook **3302** has a diameter **3405** that is wide enough so that it can easily slip over, and situate upon, the arch (**3210**) spanning the first flat side (**3208**) and the second flat side (**3209**) of a track (**3207**). Accordingly, in one or more embodiments the diameter **3405** of the hanger crook **3302** is wider than the exterior surface diameter of the arch (**3210**) of the track (**3207**).

Turning now to FIG. **38**, illustrated therein is an exploded view of one illustrative hanger assembly **3800** configured in accordance with one or more embodiments of the disclosure. The hanger assembly **3800** comprises the hanger **3300** of FIGS. **33-37**, a first backer plate **3801** and a second backer plate **3802**.

In this illustrative embodiment, the first backer plate **3801** is substantially planar and has a flat major face **3803** that abuts the flat surface **3304** of the hanger arm **3301** between

the mechanical stop **3404** and the hanger footer **3303**. A partial assembly **3901** showing how this occurs is illustrated in FIG. **39**.

In one or more embodiments, the first backer plate **3801** comprises one or more fasteners **3804** that can be used to couple the first backer plate **3801** to the second backer plate **3802** around the hanger arm **3301** of the hanger **3300**. In this illustrative embodiment, the one or more fasteners **3804** comprise a plurality of rivets that are molded into the first backer plate **3801**. While rivets are one example of fasteners suitable for use with embodiments of the disclosure, other types of fasteners can be substituted as will be obvious to those of ordinary skill in the art having the benefit of this disclosure. Also, it should be noted that the rivets shown in FIGS. **38-41** and **43** are shown in their unriveted configuration. Accordingly, they protrude from the first backer plate **3801** in these figures. When riveted with a flexible substrate **3805** situated between the first backer plate **3801** and the second backer plate **3802**, they will become flat.

In one or more embodiments, a flexible substrate **3805** is positioned between the flat major face **3803** of the first backer plate **3801** and the flat surface **3304** of the hanger arm **3301**. In one or more embodiments, the flexible substrate **3805** comprises a curtain. In one or more embodiments, the one or more fasteners **3804** pass through the flexible substrate **3805** and maintain a fixed coupling between the first backer plate **3801** and the flexible substrate **3805**.

The second backer plate **3802** then includes a contoured surface receiver **3806**. In one or more embodiments, the contoured surface receiver **3806** is configured to situate around the contoured surface **3401** of the hanger arm **3301**. In one or more embodiments, the contoured surface receiver **3806** is situated between two planar extensions (one planar extension **3810** is shown in FIG. **38** and the other is shown in FIG. **41**).

In one or more embodiments, each of the two planar extensions define one or more apertures through which the one or more fasteners **3804** may pass, and optionally with which the one or more fasteners **3804** may engage, when the first backer plate **3801** is placed against the flat surface **3304** of the hanger arm and the contoured surface **3401** of the hanger arm **3301** is situated within the contoured surface receiver **3806** of the second backer plate **3802** such that the two planar extensions abut the flat major face **3803** of the first backer plate **3801**, or alternatively abut the flexible substrate **3805** situated between the two planar extensions about the flat major face **3803** of the first backer plate **3801**.

Turning now to FIG. **39**, to assemble the hanger assembly (**3800**), the flat major face **3803** of the first backer plate **3801** is placed against a surface of the flexible substrate **3805**. In one or more embodiments, when this occurs, the one or more fasteners **3804** pass through, or form, apertures in the flexible substrate **3805** so as to retain the first backer plate **3801** at a fixed location along the flexible substrate **3805**. Once this occurs, the portion of the flexible substrate **3805** situated between the one or more fasteners **3804** can be positioned such that it abuts the flat surface **3304** of the hanger arm **3301** between the mechanical stop **3404** and the hanger footer **3303**. This results in the flexible substrate **3805** being sandwiched between the flat major face **3803** of the first backer plate **3801** and the flat surface **3304** of the hanger arm **3301** between the mechanical stop **3404** and the hanger footer **3303**.

Turning now to FIG. **40**, the second backer plate **3802** can then be positioned such that the contoured surface **3401** of the hanger arm **3301** situates within the contoured surface receiver **3806** of the second backer plate **3802**, with the

second backer plate **3802** positioned between the mechanical stop **3404** and the hanger footer **3303**. In one or more embodiments, this results in the one or more fasteners **3804** passing through, and engaging, the one or more apertures **3809** of the second backer plate **3802**. Where, for example, the one or more fasteners **3804** are rivets, the rivets can be expanded so that they are wider than the one or more apertures **3809**, thereby retaining the first backer plate **3801** to the second backer plate **3802**, as well as the flexible substrate (**3805**) not shown in FIG. **40** for clarity) between the first backer plate **3801** and the second backer plate **3802**. In one or more embodiments, where so configured, the first backer plate **3801**, the second backer plate **3802**, and the flexible substrate (**3805**) can translate vertically along the hanger arm **3301** between the mechanical stop **3404** and the hanger footer **3303**. The completed hanger assembly **3800** is shown in FIGS. **41-42**.

Turning now to FIG. **43**, illustrated therein is one explanatory assembled system **4300** in accordance with one or more embodiments of the disclosure. As shown in FIG. **43**, the first base member portion **1508** is positioned such that its first half rail clamp **1512** situates about a rail **2207**. The second base member portion **1509** is positioned such that its second half rail clamp **1513** situates about the rail **2207**. The head (**1605**) of the hanger **1501** is inserted into the head receiver formed by the first half head receiver (**2015**) extending distally from the bottom of the first base member portion **1508** and the second half head receiver (**2016**) extending distally from the bottom of the second base member portion **1509** until its upper surface abuts the rail **2207**.

The coupler **1511** is threaded onto the threads defined by the exterior surfaces (**2019,2020**) of the first half head receiver (**2015**) and the second half head receiver (**2016**) to couple the first base member portion **1508** and the second base member portion **1509** together. The key **2700** is attached to the distal end (**1601**) of the hook **1504**. The track **3207** is attached to the key **2700**.

The hanger assembly **3800** can then be placed upon the track **3207**. The one or more hanger assembly **3800** can be coupled to a flexible substrate (**3805**) such as a curtain or other substrate, thereby suspending the flexible substrate (**3805**) from the track **3207**.

Embodiments of the disclosure contemplate that it can be tedious to place each hanger assembly **3800** upon the track **3207**, one by one, when there are many hanger assemblies, or when the track **3207** is very long. To make this process easier, embodiments of the disclosure include an optional tool that allows a plurality of hanger assemblies to be quickly and easily placed upon the track **3207**. Turning now to FIGS. **44-45**, illustrated therein is one such hanger tool **4400**. FIG. **44** shows a rear perspective view of the hanger tool **4400**, while FIG. **45** shows a front perspective view of the hanger tool **4400**.

The hanger tool **4400** of FIGS. **44-45** includes a track grabber **4401** that extends along a major axis of the hanger tool **4400**. In one or more embodiments, the track grabber **4401** is configured as a half U-shape with an arch **4402** and a major face **4403** extending distally downward from the arch **4402** such that the end of the hanger tool **4400** looks like a half-U with one upright missing.

In one or more embodiments, a hanger stop **4404** is situated at a first end of the track grabber **4401**, while the other end of the track grabber **4401** is open, and includes no stop. The inclusion of the hanger stop **4404** provides a mechanical wall against which hangers will not slide off the track grabber **4401**.

In one or more embodiments, the diameter of the arch **4402** of the track grabber **4401** gets larger at a resistance point **4405** that is located about seventy-five percent of the way down the length of the track grabber **4401** from the hanger stop **4404**. The resistance point **4405** helps to retain hangers between the hanger stop **4404** and the resistance point **4405** until the arch **4402** of the track grabber **4401** is placed atop a track **3207** and the hanger tool **4400** is tipped toward the open end, i.e., the end of the track grabber **4401** opposite the hanger stop **4404**, to a sufficient angle that any hangers situated along the track grabber **4401** slide over the resistance point **4405**, off the track grabber **4401**, and onto the track **3207**.

The hanger tool **4400** of FIGS. **44-45** facilitates quick and easy mounting of hanger assemblies on a track **3207** by allowing many hanger assemblies to be “scooped up” by the track grabber **4401** and placed between the hanger stop **4404** and the resistance point **4405**. Thereafter, the hanger tool **4400** can be placed such that the track grabber **4401** sits atop the track **3207**. The hanger tool **4400** can be tipped, with the hanger stop **4404** situated above the open end of the track grabber **4401** such that the hanger stop **4404** is a sufficient distance away from the track **3207** to allow gravity to cause the hanger assemblies to pass over the resistance point **4405** in one or more embodiments. In other embodiments, the hangers can be manually slid off the track grabber **4401** onto the track **3207** without tipping. For example, where the hanger assemblies are attached to a curtain, a person can place the track grabber **4401** atop the track **3207** and simply pull the curtain away from the hanger stop **4404** to slide the hangers over the resistance point **4405** and onto the track **3207**.

In one or more embodiments, the hanger tool **4400** includes goal post arms **4406,4407** that can be attached to a shaft or pole so that the hanger tool **4400** can be elevated to reach a track **3207**. In one or more embodiments, an internal snap protrusion can couple one or more apertures **4408** to one or more apertures of the shaft or pole. In one or more embodiments, the shaft or pole can be a mop handle so that a worker in a hospital can utilize a tool they already have to attach to the hanger tool **4400**.

Thus, as shown and described with reference to FIGS. **44-45**, in one or more embodiments a track hanger configured in accordance with one or more embodiments of the disclosure includes a hanger tool **4400**. In one or more embodiments, the hanger tool **4400** comprises a track grabber **4401**. In one or more embodiments, the track grabber **4401** is situated atop the track **3207**. In one or more embodiments, the track grabber **4401** defines a half-U shape comprising an arch **4402** and a flat side extending distally from the arch **4402**. In FIGS. **44-45**, the flat side is shown as a major face **4403** of the track grabber **4401**. In one or more embodiments, a diameter of the arch **4402** is wider at a resistance point **4405** of the track grabber **4401** than it is at other portions of the track grabber **4401** to retain hangers placed along the track grabber **4401** between the resistance point **4405** and the hanger stop **4404** until removal of the hangers from the hanger tool **4400** is desired.

Turning now to FIGS. **46-47**, illustrated therein are two track inserts **4600,4700** configured in accordance with one or more embodiments of the disclosure. The track insert **4600** of FIG. **46** is straight, while the track insert **4700** of FIG. **47** is curved. Where a track becomes a long continuous piece, the track insert **4600** can be inserted into the track to provide added rigidity, thereby preventing waviness and/or bouncing hooks as the curtain slides along the track. The track insert **4700** of FIG. **47** can be inserted into curved

portions of the track for the same purpose, namely, providing additional support at corners of the track where the supporting ceiling hooks tend to be situated further apart.

In one or more embodiments, each track connector **4600,4700** includes a first major surface **4601,4701** and a second major surface. While not shown in FIGS. **46-47**, in one or more embodiments the second major surface is simply a flat surface, with the flat surface being bent into a contour in accordance with the curvilinear shape of the track connector **4700** of FIG. **47**, and substantially planar with the track connector **4600** of FIG. **46**.

In one or more embodiments, the first major surface **4601,4701** defines one or more barbs **47602,4702**. In one or more embodiments each barb comprises a hemi-triangular protrusion with the base of each hemi-triangular protrusion being disposed beneath the side of the hemi-triangular protrusion.

In one or more embodiments, sections of track (**106,3207**) attach to the track connector **4600,4700** so that the sections of track (**106,3207**) can be coupled together to make a longer track. For example, in one or more embodiments the track (**106**) has a first flat side (**407**) and a second flat side (**408**), with an arch (**409**) spanning between ends of the first flat side (**407**) and the second flat side (**408**). The track (**106**) can be flexible so that it can be shaped into different contours when attached the track connectors **4600,4700**.

In one or more embodiments, the inclusion of the first flat side (**407**), the second flat side (**408**), and the arch (**409**) define a peninsular indentation (**410**) into which the track connectors **4600,4700** may be inserted. The track (**106**) can comprise one or more complementary barbs (**405**). Each of the one or more complementary barbs (**405**) is, in one or more embodiments, complementary in shape to the one or more barbs **4602,4702** of the track connectors **4600,4700**. For example, each complementary barb can comprise an inverted hemi-triangular protrusion with a base of each inverted hemi-triangular protrusion being disposed above a side of the inverted hemi-triangular protrusion.

Using this configuration, the one or more barbs **4602,4702** of the track connectors **4600,4700** facilitate one-way penetration of the track connectors **4600,4700** into the track (**106**). In one or more embodiments, when the track connector **4600,4700** is inserted into the peninsular indentation (**410**), the first flat side (**407**) of the track (**106**) flexes so that the one or more complementary barbs (**405**) of the track (**106**) pass over the one or more barbs **4602,4702** of the track connectors **4600,4700**. Once the track connector **4600,4700** is fully inserted into the peninsular indentation (**410**) of the track (**106**), first flat side (**407**) of the track (**106**) flex back toward the track connector **4600,4700**, thereby causing the one or more complementary barbs (**405**) to engage the one or more barbs **4602,4702** of the track connectors **4600,4700**. This results in the track connectors **4600,4700** being frictionally retained within the track (**106**) in one or more embodiments.

Turning now to FIG. **48**, illustrated therein are various embodiments of the disclosure. Beginning at **4801**, a hanger comprises a hanger arm. At **4801**, the hanger comprises a hanger crook. At **4801**, the hanger comprises a hanger footer. At **4801**, the hanger comprises a mechanical stop.

At **4801**, the hanger arm is situated between the hanger crook and the hanger footer. At **4801**, the mechanical stop is coupled to the hanger between the hanger crook and the hanger arm.

At **4802**, the hanger arm of **4801**, the hanger crook, and the hanger footer are manufactured as a singular, unitary component. At **4803**, the hanger arm of **4801** defines a flat

surface and a contoured surface. At **4804**, the contoured surface of **4803** comprises a central arch situated between a first minor convex curvature and a second minor convex curvature.

At **4805**, the mechanical stop of **4804** is coupled to, and extends distally from, the central arch. At **4806**, the hanger footer of **4805** defines a radius that is greater than a height of the hanger footer. At **4807**, the hanger footer of **4805** comprises a hemi-cylindrical member having another flat surface that is coplanar with the flat surface of the hanger arm.

At **4808**, the hanger footer of **4807** defines one or more apertures. At **4809**, the one or more apertures of **4808** pass through the hanger footer along an axis that is oriented parallel to a diameter of the hanger footer. At **4810**, the axis of **4809** is oriented parallel to the other flat surface.

At **4811**, the hanger of **4801** comprises a first backer plate coupled to a second backer plate around the hanger arm between the mechanical stop and the hanger footer. At **4812**, the first backer plate of **4811** comprises one or more fasteners passing through, and engaging, one or more apertures defined by the second backer plate.

At **4813**, the hanger of **4812** comprises a flexible substrate coupled between the first backer plate and the second backer plate. At **4814**, the first backer plate of **4813** and the second backer plate are configured to translate along the hanger arm between the mechanical stop and the hanger footer.

At **4815**, a track hanger comprises a hook, a head, and an extension. At **4815**, the extension is disposed between the head and the hook. At **4815**, the track hanger comprises a base member. At **4815**, the base member comprises a first half rail clamp and a second half rail clamp. At **4815**, the base member comprises a head receiver defining a head-receiving aperture exposing the head to the first half rail clamp and the second half rail clamp when the head is inserted into the head-receiving aperture.

At **4815**, the track hanger comprises a key that is selectively attachable to the hook. At **4815**, the key comprises a first plurality of ledges and a second plurality of ledges situated within a partially cylindrical protuberance extending from a first major face of the key and a coupler. At **4815**, the head is insertable into the head receiver and the coupler is operable to engage an exterior surface of the head receiver to move the head through the head-receiving aperture toward the first half rail clamp and the second half rail clamp.

At **4816**, the first major surface of the key of **4815** defines one or more barbs separated from the partially cylindrical protuberance by a planar portion of the first major surface. At **4817** the track hanger of **4816** further comprises a track. At **4817**, the one or more barbs facilitate one-way penetration of the key into the track.

At **4818**, the track hanger of **4817** further comprises one or more hangers hanging from the track. At **4819**, each hanger of the one or more hangers of **4818** comprises a hanger arm, a hanger crook, a hanger footer, and a mechanical stop. At **4819**, the hanger arm is situated between the hanger crook and the hanger footer and the mechanical stop is coupled to the hanger between the hanger crook and the hanger arm.

At **4820**, each hanger of **4819** further comprises a first backer plate coupled to a second backer plate around the hanger arm between the mechanical stop and the hanger footer. At **4820**, a flexible substrate is coupled between the first backer plate and the second backer plate. At **4820**, the flexible substrate abuts a flat surface of the hanger arm. At **4820**, the first backer plate comprises one or more fasteners

passing through, and engaging, one or more apertures defined by the second backer plate.

In the foregoing specification, specific embodiments of the present disclosure have been described. However, one of ordinary skill in the art appreciates that various modifications and changes can be made without departing from the scope of the present disclosure as set forth in the claims below. Thus, while preferred embodiments of the disclosure have been illustrated and described, it is clear that the disclosure is not so limited. Numerous modifications, changes, variations, substitutions, and equivalents will occur to those skilled in the art without departing from the spirit and scope of the present disclosure as defined by the following claims. Accordingly, the specification and figures are to be regarded in an illustrative rather than a restrictive sense, and all such modifications are intended to be included within the scope of present disclosure. The benefits, advantages, solutions to problems, and any element(s) that may cause any benefit, advantage, or solution to occur or become more pronounced are not to be construed as a critical, required, or essential features or elements of any or all the claims.

What is claimed is:

1. A track hanger, comprising:

a hook, a head, and an extension, the extension disposed between the head and the hook;

a key that is selectively attachable to the hook and comprises a plurality of barbs;

a track that is selectively attachable to the key and comprises a plurality of complementary barbs;

wherein:

the plurality of barbs facilitates one-way penetration of the key into the track when the plurality of complementary barbs pass over the plurality of barbs with the plurality of barbs thereafter engaging the plurality of complementary barbs to frictionally retain the track to the key.

2. The track hanger of claim 1, wherein a first major surface of the key defines the plurality of barbs.

3. The track hanger of claim 2, wherein:

the track defines a first flat side and a second flat side; and the plurality of complementary barbs are configured to pass over the plurality of barbs when the key inserts between the first flat side and the second flat side of the track.

4. The track hanger of claim 3, wherein the track further defines an arch spanning between the first flat side and the second flat side.

5. The track hanger of claim 4, wherein the key is configured to insert between the first flat side and the second flat side of the track until it abuts an interior surface of the arch.

6. The track hanger of claim 5, wherein the key further comprises a first plurality of ledges and a second plurality of ledges situated within a partially cylindrical protuberance extending from a first major face of the key.

7. The track hanger of claim 1, wherein the track is continuous.

8. The track hanger of claim 1, wherein the track comprises a plurality of segments aligned end-to-end.

9. The track hanger of claim 8, further comprising one or more track connectors coupling together the plurality of segments.

10. The track hanger of claim 1, wherein the track is malleable.

11. The track hanger of claim 1, wherein the track defines a hemi-cylindrical latching protrusion configured to wrap

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around a bottom edge of a major surface of the key when the track is frictionally retained to the key.

12. The track hanger of claim 11, wherein the bottom edge of the key is concave, thereby defining a complementary indentation into which the hemi-cylindrical latching protrusion seats when the key is fully inserted into the track.

13. The track hanger of claim 1, wherein the hook, the head, and the extension are manufactured as a singular, unitary component.

14. A track hanger, comprising:

a hook;

a key that is selectively attachable to the hook; and

a track that is selectively attachable to the key;

wherein:

the key defines a major surface having a plurality of barbs extending therefrom; and

the track defines a peninsular indentation into which the key may insert defining a plurality of complementary barbs that engage the plurality of barbs to frictionally retain the track to the key with the key is fully inserted into the peninsular indentation.

15. The track hanger of claim 14, wherein the plurality of barbs and plurality of complementary barbs facilitate one-way penetration of the key into the track.

16. The track hanger of claim 14, wherein each barb of the plurality of complementary barbs comprises an inverted hemi-triangular protrusion with a base of each inverted

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hemi-triangular protrusion being disposed above a side of the inverted hemi-triangular protrusion.

17. The track hanger of claim 14, wherein the key further comprises a first plurality of ledges and a second plurality of ledges situated within a partially cylindrical protuberance extending from a first major face of the key.

18. The track hanger of claim 14, wherein the track further defines a hemi-cylindrical latching protrusion configured to wrap around a bottom edge of the major surface of the key.

19. A track hanger, comprising:

a hook;

a key that is selectively attachable to the hook; and

a malleable track that is selectively attachable to the key;

wherein:

the key defines a major surface having a plurality of barbs extending therefrom;

the malleable track defines a peninsular indentation into which the key may insert defining a plurality of complementary barbs; and

the malleable track further defines a hemi-cylindrical latching protrusion configured to wrap around a bottom edge of a surface of the key.

20. The track hanger of claim 19, wherein the plurality of barbs are configured to engage the plurality of complementary barbs to frictionally retain engagement of the track to the key.

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